VOLKSWAGEN GROUP INDIA

T-WORKS × WOXSEN UNIVERSITY

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PORTFOLIO REPORT · INTEGRATED EDITION

CIRCULAR & CARBON ECONOMY

INDIA'S POLICY LANDSCAPE

SUPPLY CHAIN PREPAREDNESS

ELEVEN PoC INTERVENTIONS

COMPREHENSIVE INTEGRATED EDITION

DECARBONISATION · MATERIAL CIRCULARITY · DIGITAL TRACEABILITY · SUSTAINABILITY INTELLIGENCE

SUBMITTED BY T-WORKS & WOXSEN UNIVERSITY

Table of Contents

Executive Summary

70.5/100	131/200	11	INR 70–140 Cr
SCP Index Supply Chain Preparedness	**Avg PVI 2020–25** Sustained Elevated	**PoC Interventions** Across 4 Thematic Pillars	**Portfolio NPV (5yr)** Base Case

This report constitutes the definitive, integrated strategic document for Volkswagen Group India's Circular and Carbon Economy Programme - the product of a structured research collaboration between T-Works and Woxsen University's Central R&D Office. It synthesises quantitative policy analysis, primary survey data from 46 automotive supply chain professionals, eleven detailed Proof of Concept interventions with full business feasibility assessments, industry intelligence on closed-loop polymer markets, and a prioritised partner ecosystem - into a unified framework for executive decision-making and operational deployment.

The report is organised across six interconnected parts. Part I constructs the Policy Volatility Index (PVI) using the Baker-Bloom-Davis EPU methodology, revealing that India's automotive regulatory environment has been operating at an average PVI of 131 over the 2020–2025 period - consistently in the 'Elevated' band, driven by the FAME-III policy vacuum, CAFE-III 2027 uncertainty, and persistent cross-ministry divergence on Scope 3 emission disclosure mandates. Part II presents the Supply Chain Preparedness Index (SCP), derived from a structured 87-item Likert questionnaire administered to supply chain professionals across the Pune-Mumbai-Nashik and NCR clusters, recording an overall score of 70.5/100 - a finding of creditable capability in a structurally volatile environment, with the Perceived Barriers sub-dimension at 58.3/100 and Supply Chain Emissions at 65.2/100 as the highest-risk nodes.

Part III presents the eleven Proof of Concept interventions - the operational core of the report - structured across four thematic pillars: Decarbonisation (PoCs 1–4), Supply Chain Infrastructure (PoCs 5, 8, 11), Material Circularity (PoCs 7, 9, 10), and Sustainability Intelligence (PoCs 6, 11). Each PoC is elaborated with full technical specification, India-specific business feasibility analysis, OEM benchmarking from global and Indian automotive peers, detailed financial modelling, and contextualisation to SAVWIPL's Pune manufacturing operations and the Škoda Kylaq product platform.

Part IV provides three chapters of dedicated industry intelligence: a closed-loop business case analysis for PP and PET in Indian automotive manufacturing; a landscape assessment of India's pyrolysis technology sector; and a partner prioritisation matrix covering the nine recommended ecosystem partners across three functional lanes - closed-loop reprocessing, scrap marketplace and digital traceability, and pyrolysis. Part V presents a 36-month implementation roadmap, PVI-adaptive governance architecture, portfolio-level financial modelling with a base case NPV of INR 70–140 crore, carbon credit monetisation strategy under India's CCTS framework, and a risk register covering policy, technology, and strategic risk categories. Part VI provides a regulatory deep dive into India's five most material compliance developments affecting VWG India's strategy through 2030.

Central Findings

The central finding of the SCP analysis is one of structured misalignment: Indian automotive supply chain participants have built genuinely competitive sustainability capabilities, yet these capabilities operate in a policy environment that has remained persistently above the threshold at which structured strategic intervention is warranted. Three sub-dimensions - OEM & Industry Pressure (74.2), Technology & Product Transition (74.1), and Strategic Carbon Governance (72.4) - demonstrate resilience to domestic policy volatility, driven by European OEM customer requirements and ESG investor pressure that operate independently of Indian regulatory timelines.

The central finding of the PoC portfolio analysis is that VWG India possesses the brand credibility, production scale, and global group mandate to lead India's automotive circular economy transition - and that the partner ecosystem required to execute this transition is commercially ready and actively seeking OEM partnerships. No organisational capability gap prevents action. The constraint is sequencing and activation discipline.

The Portfolio: Four Pillars, Eleven Interventions

PoC	Title	Pillar	Priority	ROI Window
1	Dynamic Carbon-Aware Supplier Routing	Decarbonisation	High	18 months
2	Micro-Grid Carbon Buffering System	Decarbonisation	High	24–36 months
3	Design for Disassembly Embedded in CAD	Material Circularity	Medium	Long-term Strategic
4	AI-Based Low-Carbon Material Substitution	Decarbonisation	High	18–24 months
5	Zero-Emission Packaging Loop	Supply Chain Infra	Very High	18–36 months
6	Green UX In-Vehicle CO_2 Dashboard	Sustainability Intel	Medium	Product Cycle
7	Closed-Loop PP/ABS Reprocessing	Material Circularity	Very High	12–24 months
8	Decentralised Micro-Recycling Units	Supply Chain Infra	High	36–60 months
9	Reactive Extrusion for Mixed	Material	Medium	24–36 months

	Plastics	Circularity		
10	Plastic-to-Fuel Pyrolysis	Material Circularity	Medium	24–48 months
11	Digital Traceability & Certification	Sustainability Intel	Very High	12–18 months

Portfolio Integration Architecture

The eleven PoCs are not independent interventions - they form a deeply integrated system in which the data outputs of upstream PoCs become the operational inputs and commercial credentials of downstream PoCs. Three integration layers structure the portfolio:

The Foundation Layer - PoCs 1, 8, and 11 - creates the data infrastructure, physical ELV processing capability, and digital traceability architecture without which the material circularity PoCs cannot generate EPR credits, carbon credits, or verified recycled content certificates. These three PoCs sit on the critical path of the entire portfolio.

The Material Layer - PoCs 7, 9, and 10 - operates a cascading recovery hierarchy that maximises value extraction from ELV plastic streams: mechanical closed-loop recycling for clean mono-material PP (PoC 7); reactive extrusion compatibilisation for mixed but sortable streams (PoC 9); and thermal pyrolysis for residual, contaminated fractions (PoC 10). Each PoC in this cascade is sequenced to deploy after its predecessor, ensuring that the highest-value pathway captures material first.

The Consumer Intelligence Layer - PoCs 3, 4, 6, and 2 - translates the foundation and material layer work into product design decisions, energy management improvements, and a driver-facing sustainability dashboard that completes the circle from supply chain to showroom. PoC 6's verified sustainability narrative is directly dependent on PoC 1's Scope 3 data and PoC 11's certified material origin records.

Financial Summary

The total portfolio capital investment ranges from INR 51 to 103 crore, deployed in phases over 36 months. The combined annual benefit by Year 3 - from packaging cost savings, material cost reductions, carbon credit revenues, EPR credit generation, and energy cost savings - is estimated at INR 41 to 81 crore per year. The blended portfolio payback period is 30 to 42 months under base case assumptions.

The five-year net present value of the portfolio (at a 10% discount rate) is INR 70 to 140 crore under the base case, escalating to INR 150 to 280 crore under an optimistic scenario in which FAME-III confirmation, CCTS pricing at INR 1,000 per tonne, and CBAM enforcement against VWG India's export components all materialise within the modelling horizon. Strategic optionality value - EPR compliance headroom, first-mover rPP positioning, and ESG rating improvement - adds a conservatively estimated INR 50 to 80 crore in total enterprise value.

Immediate Next Steps

The report concludes with a 90-day activation plan:

- Days 0–30: Initiate MoU negotiations with Banyan Nation (rPP), Recykal (digital traceability platform), and CERO (ELV scrap network); deploy PoC 5 packaging loop pilot across top 20 inbound component suppliers.
- Days 30–60: Commission CIPET independent material qualification audit for Tier A rPP components (fender liners); initiate Recykal platform onboarding for EPR compliance under Plastic Waste Management Rules 2022.
- Days 60–90: Convene inaugural Circular Material Council; appoint Council Chair; activate FAME-III watch protocol; initiate VWG India SAP API specification review for PoC 11 integration.

VWG India has the brand credibility, production scale, and global group mandate to be the first premium OEM in India to close the loop on PP and PET. The partner ecosystem - anchored by Banyan Nation, Recykal, CERO, and Ganesha Ecopet - offers a shovel-ready pathway to measurable circular content in VWG India vehicles by FY 2026–27. First-mover advantage in automotive rPP is real: Tata Motors has demonstrated it and Mahindra CERO has declared intent. The window to lead rather than follow is now.

PART I

India's Automotive Policy Landscape

Policy Volatility Index (PVI) · Baker-Bloom-Davis EPU Methodology · 2015–2025

Chapter 1: India's Automotive Policy Context & Significance

1.1 The Case for a Policy Volatility Index

India's automotive sector occupies a uniquely complex position in the global transition to sustainable mobility. It is simultaneously one of the world's largest vehicle markets, a key node in global automotive supply chains, a major employer across formal and informal labour, and a sector undergoing fundamental technological transformation - from internal combustion engines to electric powertrains, from linear to circular material flows, and from fragmented to digitally integrated supply chain management.

The Policy Volatility Index (PVI) was developed specifically to provide quantitative understanding of this environment. Unlike generalised measures of economic policy uncertainty, the PVI is sector-specific, India-focused, and designed to capture the four distinct channels through which regulatory uncertainty affects automotive supply chain participants: the frequency of new regulations; the intensity of policy reversals and subsidy disruptions; the degree of cross-ministry divergence on connected policy questions; and the level of media and market uncertainty generated by ambiguous or contested policy signals.

The index follows the Baker-Bloom-Davis EPU methodology (2016), which has been applied across economies and sectors to measure policy uncertainty and its effects on investment, employment, and economic output. Adapting this methodology to the Indian automotive context required constructing sector-specific indicators and calibrating the weighting structure to the particular mechanisms through which policy uncertainty affects OEMs and suppliers across the value chain.

1.2 The Four Pillars of Automotive Policy Volatility

Pillar	Name	Measures	Weight
RF	Regulatory Frequency	Annual count of gazette notifications, CAFE/BS phase changes, FAME/PLI updates. Sources: MoRTH gazettes, BEE announcements, SIAM policy tracker	25%
PRI	Policy Reversal Intensity	Frequency × magnitude of U-turns: subsidy cuts, deadline changes, scope reversals. Each reversal scored 1–3 by impact	25%
CMD	Cross-Ministry Divergence	Conflicts between MoRTH, MHI, BEE, Finance Ministry, NITI Aayog on EV/emission policy. Scored by documented disagreements	25%
MNU	Media & News Uncertainty	Relative article frequency in ET, Business Standard, The Hindu using EPU keyword search. Normalised to baseline mean	25%

The Regulatory Frequency pillar captures the volume of new rules, amendments, and policy updates that supply chain participants must absorb each year. High regulatory frequency, even when directionally appropriate, imposes compliance costs, planning burdens, and investment uncertainty - particularly on smaller Tier-2 and Tier-3 suppliers that lack the regulatory monitoring infrastructure of large OEMs.

The Policy Reversal Intensity pillar is particularly consequential in the Indian context, where the FAME-II subsidy cuts of 2022, the BS-IV to BS-VI compression, and the retracted diesel tax threat of 2023 are canonical high-PRI events that fundamentally altered investment planning horizons across the sector.

Cross-Ministry Divergence reflects a structural feature of the Indian policy environment: the automotive and EV policy space spans at least five central ministries - MoRTH, MHI, BEE/MoP, Finance, and NITI Aayog - whose priorities and timelines do not always align. The Media and News Uncertainty pillar captures the ambient uncertainty through systematic keyword-based article tracking across leading financial press outlets.

1.3 Score Interpretation Framework

PVI Band	Score Range	Strategic Implication
Stable	< 80	Low regulatory flux; high predictability; investment commitments can be made with confidence
Moderate	80–120	Normal rulemaking; manageable uncertainty; standard scenario planning sufficient
Elevated	121–160	Active phase transitions; conflicting signals; market caution warranted; dual-track planning required
Volatile	> 160	Sudden reversals; cross-ministry conflict; investment freeze risk; only policy-neutral capex advisable

The PVI is not a static measurement - it is an early-warning management tool. When PVI exceeds 130, supply chain participants are advised to activate policy-neutral investment tracks and defer discretionary capex that depends on specific policy outcomes such as FAME subsidy continuation or CAFE target stability.

1.4 Strategic Relevance to VWG India

The PVI analysis directly contextualises Volkswagen Group India's strategic environment. The 2025 Volkswagen Group Annual Report confirms a 34.9% year-on-year demand increase in the Indian passenger car market in FY 2025, driven significantly by the Škoda Kylaq and Volkswagen Virtus. This growth trajectory makes policy stability in EV transition, emissions compliance, and circular economy regulation increasingly material to VWG India's long-range investment decisions.

VWG's global goTOzero strategy commits the Group to net carbon neutrality by 2050, with intermediate targets of a 55% reduction in the lifecycle carbon footprint of its vehicle fleet by 2030 (versus 2018). In India, where the PVI remains elevated, translating these global commitments into actionable supply chain investments requires exactly the kind of dual-track, policy-adaptive planning the PVI is designed to support.

Chapter 2: Policy Volatility Index - Annual Analysis (2015–2025)

2.1 Annual PVI Scores

Year	RF	PRI	CMD	MNU	PVI	Band
2015	88	72	85	80	81	Moderate
2016	95	80	90	88	88	Moderate
2017	110	90	95	105	100	Moderate
2018	118	115	120	125	120	Moderate
2019	105	108	112	118	111	Moderate
2020 ▲	145	125	118	185	143	ELEVATED - PEAK
2021	120	100	108	128	114	Moderate
2022	130	142	135	148	139	Elevated
2023	125	138	140	142	136	Elevated
2024	115	120	125	130	123	Elevated
2025	128	130	135	140	133	Elevated

2.2 Phase Analysis

Phase 1: Calibration Era (2015–2019) | Average PVI: 97

The sector experienced moderate, largely predictable policy activity. FAME-I, CAFE-I, and BS-IV provided a structured but manageable regulatory environment. Cross-ministry alignment was relatively stronger during this period as policy agendas had not yet diversified into the full complexity of the EV transition. The GST rollout of 2017 introduced a temporary spike in CMD and PRI scores, but overall regulatory communication remained coherent. Investment decisions made during this period could be planned with reasonable confidence in policy stability.

Phase 2: Shock Compression (2020) | PVI: 143 - Peak

The BS-IV to BS-VI regulatory leap - compressing a nominally four-year transition into two years - represents the single largest regulatory shock in the sector's recent history. Simultaneously, COVID-19 introduced supply chain disruptions while the policy apparatus was effectively frozen. The MNU pillar spiked to 185 in 2020, reflecting extraordinary media coverage of policy uncertainty. The Scrappage Policy, which had been building as a regulatory landmark, was stalled indefinitely - adding a policy vacuum to an already turbulent environment.

Phase 3: Sustained Elevation (2021–2025) | Average PVI: 131

Despite no single catastrophic event comparable to 2020, the sector has not returned to pre-2020 baseline levels. FAME-II's subsidy cuts and eventual closure without a confirmed successor, the CAFE-III 2027 target debate, RDE Phase 2 norms implementation, EV fire investigations, and the ongoing FAME-III design vacuum are collectively generating persistent uncertainty that appears systemic rather than episodic. The 2023 retracted diesel tax threat and the 2024 FAME-II closure without FAME-III confirmation are emblematic of the PRI and CMD dynamics that keep the index elevated.

2.3 Critical Policy Events Driving PVI Elevation

Year	Policy Event	Pillars	Impact	Reversal?
2018	NITI Aayog 100% EV by 2030 - subsequently retracted	CMD, MNU	High	Yes
2020	BS-IV→BS-VI 4-year compression executed April 1	RF, PRI	High	No
2020	COVID-19 shutdown - policy freeze & production halt	MNU, CMD	High	No
2022	FAME-II subsidy cut 33% for two-wheelers (June)	PRI, MNU	High	Yes
2022	EV fire incidents → safety probe & recalls	MNU, CMD	High	Yes
2023	Nitin Gadkari diesel tax threat (Sep, retracted)	PRI, MNU	High	Yes
2023	FAME-II closure uncertainty - no FAME-III signal	PRI, MNU	High	Yes
2024	FAME-II ends; PM E-DRIVE bridge (Rs 500 Cr)	PRI, RF	High	Yes
2025	FAME-III design uncertainty - no confirmed budget	PRI, MNU	High	Yes

> The sustained elevation of India's PVI since 2020 is not merely a risk to manage - it is a strategic differentiation opportunity. OEMs that build policy-adaptive supply chain infrastructure now will hold structural cost and compliance advantages when the regulatory environment clarifies.

PART II

Supply Chain Preparedness

SCP Index · PVI Comparative Analysis · n=46 Supply Chain Professionals · March 2026

Chapter 3: Supply Chain Preparedness - Analytical Framework & Results

3.1 The Supply Chain Preparedness Index (SCP)

The Supply Chain Preparedness Index (SCP) was developed to provide a rigorous, quantitative assessment of how well Indian automotive supply chain participants are positioned to manage sustained policy volatility while advancing their decarbonisation and circularity commitments. The index is derived from a structured questionnaire administered to 46 automotive supply chain professionals in India (March 2026), covering OEMs, Tier-1 suppliers, and Tier-2 component manufacturers across the Pune-Mumbai-Nashik and NCR clusters.

Respondents scored 87 Likert-scale items (1–7) across two major pillars: the Decarbonisation Index (50% weight), covering 13 sub-dimensions spanning carbon governance, technology transition, manufacturing operations, supply chain disclosure, and green innovation; and the Material Circularity Index (50% weight), covering 5 sub-dimensions encompassing circular design integration, lifecycle value management, material innovation, and closed-loop collaboration. All raw scores were normalised to a 0–100 scale.

3.2 Overall SCP Results

70.5 **Overall SCP Index / 100** Good - Structured Misalignment	**71.7** **Decarbonisation Index / 100** Good	**69.3** **Material Circularity / 100** Good
123 **PVI (2024 Estimate)** Elevated Band	**58.3** **Perceived Barriers** Lowest Score - Critical Risk	**65.2** **SC Emissions** High Risk Node

3.3 Core Finding: Structured Misalignment

The core finding is one of structured misalignment. Supply chain participants have built genuinely competitive sustainability capabilities - a 70.5/100 SCP Index in a developing market context represents meaningful organisational investment. Yet the PVI has remained consistently elevated above 120 since 2020 - a level that warrants active strategic intervention.

Three SCP sub-dimensions demonstrate relative resilience to policy volatility - OEM & Industry Pressure (74.2), Technology & Product Transition (74.1), and Strategic Carbon Governance (72.4) - suggesting that private-sector demand signals and ESG investor pressure from European OEM customers are providing investment momentum independent of domestic policy clarity. These represent the 'resilience quadrant' of the SCP framework.

3.4 PVI vs SCP Comparative Matrix

Year	PVI	Policy Pressure Score	SCP Index	Prep. Gap	Status
2015	81	84	70.5	−13	Exposed
2016	88	78	70.5	−7	At Risk
2017	100	69	70.5	+2	Aligned
2018	120	54	70.5	+17	Well-Positioned
2019	111	61	70.5	+10	Aligned
2020	143	36	70.5	+35	Well-Positioned
2021	114	58	70.5	+13	Well-Positioned
2022	139	39	70.5	+32	Well-Positioned
2023	136	42	70.5	+29	Well-Positioned
2024	123	52	70.5	+19	Well-Positioned
2025	133	44	70.5	+27	Well-Positioned

Note: SCP Index reflects the 2026 survey cohort score applied uniformly across years for comparison. $PPS = ((190 - PVI) / 130) \times 100$. A positive Preparedness Gap ($Gap > 0$) indicates SCP exceeds policy pressure; $Gap < 0$ indicates policy pressure exceeds preparedness.

Chapter 4: Sub-Dimension Analysis & Strategic Recommendations

4.1 Decarbonisation Index Sub-Dimensions

Sub-Dimension	Score	PVI Exposure Risk
Strategic Carbon Governance	72.4	Medium - governance misaligns if CAFE/FAME policy reverses
Technology & Product Transition	74.1	High - BEV timelines sensitive to PLI/EV policy
Low-Carbon Manufacturing	70.3	Low-Medium - internal operations less policy-dependent
Supply Chain Decarbonisation	68.9	High - Scope 3 rules under cross-ministry dispute
Supply Chain Emissions	65.2 ⚠️	High - No finalised Scope 3 disclosure mandate
External & Investor Pressure	73.8	Medium - ESG pressure partially decoupled from domestic policy
Production Mix & Performance	71.6	High - CAFE-III 2027 target directly affects mix decisions
Carbon Measurement Capability	69.5	Medium - capability built, reporting standards shifting
Emissions Monitoring	67.4	Medium - BS-VI/RDE norms drive compliance investment
Operational Decarbonisation	70.8	Low - operational efficiency largely policy-independent
OEM & Industry Pressure	74.2	Medium - OEM requirements cascade independently
Perceived Barriers	58.3 ⚠️	Very High - barriers score mirrors PVI directly
Green Innovation Capability	73.1	Medium - PLI-linked but partially autonomous

4.2 Material Circularity Index Sub-Dimensions

Sub-Dimension	Score	Policy Linkage
Circular Economy Integration	71.2	Battery Waste Management Rules 2022 - direct mandate
Lifecycle Value	68.4	ELV recycling rules pending; no enforceable standard yet
Material Innovation	67.8	Green steel/aluminium; no procurement policy mandate
Circular Collaboration	70.6	OEM take-back; PM E-DRIVE partially incentivises
Circular Economy Practices	68.5	PLI linkages partial; FAME did not cover remanufacturing

4.3 High-Exposure Risk Nodes

Cross-mapping the lowest SCP sub-dimension scores against the most volatile PVI pillars reveals five high-priority risk nodes that merit immediate strategic attention:

- Perceived Barriers (58.3) - PVI Pillars: RF + PRI - Risk: Critical. The lowest-scoring sub-dimension directly mirrors the PVI. Investment cost barriers, green technology access, and policy uncertainty are explicit barrier items in the questionnaire. An elevated PVI structurally suppresses this score.
- Supply Chain Emissions (65.2) - PVI Pillars: CMD - Risk: High. Scope 3 quantification requires stable disclosure mandates. Cross-ministry divergence between BEE and MoRTH on lifecycle emission accounting creates a regulatory vacuum that allows firms to defer Scope 3 investment.
- Lifecycle Value (68.4) - PVI Pillars: RF + CMD - Risk: High. End-of-life vehicle recycling rules remain pending enforcement. Without a finalised ELV policy, firms cannot plan closed-loop material systems with confidence.
- Technology & Product Transition (74.1) - PVI Pillars: PRI + RF - Risk: Medium. Despite a healthy score, BEV production mix planning is acutely sensitive to FAME-III design and CAFE-III targets.
- Supply Chain Decarbonisation (68.9) - PVI Pillars: CMD + MNU - Risk: Medium-High. Tier-1 supplier emission disclosure requirements are stalled by cross-ministerial ambiguity.

4.4 Strategic Recommendations

For Supply Chain Firms

- Build Policy-Decoupled Roadmaps: Develop 3-year decarbonisation investment plans with policy-accelerated and policy-neutral scenarios. When PVI exceeds 130, activate the slower investment track.
- Prioritise Perceived Barriers Reduction: Establish internal green technology access programmes, industry consortia for shared green steel procurement, and dedicated regulatory monitoring cells.
- De-risk Scope 3 Investment: Adopt the BRSR Core framework voluntarily rather than waiting for finalised CMD-aligned Scope 3 mandates. Early adoption reduces exposure when mandatory standards arrive.
- Build ELV Circularity Ahead of Regulation: Firms establishing reverse logistics and material recovery partnerships now will hold structural cost advantages when the ELV recycling mandate is finalised.

For Policy Makers

- Publish a 5-Year Automotive Policy Stability Compact: Formally commit to no retroactive changes to CAFE or FAME structures within 24 months of announcement - directly reducing the PRI pillar.

- Establish an Inter-Ministerial Automotive Coordination Committee to align MoRTH, MHI, BEE, and Finance Ministry positions before public announcements - reducing CMD volatility.
- Resolve FAME-III Design Before Q3 2026: The FAME vacuum is the single largest driver of current PVI elevation. A clear, multi-year successor scheme would immediately reduce RF and MNU scores.

PART III

The PoC Portfolio - Eleven Proof of Concept Interventions

Volkswagen Group India · T-Works & Woxsen University · April 2026

The eleven PoCs form an integrated system. Each PoC's outputs inform or enable one or more others. The portfolio is designed to be sequenced: early wins in packaging loop closure and supplier carbon visibility provide the foundation upon which more complex interventions in reactive extrusion, pyrolysis, and design-for-disassembly can be built. The table below maps the cross-PoC linkages that are fundamental to understanding the portfolio as a system rather than a collection of independent initiatives.

PoC	Key Output	Enables	Depends On
1 - Supplier Routing	Scope 3 carbon data per supplier	PoC 4 (material intel), PoC 6 (dashboard), PoC 11 (traceability)	None - foundational
2 - Micro-Grid	Reduced Scope 2 intensity	PoC 7 (green reprocessing energy), PoC 9 (extrusion energy)	None - foundational
3 - DfD in CAD	Higher recyclability vehicle designs	PoC 7 (cleaner ELV PP), PoC 10 (less contaminated residue)	PoC 4 (material intelligence)
4 - AI Material Sub	Low-carbon material recommendations	PoC 3 (DfD material choices), PoC 7 (rPP spec)	PoC 1 (carbon data)
5 - Packaging Loop	Single-use packaging eliminated	Portfolio momentum funding	None - standalone
6 - Green UX	Verified consumer carbon story	Brand equity; ESG narrative	PoC 1 + PoC 11 data
7 - Closed-Loop PP	Automotive-grade rPP	PoC 9 (mixed stream residue), PoC 10 (residual fraction)	PoC 8 (ELV feedstock), PoC 11 (RCC)
8 - Micro-Recycling	ELV feedstock at scale	PoC 7 (rPP supply), PoC 9 (mixed plastic), PoC 10 (residual)	PoC 11 (EPR credits)
9 - Reactive Extrusion	Upcycled mixed recyclate	PoC 10 (handles remaining residue)	PoC 8 (feedstock), PoC 7 (precedent)
10 - Pyrolysis	Pyrolysis oil + carbon credits	PoC 10c: circular cracker feedstock	PoC 8 (feedstock), PoC 11 (ISCC cert)

11 - Traceability	EPR credits, RCC, CBAM data	All PoCs requiring certification	PoC 8 (physical records)

PoC 1 · Decarbonisation

Dynamic Carbon-Aware Supplier Routing

Priority: High · ROI Window: 18 months

5.1 What Is This PoC?

Dynamic Carbon-Aware Supplier Routing is a digital procurement intelligence intervention that embeds real-time and periodic carbon emission data from Tier-1 and Tier-2 suppliers directly into Volkswagen Group India's sourcing and supplier selection decisions. Rather than treating carbon performance as a compliance reporting afterthought, this PoC makes Scope 3 supply chain emissions a live operational variable - visible, measurable, and actionable in day-to-day procurement processes.

At its core, the PoC consists of three integrated components: (1) a Supplier Carbon Readiness Kit (SCRK) - a structured data collection and onboarding toolkit deployed to suppliers to capture emission intensities per production unit; (2) a Scope 3 Emissions Dashboard - a real-time visualisation layer built into VWG India's procurement management system that maps carbon performance across the supplier base; and (3) a Carbon-Weighted Sourcing Algorithm - a rule-based decision support module that incorporates carbon scores alongside traditional procurement variables such as price, quality, and lead time.

The system begins with the deployment of the SCRK to a pilot cohort of 10 to 15 Tier-1 suppliers at VWG India's Pune manufacturing hub. Each supplier is onboarded through a structured questionnaire covering energy mix, fuel consumption, production volume, logistics distance, and material input carbon intensity. These inputs are processed through a standardised emission factor database - aligned with India's National Greenhouse Gas Inventory methodology and the GHG Protocol Scope 3 Category 1 standard - to produce a verified emission intensity score expressed in kg CO_2-e per unit of component supplied.

The dashboard aggregates these scores across the supplier base and presents a ranked, filterable view of Scope 3 exposure by supplier, component category, and material type. Procurement teams can see, at a glance, which supplier relationships represent the highest carbon cost - and where switching to a lower-carbon alternative supplier is technically and commercially feasible. The sourcing algorithm then applies a carbon weighting factor to the total cost of supply calculation, making carbon intensity visible in the financial modelling of supplier selection decisions without overriding commercial judgment.

5.2 Why This PoC - Impact on Circularity, Decarbonisation, and CO_2 Reduction

Scope 3 emissions represent an estimated 65 to 80 percent of VWG India's total lifecycle carbon footprint. For a manufacturing operation that sources components from hundreds of Tier-1 suppliers - each drawing on their own energy mix, logistics networks, and production processes - the carbon embedded in purchased parts is the single largest lever available to VWG India for decarbonisation. Plant-level efficiency improvements in Scope 1 and Scope 2 emissions, while important, operate on a much smaller fraction of total lifecycle emissions.

India's Carbon Credit Trading Scheme (CCTS), expected to go live in 2026, will progressively attach a financial cost to Scope 3 emissions disclosure gaps. The EU Carbon Border Adjustment Mechanism (CBAM) - while currently covering steel, aluminium, cement, and fertilisers - is widely expected to expand to automotive components by 2028 to 2030. VWG India's exposure to both mechanisms creates a direct financial rationale for building Scope 3 visibility now, rather than facing a retroactive compliance scramble when mandatory reporting arrives.

A 10% reduction in Scope 3 intensity across VWG India's Tier-1 supplier base - achievable through targeted low-carbon sourcing switches - is equivalent to removing approximately 8,000 to 12,000 tonnes of CO_2-equivalent per year from VWG India's lifecycle footprint. At an emerging CCTS price of INR 1,000 per tonne CO_2-e, this represents INR 8 to 12 crore per year in avoided carbon cost liability.

From a circularity perspective, the supplier routing PoC is foundational: it creates the data infrastructure upon which recycled material content claims can be verified. A supplier that switches from virgin polypropylene to Banyan Nation's recycled PP (rPP) will show a measurable reduction in their emission intensity score - incentivising circular material adoption at the supplier level rather than requiring VWG India to mandate it contractually. This creates a market mechanism for circularity acceleration across the supply base.

The CO_2 impact pathway operates through three channels: direct emission intensity reduction from switching to lower-carbon suppliers where feasible; logistics optimisation, where the carbon routing algorithm factors in transport distance and modal choice as part of the carbon score; and supplier capability improvement, where the SCRK onboarding process itself raises awareness and capability among smaller Tier-2 suppliers who have never previously measured their production-related emissions.

5.3 Impact Metrics

Impact Dimension	Baseline (2026)	Target (18 months)	Mechanism
Scope 3 Coverage	0% - no verified data	50% Tier-1 by volume	SCRK deployment

Scope 3 Intensity Reduction	Not measurable	8–15% reduction	Carbon-weighted sourcing switches
CO_2-e Avoided (Annual)	Not quantified	8,000–12,000 tCO_2-e	Low-carbon supplier preference
CBAM Readiness	0% data coverage	80% Tier-1 CBAM-ready	Verified embedded carbon data
Recyclability Impact	No supplier-level data	rPP adoption incentivised	Carbon score rewards circular inputs

5.4 Deployment - Speed, TRL Level, and Development Readiness

This PoC sits at Technology Readiness Level 6 to 7, meaning the core technology components - supplier carbon data collection, emissions dashboards, and carbon-weighted procurement algorithms - are not experimental. They have been validated in analogous industrial contexts globally and are commercially available in various forms. The innovation at VWG India is the integration and India-specific calibration, not the underlying technology.

Deployment speed is rated as Fast-to-Medium. Phase 1 (supplier kit deployment and dashboard build) can be completed within six months, making this one of the faster-to-deploy interventions in the PoC portfolio. The primary constraint is not technology but change management: supplier onboarding requires relationship management, data validation support, and some supplier capability building for smaller Tier-2 firms.

Parameter	Status
TRL Level	TRL 6–7 - Integration-ready; commercially validated globally
Deployment Speed	Fast - Phase 1 in 6 months; Phase 2 in 12 months; fully operational in 18 months
Infrastructure Required	Procurement system API, supplier data portal, dashboard hosting - cloud-deployable
Key Dependencies	Supplier participation rate; VWG India SAP procurement integration; GHG Protocol alignment
Regulatory Readiness	BRSR Core (SEBI 2023) mandates Scope 3 disclosure for top 150 listed entities - aligned
India Adaptation Required	Emission factors need India-specific grid intensity and logistics data calibration
Pilot Cohort	10–15 Tier-1 Pune suppliers; expandable to 40+ in Phase 2
Global Analogues	BMW Group Supplier Sustainability Program; VW Group Scope 3 Steering (Germany); BMW i3 supply chain carbon mapping

5.5 Business Feasibility

5.5.1 Use Cases in the Automotive Industry

Carbon-aware supplier routing is not a theoretical concept - it is already operationalised by leading global OEMs and represents the next frontier for India-market implementation. The following benchmarks demonstrate feasibility at commercial scale.

- BMW Group's Supplier Sustainability Program: BMW has integrated supplier-level CO_2 data into its procurement scoring system across 300+ Tier-1 suppliers globally, reporting an average 25% reduction in Scope 3 supply chain emissions between 2019 and 2023. India-based suppliers to BMW are progressively being brought into this framework as BMW's Chennai plant scales production.
- Volkswagen Group's Own Scope 3 Programme: The VW Group Annual Report 2025 explicitly targets a 55% reduction in lifecycle carbon footprint by 2030 versus 2018, with Scope 3 supply chain emissions identified as the primary lever. The Group's global procurement policy requires Scope 3 data from all Tier-1 direct material suppliers - making PoC 1 a direct mandate fulfilment exercise for VWG India, not an optional innovation.
- Tata Motors' Supply Chain Carbon Visibility: Under the Tatva circularity framework, Tata Motors has deployed a supplier sustainability assessment module across 200+ Tier-1 suppliers, integrating carbon intensity data into its supplier performance management system. This has informed sourcing switches in the steel and aluminium categories.
- Maruti Suzuki's Vendor Sustainability Programme: Maruti has rolled out a supplier environmental assessment covering energy, water, and emissions across its Tier-1 vendor base in the Gurgaon-Manesar cluster, creating a carbon readiness benchmark that VWG India's Pune-based programme can reference and exceed.

5.5.2 Cost Impact

Cost Category	Estimate	Basis
PoC 1 Capex	INR 1–3 crore	Dashboard development, API integration, procurement system customisation
Annual OpEx	INR 0.5 crore	Platform maintenance, supplier support team, data validation
Carbon Cost Avoided (Year 3)	INR 2–4 crore/year	CCTS credits at INR 1,000/tonne; 2,000–4,000 tCO_2-e reduction
Procurement Efficiency Gain	INR 1–2 crore/year	Carbon-weighted sourcing reduces total cost of supply by 3–5% on targeted components
CBAM Liability Avoided	INR 3–8 crore/year (by 2028)	Steel and aluminium components; EUR 50–100/tonne EU carbon price
Net 5-Year NPV (10% discount)	INR 6–12 crore	Conservative; excludes CBAM upside
Payback Period	18–30 months	Base case; accelerated if CCTS goes live at INR 1,000/tonne

		in 2026

5.5.3 Performance Impact

Beyond financial returns, PoC 1 delivers performance improvements across three dimensions. Supply chain resilience is enhanced by identifying suppliers with high carbon risk profiles - often correlated with high fossil energy dependence - and proactively diversifying sourcing toward lower-risk alternatives. Regulatory performance is strengthened by building the Scope 3 data infrastructure required for BRSR Core compliance, CBAM reporting, and VWG Group's internal Scope 3 steering mechanism. Supplier relationship quality improves as the SCRK onboarding process raises supplier sustainability capability, creating a more capable and aligned Tier-1 base over time.

5.5.4 Contextualisation to Škoda India (SAVWIPL)

For SAVWIPL, which produces the Škoda Kylaq, Kushaq, Slavia, and Octavia from the Pune manufacturing hub, PoC 1 has direct and immediate relevance. The Kylaq - which drove VWG India's 34.9% year-on-year growth in FY 2025 - is produced at scale in Pune and draws on a dense cluster of Tier-1 and Tier-2 suppliers in the Pune-Nashik-Aurangabad corridor.

Škoda Auto's global parent has committed to reducing the carbon footprint of the Škoda Enyaq iV by 50% in lifecycle terms by 2030 versus the Octavia diesel. While the Enyaq is not yet sold in India, SAVWIPL's product roadmap is expected to include EV variants of the Kylaq and Kushaq platforms in the 2026 to 2028 timeframe. The Scope 3 data infrastructure built through PoC 1 is directly applicable to lifecycle carbon claims for these future EV models - claims that will be required for EU export compliance and increasingly expected by Indian ESG-aware consumers.

Additionally, Škoda Auto Volkswagen India's joint venture structure - manufacturing for multiple VW Group brands - means that the carbon data infrastructure built for Škoda-brand component suppliers automatically extends to Volkswagen-brand components, amplifying the return on the PoC 1 investment.

> Business Case Summary - PoC 1: INR 1–3 Cr investment delivers INR 6–12 Cr NPV over 5 years through carbon cost avoidance, procurement efficiency, and CBAM liability mitigation. Deployment in 18 months. First-mover advantage in Scope 3 data infrastructure is a strategic necessity given VWG Group's global reporting mandates. For SAVWIPL's Pune operations, this PoC is not optional - it is the data foundation for every other decarbonisation claim VWG India will need to make through 2030.

PoC 2 · Decarbonisation

Micro-Grid Carbon Buffering System

Priority: High · ROI Window: 24–36 months

6.1 What Is This PoC?

The Micro-Grid Carbon Buffering System is an on-site distributed energy management intervention designed to reduce the carbon intensity of Volkswagen Group India's manufacturing operations by intelligently managing when, how, and from where the Pune plant draws its electrical energy. Rather than simply adding renewable energy generation capacity, this PoC takes a system-level approach: it integrates energy generation, storage, load shifting, and real-time carbon intensity forecasting into a coordinated micro-grid management platform.

The system architecture consists of three functional layers working in concert. The generation and storage layer combines rooftop solar photovoltaic (PV) arrays with battery energy storage systems (BESS) to create a flexible on-site energy buffer. The load management layer identifies the most energy-intensive operations within the manufacturing facility - specifically the paint shop, HVAC systems, compressed air generation, and large electromechanical equipment - and applies dynamic load shifting protocols that reschedule high-draw activities to periods of lower grid carbon intensity. The intelligence layer operates a real-time carbon intensity forecasting engine, using grid frequency data, renewable penetration signals from Maharashtra's DISCOM (Mahadiscom), and weather forecasting APIs to predict 6 to 24 hour ahead windows of minimum grid carbon intensity.

The net effect is that the Pune plant's actual electricity consumption profile is progressively decoupled from the grid's peak carbon intensity periods. Energy drawn from the grid during high-coal-intensity windows - typically early morning and evening peak demand periods - is minimised; energy-intensive operations are scheduled for windows when grid carbon intensity is lower, supplemented by solar generation and battery discharge. Over time, as the BESS capacity scales, the plant can sustain operations from on-site stored energy during grid carbon spikes - effectively carbon-buffering its manufacturing output.

A longer-term extension of this PoC evaluates the integration of biogas from manufacturing process waste and, conditionally on green hydrogen cost trajectories in India, hydrogen fuel cell backup generation. These extensions are classified as Phase 3 activities with a 36 to 60 month horizon.

6.2 Why This PoC - Impact on Decarbonisation and CO_2 Reduction

India's electricity grid carbon intensity averages approximately 0.71 kg CO_2 per kWh (as per India's 2023 grid emission factor published by the Ministry of Power), but this average masks significant variation by state, time of day, and season. Maharashtra's grid, which supplies the Pune plant, has a carbon intensity that can vary from 0.55 kg CO_2/kWh during high-solar afternoons to over 0.90 kg CO_2/kWh during evening demand peaks when coal plants are fully dispatched.

For VWG India's Pune manufacturing operations - which consume approximately 80 to 120 million kWh per year across all production processes - even a 15% shift in consumption timing toward lower-intensity windows represents a Scope 2 emission reduction of 8,500 to 15,000 tonnes CO_2-equivalent per year without reducing production volume or efficiency. This is equivalent to the Scope 2 footprint of approximately 4,000 to 7,000 additional Škoda Kylaq units.

Scenario	Annual Energy	Grid Carbon Factor	CO_2-e Reduction	Cost Saving
Baseline (current)	100 MWh/year (index)	0.71 kg/kWh	-	-
Phase 1: Solar + load shift	100 MWh/year	0.58 kg/kWh effective	13,000 tCO_2-e	INR 3–5 Cr/year
Phase 2: BESS + advanced forecasting	100 MWh/year	0.50 kg/kWh effective	21,000 tCO_2-e	INR 6–9 Cr/year
Phase 3: Green H_2 / biogas integration	100 MWh/year	0.40 kg/kWh effective	31,000 tCO_2-e	INR 10–15 Cr/year

From a circularity perspective, the micro-grid PoC directly reduces the carbon footprint of all manufacturing operations - including the reprocessing activities in PoC 7 (closed-loop PP) and PoC 9 (reactive extrusion). Recycled polymer production has a lower carbon footprint than virgin polymer production only if the reprocessing energy is itself low-carbon. By greening the energy supply to VWG India's manufacturing and reprocessing operations, this PoC amplifies the carbon benefits of every other circular economy intervention in the portfolio.

6.3 Deployment - Speed, TRL Level, and Development Readiness

The component technologies within this PoC span a wide TRL range: rooftop solar PV and grid-tied battery storage are fully commercial at TRL 9, while real-time carbon intensity forecasting integrated with manufacturing load management is at TRL 7 to 8 in the Indian industrial context. The green hydrogen extension is at TRL 4 to 5 for Indian industrial application at this scale.

Deployment speed is rated as Medium. Phase 1 (solar + basic load shifting) can be implemented within 12 to 18 months using standard EPC contractors available in the Pune region. The BESS and advanced forecasting layer of Phase 2 requires a 24 to 36 month deployment horizon, constrained by BESS procurement lead times and system integration complexity.

Phase	Timeline	TRL	Capex (INR)	Key Activities
Phase 1 - Solar + Load Shift	12–18 months	8–9	4–7 Cr	Rooftop solar install; load profile audit; shift scheduling software
Phase 2 - BESS + Forecasting	18–36 months	7–8	4–8 Cr	Battery storage procurement; carbon intensity API integration; advanced load optimiser
Phase 3 - Biogas / H_2 Evaluation	36–60 months	4–5 (H_2)	0.5–1 Cr (study)	Green H_2 cost modelling; biogas potential assessment; pilot integration design

6.4 Business Feasibility

6.4.1 Use Cases in the Automotive Industry

- Volkswagen Wolfsburg Plant: The VW Group's home Wolfsburg plant operates a 1.4 GW combined heat and power (CHP) plant and has invested EUR 160 million in solar and wind generation capacity at Group facilities globally. The India implementation of this PoC draws directly on the VW Group's energy management best practices, making it portfolio-aligned rather than experimental.
- BMW Leipzig 'Electric' Plant: BMW's Leipzig facility - which manufactures the i3 and i4 - is powered 100% by on-site wind turbines during production hours. While large-scale wind is not feasible at the Pune site, the load management and grid orchestration principles are directly transferable.
- Tata Motors Sanand Plant: Tata's Sanand manufacturing facility in Gujarat operates a 2 MW rooftop solar installation and has achieved a 35% reduction in grid-sourced electricity since 2019. As Gujarat's grid carbon intensity is lower than Maharashtra's, the Pune plant's delta may be larger.
- Hyundai Chennai Plant: Hyundai's Chennai manufacturing facility operates an extensive renewable energy programme with rooftop solar and Power Purchase Agreements for off-site wind energy, achieving Scope 2 intensity reductions of approximately 20% since 2021 - demonstrating that OEM-scale manufacturing in India can significantly reduce grid electricity carbon intensity within a 36-month deployment horizon.

6.4.2 Cost Impact

Financial Parameter	Value	Notes
Total Phase 1 + 2 Capex	INR 8–15 Cr	Solar PV (4–6 Cr) + BESS (4–8 Cr) + software (0.5–1 Cr)
Annual OpEx	INR 1 Cr	Maintenance, monitoring, BESS degradation reserve
Annual Energy Cost Saving (Phase 2)	INR 5–10 Cr	Grid tariff differential + RPO compliance cost avoided
CCTS Carbon Credit Revenue (Year 3)	INR 0.4–1.0 Cr	500–800 tCO_2-e at INR 800–1,200/tonne

Renewable Purchase Obligation Cost Avoided	INR 1–2 Cr/year	Mahadiscom RPO compliance penalty avoidance
5-Year NPV (10% discount)	INR 12–25 Cr	Base case; upside if energy tariffs rise as projected
Payback Period	24–36 months	Phase 1 alone payback: 18–24 months

6.4.3 Performance Impact

Energy security is a direct performance benefit: the BESS layer provides backup capacity for critical manufacturing operations during grid outages, which are non-trivial in the Pune industrial belt. Manufacturing uptime improvement from reduced unplanned outages can represent INR 2 to 5 crore per year in avoided production loss. The Scope 2 carbon intensity metric - a direct input to Volkswagen Group's global carbon dashboard and the VWG India annual sustainability report - shows measurable improvement from Phase 1, creating an immediate ESG reporting benefit even before Phase 2 is complete.

6.4.4 Contextualisation to Škoda India (SAVWIPL)

SAVWIPL's Pune manufacturing complex is the single site that produces all VW Group India vehicles - Škoda Kylaq, Kushaq, Slavia, Octavia, Volkswagen Virtus, and Taigun. The shared infrastructure means that every MWh of carbon-reduced energy consumed at the Pune plant reduces the embedded carbon intensity of all six models simultaneously. For the Kylaq - VWG India's highest-volume model - a demonstrable reduction in manufacturing carbon intensity strengthens the lifecycle carbon narrative for future export variants and EV successors.

Maharashtra's industrial solar policy offers attractive capital subsidies (up to 30% of solar capex) and accelerated depreciation benefits for manufacturing units installing renewable energy. SAVWIPL's Pune plant is well-positioned to benefit from these incentives, improving the Phase 1 payback period by 3 to 6 months.

Business Case Summary - PoC 2: INR 8–15 Cr phased investment delivers INR 12–25 Cr NPV over 5 years through energy cost savings, RPO compliance cost avoidance, and carbon credit revenues. Phase 1 (solar + load shift) payback in 18–24 months. For SAVWIPL's Pune plant, this PoC delivers both operational and strategic value: operational through direct energy cost reduction, and strategic through Scope 2 carbon intensity improvement that feeds directly into VWG Group's global goTOzero reporting.

PoC 3 · Material Circularity

Design for Disassembly Embedded in CAD

Priority: Medium · ROI Window: Long-term Strategic

7.1 What Is This PoC?

Design for Disassembly (DfD) Embedded in CAD is a product engineering intervention that integrates circularity metrics and disassembly performance indicators directly into the computer-aided design (CAD) workflow used by VWG India's engineering teams. It operates at the earliest possible point in the product lifecycle - at the moment of design - when the cost of change is lowest and the downstream impact on recyclability is highest.

The PoC takes the form of a CAD-integrated plugin and design decision checklist that alerts engineers, in real time, when design choices are being made that will impede end-of-life disassembly, reduce material recyclability, or create mixed-material contamination problems that compromise the value of ELV plastics. Instead of asking product engineers to consciously remember circularity principles, the tool embeds those principles as automated prompts and scoring metrics within their existing design environment - whether that is CATIA, Siemens NX, or Creo, all of which are in use across the VWG Group's global engineering operations.

Four primary design metrics are tracked and scored: the Recyclability Percentage (the proportion by weight of each component that can be recovered and recycled under standard ELV conditions); the Fastener Standardisation Index (the degree to which fastener types are rationalised to reduce disassembly tool changes and time); the Material Separability Score (the ease with which different polymer types can be separated without contamination or degradation); and the Disassembly Time Index (DTI - the estimated time in minutes required to remove each component from the vehicle under standard dismantling protocols).

Each component designed or modified through VWG India's CAD system receives a composite DfD Score at the end of the design process. Components that fall below a defined threshold trigger a mandatory design review flag, prompting the engineering team to consider alternative fastener systems, mono-material substitutions, or modular design approaches that improve the circularity score. The DfD scoring data is also archived to feed lifecycle assessment (LCA) models and regulatory compliance documentation for India's EPR rules and the EU's 2027 ELV Regulation.

7.2 Why This PoC - Impact on Circularity and Recyclability

The fundamental rationale for Design for Disassembly is that 80% of a product's environmental impact is locked in at the design stage. The average Indian passenger vehicle contains approximately 19 kg of plastic across 200 to 400 discrete plastic components. Of this, only 35 to 45% is currently recovered and recycled in India's informal ELV ecosystem - the remainder is landfilled, incinerated, or downcycled into low-value construction materials. This poor recovery rate is not primarily caused by inadequate recycling technology; it is caused by design decisions that mix incompatible polymer types, use multi-material assemblies, bond components with adhesives that cannot be separated, and specify coatings and flame retardants that contaminate the recyclate stream.

Each generation of vehicle designed with DfD principles improves the yield of the closed-loop PP reprocessing pathway (PoC 7), reduces the volume of contaminated plastic requiring pyrolysis (PoC 10), and increases the economic value of ELV-sourced recyclate. The compounding effect across multiple vehicle generations is transformative: a DfD-designed Škoda Kylaq of 2027 entering the ELV stream in 2042 will yield significantly higher-quality, higher-value recyclate than a conventionally designed equivalent.

Impact Category	Without DfD (Current)	With DfD (2030 target)	Enabling Mechanism
Plastic Recyclability Rate	35–45% by weight	65–75% by weight	Mono-material design; fastener standardisation
Disassembly Time (dashboard cluster)	45–60 minutes	18–25 minutes	DTI-optimised fastener system; modular design
rPP Yield from ELV (per vehicle)	~8 kg recoverable	~14 kg recoverable	Material separability improvements
Contaminated Plastic to Pyrolysis	~11 kg per vehicle	~5 kg per vehicle	Reduced mixed-material assemblies
Recyclate Grade Consistency	Variable - frequent downcycling	Consistent automotive grade	AIS-129 marking + DfD documentation

7.3 Deployment - Speed, TRL Level, and Development Readiness

DfD as a design philosophy is at TRL 9 in the broader engineering community - it is a well-established discipline with ISO standards (ISO 11469 for plastics marking, VDI 2243 for recyclability assessment) and global automotive implementations. However, the integration of DfD scoring into an automated CAD plugin environment tailored to VWG India's specific engineering software stack, Indian regulatory context, and ELV ecosystem realities places this PoC at TRL 5 to 6 for the India-specific implementation.

Deployment speed is rated as Medium-to-Slow in terms of impact realisation, because design changes only affect vehicles entering production 18 to 36 months after the design decision and only generate ELV circularity benefits 10 to 15 years later when those vehicles retire. However, the deployment of the tool itself is relatively fast: the CAD plugin can be developed and piloted within 12 months. The strategic investment case is therefore forward-looking - VWG India is building circularity equity in future vehicle generations.

Parameter	Specification

TRL Level	TRL 5–6 for India-specific CAD integration; underlying DfD principles at TRL 9
Deployment Timeline	Plugin development: 6–9 months; Pilot on 2 component families: 12 months; Full VWG India design integration: 24 months
CAD Platforms	CATIA V5/V6 (Skoda/VW primary); Siemens NX (VWG Group standard); Creo (Tier-1 suppliers)
Pilot Component Families	Dashboard cluster (highest PP content + complexity); Door panel (multi-material; high DfD improvement potential)
Regulatory Alignment	India EPR Rules 2025; EU ELV Regulation 2027; AIS-129 ELV traceability marking requirements
Key Capability Required	LCA data for Indian material and logistics conditions; CIPET test data for rPP component performance
Partner for Validation	CIPET (Central Institute of Plastics Engineering and Technology) for component testing and specification validation

7.4 Business Feasibility

7.4.1 Use Cases in the Automotive Industry

- Renault-Nissan Alliance DfD Programme: Renault's Recycling Design Index, deployed across all models since 2015, requires 95% theoretical recyclability for all new vehicle designs. The index is calculated during the design phase using Renault's proprietary ECOCALC tool - a direct analogue to the CAD plugin proposed in this PoC. Renault's Koleos, sold in India, carries a 95% recyclability rating derived from this process.
- BMW iX Circular Design: BMW's iX electric SUV was designed with explicit DfD principles, using snap-fit connections in the interior rather than adhesives, standardised polymer types across interior surfaces, and colour-coded material marking for dismantlers. The iX achieves a 90% recyclability rate by weight - significantly higher than the industry average of 75 to 80%.
- Toyota's 3R Programme: Toyota has mandated Design for Recyclability across its global vehicle platform since 2015, achieving a target of 95% recyclability in production vehicles. The GD platform used in the Innova Crysta (manufactured in India) demonstrates 91% recyclability - evidence that DfD is technically feasible in India-market, India-manufactured vehicles.
- Tata Motors Altroz Eco Design: Tata's Altroz incorporates DfD principles in its interior design, using standardised fastener families and material marking consistent with AIS-129, enabling higher-quality disassembly at Tata's Re.Wi.Re network.

7.4.2 Cost Impact

Cost Category	Value	Notes
CAD Plugin	INR 1–2 Cr	Software development; LCA database

Development Capex		integration; regulatory mapping
Annual OpEx	INR 0.3 Cr	Platform maintenance; database updates; engineer training
ELV Compliance Cost Avoided (2027+)	INR 3–8 Cr/year	EU ELV Regulation recycled content mandate compliance headroom
rPP Yield Improvement Value	INR 2–5 Cr/year (2030+)	Higher yield from ELV → lower procurement cost for PoC 7 rPP loop
EPR Credit Generation	INR 2–4 Cr/year (2030+)	Higher recyclability rate → more EPR credits per vehicle processed

7.4.3 Performance Impact

The performance impact of DfD is disproportionately large relative to the investment cost because it operates at the root of the material circularity problem. Every kilogram of plastic recovered at automotive grade from a DfD-compliant vehicle displaces approximately 1.8 kg of CO_2-equivalent in virgin polymer production. At 14 kg of recoverable plastic per DfD-compliant vehicle versus 8 kg from a standard design, the annual carbon benefit across VWG India's total vehicle production of approximately 180,000 units per year (by 2030) could exceed 60,000 tonnes CO_2-equivalent per year from plastic circularity alone. The operational performance benefit is equally significant: DfD-designed components reduce end-of-line repair and warranty costs by simplifying disassembly for servicing, not only end-of-life recovery.

7.4.4 Contextualisation to Škoda India (SAVWIPL)

For SAVWIPL, DfD integration is most immediately relevant to the next-generation India-specific product development cycles. The Škoda Kylaq, launched in 2025, was designed primarily in the Czech Republic - DfD retrofitting on an existing platform is possible but limited. The opportunity for transformative impact is in the India-specific EV variants planned for 2027 to 2029. Designing these platforms from inception with DfD principles embedded in the CAD workflow will ensure that VWG India's next generation of India-market vehicles enters the ELV stream in the 2040s with a 65 to 75% recyclability profile - the highest in India's passenger car segment.

SAVWIPL's India-based R&D capability at the Pune Technical Centre is a critical enabler: local engineers working on India-specific adaptations of global platforms can integrate the DfD plugin into their workflow immediately, applying DfD principles to India-specific component specifications, local material substitutions, and India-market manufacturing processes.

Business Case Summary - PoC 3: INR 1–2 Cr investment with long-dated but high-value returns. The DfD CAD plugin is the lowest-capex, highest-strategic-leverage intervention in the portfolio. VWG India's 2025 vehicle designs determine the recyclability of its 2040 ELV stream. Starting DfD integration now positions SAVWIPL to meet EU ELV 2027 recycled content mandates, India EPR targets, and Volkswagen Group GMK material standards through the 2030s.

PoC 4 · Decarbonisation

AI-Based Low-Carbon Material Substitution

Priority: High · ROI Window: 18–24 months

8.1 What Is This PoC?

AI-Based Low-Carbon Material Substitution is an intelligence and decision support system that enables VWG India's procurement and engineering teams to identify, evaluate, and implement substitutions of high-carbon materials with lower-carbon alternatives - systematically, at scale, and with full dual optimisation across both carbon intensity and cost. The system consists of an India-specific material database, a carbon footprint calculation engine, and an AI-powered recommendation layer that generates ranked substitution options for specific components.

The material database is the foundational asset: it catalogues over 500 material variants across seven primary categories relevant to automotive manufacturing (steel, aluminium, polypropylene, ABS, polyamide, glass fibre composites, and PET), with each entry containing the material's embedded carbon footprint under Indian production conditions; cost per kilogram from India-based suppliers; mechanical performance specifications (tensile strength, impact resistance, heat deflection temperature); and supplier availability mapped by geographic cluster.

The AI optimisation layer uses a multi-parameter genetic algorithm to search the substitution space for each target component, generating ranked recommendations that meet the component's performance specification while minimising a cost-carbon composite score. The algorithm can be tuned by the procurement team to weight cost versus carbon more or less heavily depending on the project brief - allowing the same tool to serve both cost-reduction and ESG-driven substitution programmes.

The PoC operates in three deployment phases. Phase 1 builds the database and validates it against VWG India's current Bill of Materials (BOM) for the Škoda Kylaq and Volkswagen Virtus. Phase 2 pilots the AI recommendation engine on five to eight target component families, generating validated substitution proposals tested against VWG Group GMK material standards. Phase 3 scales the tool across VWG India's full production BOM and integrates it with the procurement management system as a standard sourcing decision support layer.

8.2 CO_2, Decarbonisation, and Recyclability Impact

Material choice is the most consequential single variable in automotive lifecycle carbon management. The embodied carbon in the materials that make up a vehicle - particularly the steel, aluminium, and polymer content - accounts for approximately 20 to 30% of total lifecycle emissions for an ICE vehicle and rises to 40 to 60% for a battery electric vehicle where tailpipe emissions are eliminated. Shifting material choices toward lower-carbon alternatives at the design and procurement stage is therefore a high-leverage decarbonisation pathway.

In India, the gap between global material carbon databases (such as the ecoinvent database or PE International's GaBi) and the actual carbon intensity of locally produced materials is significant. Indian steel production, for example, averages approximately 2.5 tonnes of CO_2 per tonne of steel - compared to a global average of 1.85 tonnes and an EU electric arc furnace average of under 0.5 tonnes. Using global average factors to calculate the carbon footprint of VWG India's Pune-produced vehicles would systematically understate their actual embedded carbon by 25 to 40%. This PoC corrects for this gap by building India-calibrated emission factors into the substitution analysis.

Material Category	Current Carbon Intensity	Low-Carbon Alternative	CO_2-e Reduction	Cost Delta
Virgin PP (Haldia Petrochemicals)	2.1 kg CO_2-e/kg	Banyan Nation rPP	−1.8 kg CO_2-e/kg (−86%)	−30 to −40%
Virgin ABS (imported)	4.5 kg CO_2-e/kg	Post-industrial rABS (domestic)	−2.8 kg CO_2-e/kg (−62%)	−15 to −25%
Primary aluminium (smelter)	8.0 kg CO_2-e/kg	Secondary/recycled aluminium	−6.5 kg CO_2-e/kg (−81%)	−10 to −20%
BF/BOF steel (SAIL, Tata Steel)	2.5 kg CO_2-e/kg	EAF recycled steel (JSW, JSPL)	−1.5 kg CO_2-e/kg (−60%)	−5 to −15%
Virgin PET (Reliance)	2.9 kg CO_2-e/kg	Ganesha Ecopet rPET	−2.1 kg CO_2-e/kg (−72%)	−25 to −35%

8.3 Deployment - Speed, TRL Level, and Development Readiness

The AI recommendation engine technology is at TRL 7 to 8 - multi-parameter optimisation algorithms for material selection are commercially available from several sustainability software providers (including SimaPro, GaBi, and emerging India-based platforms). The India-specific material database is the primary development effort, requiring 6 to 9 months of data collection, verification, and calibration. Total time to Phase 1 operational readiness is estimated at 12 to 18 months.

Parameter	Specification
TRL Level	TRL 7–8 - AI optimisation engine mature; India DB calibration required
Time to Phase 1 Operational	12–18 months
Database Build Time	6–9 months - India supplier carbon data collection + LCA calibration
AI Layer Deployment	3–6 months after database validation
Key Partners	CIPET (material testing); CSIR-NCL (polymer science); Banyan Nation / Ganesha Ecopet (rPP/rPET specs)

Integration Requirements	VWG India SAP procurement BOM; VWG Group GMK material qualification standards
Regulatory Alignment	CAFE III (2027) - material substitution reduces fleet CO_2 intensity; EU ELV Regulation (recycled content)

8.4 Business Feasibility

8.4.1 Use Cases in the Automotive Industry

- Volkswagen Group's Own MaDeIn Tool: VWG's global Material Data & Innovation (MaDeIn) platform maintains a central material database for all Group brands, enabling cross-brand material knowledge sharing and lifecycle assessment. PoC 4 is the India-market equivalent and complement to MaDeIn, filling the gap created by India-specific production conditions that global averages do not capture.
- BMW's i3 Material Intelligence: BMW's i3 development team used a multi-parameter material optimisation process to identify carbon fibre reinforced plastic (CFRP) as the primary structural material - a decision validated by lifecycle carbon modelling showing CFRP's weight reduction benefit outweighing its high embodied carbon over a 200,000 km vehicle life.
- Tata Motors' Material Sustainability Roadmap: Under the Tatva programme, Tata Motors has developed an internal sustainability scorecard for material procurement that weights carbon intensity alongside cost. The Nexon EV's interior panels use 10 to 15% recycled polymer content - identified and validated through a material substitution analysis analogous to PoC 4.
- Ford's Sustainability Material Database: Ford maintains an internal database of approximately 12,000 material variants with associated lifecycle carbon data, enabling engineers to select lower-carbon alternatives at the design stage. Ford India (Chennai) draws on this database, demonstrating that India-market OEM operations can integrate with global material intelligence platforms.

8.4.2 Cost Impact

Financial Parameter	Value	Notes
System Build Capex	INR 2–4 Cr	Database development; AI layer licensing; integration with VWG SAP BOM
Annual OpEx	INR 0.8 Cr	Database updates; AI model retraining; CIPET testing fees
Annual Material Cost Saving (Year 3)	INR 3–6 Cr	rPP/rPET substitution at 25–40% unit cost saving across targeted components
Annual CO_2 Reduction Value (CCTS)	INR 0.8–2 Cr	800–2,000 tCO_2-e avoided at INR 1,000/tonne
CAFE III Compliance Contribution	INR 2–5 Cr avoided penalty	Material lightweighting reduces fleet CO_2; avoids CAFE non-compliance levy
5-Year NPV (10%	INR 7–14 Cr	Conservative; CAFE penalty avoidance not

discount rate)		fully included
Payback Period	24–36 months	Base case

8.4.3 Performance Impact

AI-driven material substitution does not inherently trade performance for carbon reduction - its explicit design objective is to find substitutions that meet or exceed performance specifications while reducing carbon intensity. The AI layer's constraint engine rejects any recommendation that does not meet VWG Group GMK requirements for the component in question. In practice, the most impactful substitutions - virgin PP to rPP for interior cladding, virgin PET to rPET for under-hood reservoirs - have been validated at equivalent performance by Banyan Nation and Ganesha Ecopet respectively. The secondary performance benefit is supply chain resilience: domestic recycled polymers are less volatile in price than virgin polymers, which are subject to global petrochemical commodity cycles.

8.4.4 Contextualisation to Škoda India (SAVWIPL)

The Škoda Kylaq is a high-volume model manufactured predominantly for the Indian domestic market, with a production profile optimised for India's cost-sensitive pricing environment. The AI material substitution tool is directly applicable to the Kylaq's BOM because the primary cost driver for Indian-market vehicles is material and logistics cost. A 5% reduction in embodied material cost across the Kylaq's polymer content represents approximately INR 800 to 1,200 per vehicle - at 100,000 units per year, this is INR 8 to 12 crore in annual material cost savings from a single model.

The Kylaq's interior architecture - with its prominent dashboard cluster, door panels, and trim elements - is precisely the component category where rPP substitution is most technically and commercially mature. SAVWIPL's engineering team piloting the AI tool on Kylaq component families would generate validated substitution recommendations applicable to the Kushaq, Slavia, Virtus, and Taigun with minimal incremental effort, amplifying returns across the full production portfolio.

Business Case Summary - PoC 4: INR 2–4 Cr investment generates INR 7–14 Cr NPV over 5 years. The India-specific material database, once built, is a durable competitive asset. For SAVWIPL's Pune operations, material substitution toward rPP and rPET on the Kylaq platform alone could deliver INR 8–12 Cr annual savings - making this PoC self-funding within 6 months of full deployment.

PoC 5 · Supply Chain Infrastructure

Zero-Emission Packaging Loop

Priority: Very High · ROI Window: 18–36 months

9.1 What Is This PoC?

The Zero-Emission Packaging Loop is the highest-priority, fastest-to-deploy, and most commercially straightforward intervention in the entire PoC portfolio. It addresses a specific, measurable, and largely invisible cost and emission source in VWG India's inbound supply chain: single-use packaging waste. India's automotive supply chains depend heavily on cardboard cartons, wooden pallets, expanded polystyrene foam, and plastic shrink wrap to protect components in transit from Tier-1 and Tier-2 suppliers to the manufacturing plant. Each shipment arrives in packaging that is used once and discarded - creating a continuous stream of packaging waste at the receiving dock and generating upstream emissions from the manufacturing and disposal of this packaging material.

This PoC replaces single-use inbound packaging with a closed-loop system of standardised, reusable, RFID-tracked crates across VWG India's top 20 highest-frequency inbound component flows. The system comprises four elements: standardised collapsible crate families, custom-designed to the geometry of specific high-volume components (engine parts, stampings, electronic modules, injection-moulded plastic components); RFID tagging embedded in every crate, with scanners at all dock entry and exit points for real-time tracking; an IoT logistics management dashboard that monitors crate location, utilisation rate, return journey routing, and damage status across the entire inbound supply chain; and a return logistics protocol defining the collection cycle, cleaning procedure, damage assessment process, and supplier crate credit accounting.

The crate families are designed for a minimum 100-cycle life, with a 15-year operational horizon. Collapsible design means return logistics are efficient: empty crates collapse to approximately 30% of their loaded volume, dramatically reducing return freight cost and carbon. The RFID-IoT layer enables live dashboarding of crate utilisation, return rates, and loss incidents - creating data visibility that conventional single-use packaging systems entirely lack.

9.2 Why This PoC - Impact on Decarbonisation, Circularity, and CO_2

Packaging waste is a systemic inefficiency that sits at the intersection of all three sustainability dimensions: it generates CO_2 emissions from its production and disposal; it creates solid waste that is largely unrecyclable in mixed-use waste streams; and it represents a direct, quantifiable financial cost that is charged to manufacturing operations. The transition to a closed-loop packaging system eliminates the upstream emissions from packaging production, eliminates the downstream waste management cost, and reduces logistics-related Scope 3 emissions through optimised return routing.

Impact Category	Current State	Post-PoC State	Mechanism

		(Year 3)	
Single-use packaging waste	~800–1,200 tonnes/year	<100 tonnes/year	Reusable crates replace 90%+ of single-use packaging
Packaging CO_2-e (upstream production)	Est. 1,200–2,000 tCO_2-e/year	<200 tCO_2-e/year	Packaging production emissions eliminated for reusable units
Packaging material cost	INR 3–6 Cr/year (ongoing)	INR 0.3–0.6 Cr/year (maintenance)	One-time crate capex replaces recurring material cost
Return logistics Scope 3	Not measured	5–10% reduction vs baseline	IoT-optimised return routing reduces empty truck kilometres
Landfill diversion (packaging waste)	~0%	>90%	Closed-loop system eliminates packaging at end-of-cycle

The circularity impact of this PoC is straightforward and immediate: the single-use packaging loop - which follows the linear take-make-dispose model - is replaced by a genuine closed loop in which packaging assets circulate indefinitely within the supply chain. This is the purest expression of circular economy principles in the PoC portfolio: no complex technology, no regulatory dependency, no new material science - simply the intelligent design of a returnable packaging system.

9.3 Deployment - Speed, TRL Level, and Development Readiness

This is the only PoC in the portfolio at TRL 8 to 9 - fully commercially proven globally and requiring only India-market adaptation. Toyota's global returnable packaging system covers over 90% of its inbound component flows across Japanese, European, and North American plants. BMW's Reusable Packaging Programme has achieved payback within 24 months across multiple European and Chinese plants. Maruti Suzuki's Manesar facility has implemented a returnable crate system for body stampings and engine components that serves as the closest India-market analogue.

Parameter	Specification
TRL Level	TRL 8–9 - Commercially proven globally and regionally (Maruti Suzuki Manesar)
Deployment Timeline	6–9 months to full deployment across target component set
Crate Families Required	6–10 standard crate sizes covering 80% of component volume variation
RFID Infrastructure	Dock scanners at 4–6 entry/exit points; IoT platform (e.g., Recykal) for tracking
Supplier Coverage (Phase 1)	Top 20 highest-frequency inbound component suppliers in Pune

	cluster
Key Procurement Requirement	RFID-embedded collapsible crates; standard spec from 3 to 5 India-based crate manufacturers
Regulatory Alignment	Plastic Waste Management Rules 2022 (single-use plastic restrictions); EPR packaging obligations
Investment Requirement	INR 2–4 Cr total capex - fastest payback in portfolio at 18–36 months

9.4 Business Feasibility

9.4.1 Use Cases in the Automotive Industry

- Toyota Production System Returnable Packaging: Toyota's global TPS standard specifies returnable packaging for all direct material inbound flows above a defined frequency threshold. The Toyota Kirloskar plant in Bidadi, Karnataka, operates a returnable kanban crate system for local Tier-1 suppliers that has eliminated cardboard packaging entirely from approximately 60% of inbound component flows.
- Volkswagen Group Global Returnable Packaging Standard: VWG's global logistics policy (VW Standard 99000) specifies returnable load carriers (RLCs) for high-frequency supply routes. VWG India's implementation of PoC 5 would align with this global standard, enabling VWG India to report the India packaging loop as a contribution to the Group's global waste reduction targets.
- BMW's SmartFreight Programme: BMW has integrated IoT tracking into its global returnable packaging fleet, achieving 98% return rate and near-zero packaging loss. The real-time visibility dashboard has enabled a 12% reduction in return freight costs through route optimisation - directly analogous to the IoT layer in this PoC.
- Maruti Suzuki Manesar Returnable Crates: As India's highest-volume OEM, Maruti has the most mature India-market returnable packaging implementation. Its Manesar plant covers major stampings and mechanical components with returnable steel racks and foldable plastic crates. VWG India can benchmark directly against Maruti's implementation, leveraging the same crate supplier ecosystem and logistics infrastructure.

9.4.2 Cost Impact

Financial Parameter	Value	Notes
Capex - Crate Procurement	INR 1.5–3.0 Cr	2,000–5,000 crates at INR 3,000–6,000 per unit (collapsible + RFID)
Capex - RFID + IoT Infrastructure	INR 0.5–1.0 Cr	Dock scanners; dashboard software; supplier portal
Annual OpEx	INR 0.5 Cr	Maintenance; crate replacement reserve; logistics coordination

Annual Packaging Cost Saving	INR 3–5 Cr/year	Replaces recurring single-use packaging procurement cost
Annual Waste Management Saving	INR 0.5–1.0 Cr/year	Avoided cardboard/foam disposal charges and landfill fees
Annual Logistics Optimisation Saving	INR 0.3–0.8 Cr/year	IoT return routing reduces empty truck kilometres by 8–12%
Payback Period	18–36 months	Fastest ROI in portfolio; base case 24 months
5-Year NPV (10% discount)	INR 8–12 Cr	High-confidence - no regulatory dependency

9.4.3 Performance Impact

Component protection quality typically improves with purpose-designed reusable crates compared to cardboard-and-foam packaging: crates are rigid, provide consistent clearance between components, and are replaced at defined wear thresholds rather than degrading unpredictably as cardboard does. Component damage rates in transit - which generate warranty costs, production line stoppages, and supplier reorder cycles - typically fall by 15 to 25% following a returnable packaging transition in similar OEM implementations. For VWG India, a 1% reduction in inbound component damage rate is worth approximately INR 2 to 5 crore per year in avoided production disruption costs.

Dock receiving efficiency also improves: standardised crate geometry reduces unloading time per shipment, standardised RFID scanning replaces manual paper-based goods receipt, and the IoT dashboard provides advance visibility of inbound shipment status that enables proactive scheduling of receiving staff.

9.4.4 Contextualisation to Škoda India (SAVWIPL)

SAVWIPL's Pune manufacturing hub receives components from approximately 150 to 200 Tier-1 suppliers across the Pune-Nashik-Aurangabad corridor. The inbound logistics volume for a ~180,000 unit annual production rate generates an estimated 800 to 1,200 tonnes of single-use packaging waste per year - predominantly cardboard and expanded polystyrene. For SAVWIPL's sustainability reporting, packaging waste is a directly measurable and reportable metric under the BRSR Core framework (SEBI, 2023).

The Plastic Waste Management Rules 2022 prohibition on specific categories of single-use plastic packaging (effective from December 2022) creates a compliance imperative that makes this PoC not merely attractive but obligatory for components currently packaged in prohibited single-use plastic formats. Early implementation of PoC 5 converts a compliance obligation into a proactive sustainability leadership story.

Business Case Summary - PoC 5: INR 2–4 Cr investment generates INR 8–12 Cr NPV over 5 years with 18–36 month payback - the fastest and most reliable ROI in the portfolio. TRL

9 technology; no regulatory risk; no partner complexity. This PoC should be activated immediately as the portfolio's financial anchor - the early returns it generates can partially fund subsequent, more complex interventions. For SAVWIPL, packaging loop closure is also a direct regulatory compliance action under Plastic Waste Management Rules 2022.

PoC 6 · Sustainability Intelligence

Green UX In-Vehicle CO_2 Impact Dashboard

Priority: Medium · ROI Window: Product Cycle

10.1 What Is This PoC?

The Green UX In-Vehicle CO_2 Impact Dashboard is the consumer-facing sustainability intelligence layer of VWG India's circular and carbon economy programme. It translates the supply chain decarbonisation work of PoCs 1 through 5 and the material traceability data of PoC 11 into a tangible, engaging, and personally meaningful driver experience - making VWG India's sustainability credentials visible and interactive at the point of the product that consumers interact with every day.

The dashboard is a software feature integrated into the vehicle's infotainment system or instrument cluster. For EV platforms, it extends the existing energy management and range visualisation screens with a dedicated sustainability intelligence layer. For ICE and hybrid platforms, it introduces a standalone eco-impact module accessible through the central display. The core UX concept translates real vehicle usage data into ecological impact metrics expressed in terms that drivers find personally meaningful rather than abstractly scientific.

Four interaction layers constitute the dashboard experience. The Carbon Savings Counter displays cumulative CO_2 savings achieved by the driver relative to a comparable conventional vehicle - expressed both in tonnes of CO_2-equivalent and in relatable equivalents such as trees preserved, kilometres of travel offset, and litres of water saved. The Real-Time Efficiency Score provides continuous feedback on driving style's carbon impact, gamifying efficient driving behaviour through a running score that changes with throttle application, regenerative braking events, and speed management. The Route Carbon Intelligence module leverages GPS and grid carbon intensity API data to display the estimated carbon footprint of the current journey, with comparisons to public transport alternatives. The Manufacturing Story module - dependent on PoC 11 traceability data - presents the recycled content narrative of the specific vehicle, showing what percentage of its interior materials came from recycled sources and where they were processed.

10.2 Why This PoC - Decarbonisation and Sustainability Intelligence Impact

The behavioural research on eco-feedback mechanisms is unambiguous: real-time feedback on the environmental impact of consumption behaviour produces measurable reductions in that impact. Studies from Toyota's Prius, Honda's Insight, and BMW's i3 programmes all demonstrate that drivers with access to real-time fuel efficiency and carbon feedback consume 5 to 15% less fuel than drivers without such feedback - even when controlling for vehicle type and route. For VWG India, a 10% reduction in average fuel consumption across its ICE fleet corresponds to a reduction of approximately 8,000 tonnes of CO_2-equivalent in Scope 3 use-phase emissions per year.

The strategic value of this PoC extends beyond the direct emission reduction from changed driver behaviour. In the Indian market - where sustainability is an emerging but rapidly growing purchase decision criterion among the urban, educated consumers who comprise VWG India's primary target demographic - the in-vehicle sustainability dashboard creates a differentiating product feature that competitors cannot immediately replicate without the underlying data infrastructure. No Indian OEM currently offers a verified, real-time CO_2 impact dashboard that draws on independently certified supply chain carbon data.

Impact Dimension	Value	Dependency
Driver behaviour CO_2 reduction	5–15% fuel consumption reduction	No dependencies - immediate on deployment
Brand differentiation (premium positioning)	INR 3–8 Cr brand equity value	Consumer research validates sustainability premium
Verified supply chain story	First Indian OEM with certified material story	Requires PoC 1 + PoC 11 data infrastructure
EV adoption signal	Accelerates EV consideration among eco-aware buyers	Requires EV platform deployment (2027+)
ESG narrative for investor communications	Direct VWG India ESG report content	No dependency - dashboard data feeds reporting

10.3 Deployment - Speed, TRL Level, and Development Readiness

The underlying technology - infotainment software development, real-time vehicle telematics, and grid carbon intensity APIs - is at TRL 8 to 9. The innovation is the integration and contextualisation for the Indian market, the design of the UX around Indian consumer research insights, and the verified supply chain data layer from PoC 11. Deployment speed is rated as Medium, with a primary dependency on PoC 1 and PoC 11 data availability before the most differentiating features can be launched.

Parameter	Specification
TRL Level	TRL 7–8 for core dashboard; TRL 5–6 for verified supply chain data integration
Deployment Phases	Phase 1 (Driver eco-feedback): 12–18 months; Phase 2 (Supply chain story): 24–36 months (after PoC 11)
Platform Priority	EV platforms first (2027+); ICE via OTA software update for current models
Key Data Dependencies	PoC 1 (Scope 3 supplier data); PoC 11 (recycled content certificates)
UX Design Partner	Internal VWG India design team + Pune UX studio for Indian consumer research

Regulatory Notes	BRSR Core consumer disclosure requirements; future EU Digital Product Passport alignment

10.4 Business Feasibility

10.4.1 Use Cases in the Automotive Industry

- BMW i3 and iX Eco Pro Mode: BMW's eco-coaching dashboard in the i3 and iX series provides real-time efficiency feedback, regenerative braking visualisation, and cumulative CO_2 savings display. BMW reports that i3 drivers using Eco Pro mode achieve an average 12% greater range than those not using it - corresponding to a 12% reduction in energy consumption and proportional carbon savings.
- Volkswagen ID.4 Travel Assist CO_2 Display: The VW ID.4 already includes a CO_2 savings counter relative to an equivalent ICE vehicle, updated in real time based on grid carbon intensity data. The VWG India implementation of PoC 6 would bring this feature - already in the VW Group's global software library - to India-market vehicles with Indian-specific calibration.
- Toyota Prius Eco Score: Toyota's Prius eco-driving score system is credited with contributing to the model's 20% better-than-EPA real-world fuel economy in global studies. The behavioural economics literature confirms that visible scoring and feedback loops are among the most cost-effective interventions for reducing consumer energy consumption.
- Tata Nexon EV's EV Score: Tata Motors' Nexon EV includes a driving efficiency score that has become one of the model's most appreciated features among Indian EV buyers. VWG India's Green UX dashboard can extend this concept significantly by adding verified supply chain content and a circular material origin story - features that Tata currently lacks.

10.4.2 Cost Impact

Financial Parameter	Value	Notes
Development Capex	INR 2–5 Cr	UX development; API integration; consumer research; testing
Annual OpEx	INR 0.5 Cr	Platform maintenance; API subscriptions; software updates
Revenue from Feature Premium	INR 3–8 Cr/year	INR 3,000–8,000 premium per vehicle for EV variants with verified sustainability dashboard
Brand Equity Value	INR 5–15 Cr (estimated NPV)	Consumer willingness-to-pay premium for sustainability-verified vehicles
ESG Rating Impact	Qualitative - investor relations value	Demonstrable consumer-facing sustainability investment
5-Year NPV	INR 10–25 Cr	Primarily driven by revenue premium and

	(inclusive of brand value)	brand equity

10.4.3 Contextualisation to Škoda India (SAVWIPL)

For SAVWIPL, the Green UX Dashboard is most powerful as a feature on the future Škoda EV platform anticipated for the Indian market in 2027 to 2028. Škoda Auto globally has positioned its Enyaq iV with a strong sustainability narrative - including the EV's lifecycle carbon calculation relative to ICE vehicles. Bringing this narrative to the India market through an in-vehicle dashboard, backed by India-specific supply chain carbon data from PoC 1 and recycled content certification from PoC 11, creates a premium product differentiation that aligns precisely with the brand values Škoda Auto has established globally.

For the Kylaq - VWG India's highest-volume current model - an OTA (over-the-air) software update delivering a basic eco-coaching dashboard can be deployed within 12 months, reaching the existing Kylaq owner base immediately. This creates a direct consumer touchpoint that reinforces VWG India's sustainability positioning without requiring a new vehicle purchase.

> Business Case Summary - PoC 6: INR 2–5 Cr investment with INR 10–25 Cr NPV over 5 years through revenue premium, brand equity, and ESG positioning. Phase 1 (eco-coaching module) can be deployed immediately on existing models via OTA update. For SAVWIPL, this PoC converts the entire upstream sustainability investment into a visible, tangible consumer proposition - completing the circle from supply chain to showroom.

PoC 7 · Material Circularity

Closed-Loop PP/ABS Reprocessing

Priority: Very High · ROI Window: 12–24 months

11.1 What Is This PoC?

Closed-Loop PP/ABS Reprocessing is the material circularity anchor of the PoC portfolio - the intervention that physically closes the loop on automotive polypropylene by recovering, reprocessing, and re-using ELV plastic in new vehicle production. It represents the most direct expression of circular economy principles in the portfolio: the same PP material that leaves VWG India's Pune plant in a finished vehicle returns, through a structured reprocessing chain, to the same plant as recycled PP (rPP) compound, ready to re-enter the production BOM.

The PoC targets three PP component families in a sequenced approach that begins with the lowest-complexity, most commercially mature application and progressively scales to higher-value, more technically demanding applications. Fender liners (PP-NF/rPP blend, approximately 2.5 kg per vehicle) are the Phase 1 starting point: mono-material, minimal contamination, commercially validated by Maruti Suzuki Fronx and Tata Motors. Door panels (PP-GF10, approximately 5 kg per door) represent Phase 2: multi-material substrate requiring delamination, with UV skin degradation requiring stabiliser management, targeted for B-surface trim carrier applications. Dashboard clusters (PP-TD20, approximately 12 kg per vehicle) are the Phase 3 prize: highest PP content, highest value recovery, most technically complex due to flame retardant contamination requiring NIR sorting - targeted at 70% rPP plus 30% virgin blend for pilot batch production.

11.2 Six-Phase Closed Loop Process

Phase	Activity	Key Technical Step
1 - ELV Procurement	Sourcing from informal dismantling networks with piece-rate incentives and lot-ticket traceability	Formalising the informal recycler ecosystem without displacing it
2 - NIR/XRF Sorting	Grade verification and FR contamination screening at intake	Polymer classification; contaminant and flame retardant removal
3 - Pre-processing	Shredding, washing, and density separation to produce clean PP feedstock	Wash cycle efficiency; paint and adhesive residue management
4 - Compounding	MAPP compatibiliser and HALS/Irganox stabiliser system restoration	Mechanical property restoration to near-virgin automotive spec
5 - Re-use in	Injection-moulded parts verified by CIPET	BOM integration; production

OEM Parts	against VWG Group GMK standards	quality assurance
6 - AIS-129 ELV Marking	End-of-life marking on components closes the next-generation loop	Enables next-generation ELV traceability and recovery efficiency

11.3 CO_2 and Circularity Impact

The business case for closed-loop PP reprocessing in India is exceptional. Virgin PP sourced from domestic producers (Reliance, GAIL, HPCL-Mittal Energy) currently trades at INR 140 to 155 per kg. Automotive-grade rPP from Banyan Nation - the only Indian recycler to have demonstrated bumper-to-bumper automotive PP circularity at commercial scale - is available at INR 95 to 110 per kg. This 25 to 35% unit cost differential means that every kilogram of virgin PP replaced by rPP in VWG India's production BOM generates direct material cost savings alongside the environmental benefit.

The carbon impact is equally compelling. Each tonne of virgin PP replaced by rPP avoids approximately 1.8 to 2.0 tonnes of CO_2-equivalent, because rPP requires approximately 85% less energy to produce than virgin PP from petrochemical feedstock. At VWG India's potential rPP consumption volume of 500 to 2,000 tonnes per year by Year 3 of deployment, the annual CO_2 avoidance is 900 to 4,000 tonnes CO_2-equivalent - a significant and independently verifiable Scope 3 reduction.

Metric	Year 1 (Fender)	Year 2 (+ Door Panels)	Year 3 (+ Dashboard)	Mechanism
rPP volume (tonnes/year)	50–100 TPY	200–400 TPY	500–1,000 TPY	Scaled production BOM integration
CO_2-e avoided (tonnes/year)	90–200 tCO_2-e	360–800 tCO_2-e	900–2,000 tCO_2-e	1.8 tCO_2-e per tonne rPP vs virgin
Material cost saving (INR/year)	0.2–0.5 Cr	0.8–2.0 Cr	2.0–5.0 Cr	INR 45/kg rPP vs virgin differential
EPR credit generation	50–100 credits	200–400 credits	500–1,000 credits	1 credit per MT verified recycling
Recycled content in BOM (% plastic)	0.5–1%	2–4%	5–8%	Phased component family rollout

11.4 Deployment - Speed, TRL Level, and Development Readiness

The Phase 1 fender liner application is at TRL 8 to 9 - commercially validated by Maruti Suzuki Fronx and Tata Motors at automotive production scale. The primary deployment task for VWG India is partner engagement (Banyan Nation MoU), material specification alignment (CIPET GMK validation), and BOM integration - not technology development. This makes Phase 1 deployable within 9 to 12 months.

Phase 2 (door panels) is at TRL 6 to 7, requiring process optimisation for the specific door panel geometry of VWG India's current models. Phase 3 (dashboard PP-TD20) is at TRL 5, requiring NIR sorting investment and formulation development.

Phase	Component	TRL	Timeline	Partner	Key Technical Challenge
Phase 1	Fender lining (PP-NF)	TRL 8–9	9–12 months	Banyan Nation	Material specification; GMK validation by CIPET
Phase 2	Door panel B-surface (PP-GF10)	TRL 6–7	18–24 months	Banyan Nation + CIPET	UV skin delamination; multi-material substrate management
Phase 3	Dashboard cluster (PP-TD20)	TRL 5	30–42 months	Banyan Nation + CSIR-NCL	FR contamination; NIR sorting investment; 70/30 rPP/virgin blend optimisation

11.5 Business Feasibility

11.5.1 Use Cases in the Automotive Industry

- Tata Motors Re.Wi.Re Programme - India's Proof of Concept: Tata Motors, through its Re.Wi.Re (Recycle with Respect) programme and partnership with Banyan Nation, has successfully produced bumpers from recycled PP at automotive specification grade - the only Indian OEM to have demonstrated this at commercial scale. Tata's Nexon and Harrier platforms now incorporate rPP bumper fascia at 8 to 12% recycled content.
- BMW's Secondary Polymer Programme: BMW's global manufacturing standard mandates a minimum 20% recycled polymer content in all plastic components by 2030. BMW's Munich plant produces door panels with 25% rPP content using recycled PP from automotive ELV sources - the precise application targeted in PoC 7's Phase 2.
- Renault's Recycled Content Leadership: Renault's Mégane E-Tech Electric incorporates 24% recycled material by weight - including 5 kg of recycled PP in the interior. Renault's partnership with Plastic Energy for chemical recycling complements its mechanical rPP programme, providing a roadmap VWG India can adapt to the Indian context.
- Toyota RAV4 rPP Integration: Toyota uses recycled polypropylene from automotive ELV sources for the RAV4's under-body splash guards and wheel arch liners - applications comparable to VWG India's Phase 1 fender liner target. Toyota's documentation of this application validates both technical feasibility and the regulatory approval pathway.

11.5.2 Cost Impact

Financial Parameter	Year 1	Year 2	Year 3	Notes
PoC Setup Capex (one-time)	INR 3–6 Cr	-	-	CIPET testing, BOM integration, Banyan Nation supply agreement
rPP Cost vs Virgin PP (INR Cr saving)	0.2–0.5	0.8–2.0	2.0–5.0	INR 45/kg saving at scaled volume
EPR Credit Revenue (INR Cr)	0.04–0.1	0.16–0.40	0.50–1.0	At INR 800–1,200 per credit
CCTS Carbon Credit Revenue (INR Cr)	0.07–0.2	0.29–0.80	0.72–2.0	900–2,000 tCO_2-e at INR 800–1,200/tonne
Total Annual Benefit (INR Cr)	0.3–0.8	1.3–3.2	3.2–8.0	Conservative; CAFE contribution excluded
5-Year NPV (10%)	-	-	INR 10–18 Cr	High confidence given Banyan Nation commercial readiness

11.5.3 Performance Impact

The performance imperative for automotive-grade rPP is non-negotiable: if rPP does not meet VWG Group GMK specifications for the targeted component, it cannot be used. Banyan Nation's rPP has been validated against Tata Motors' equivalent component specifications, providing a strong proxy for VWG Group GMK alignment. CIPET's independent material qualification process will provide the formal validation for SAVWIPL's BOM integration. Banyan Nation's USFDA-certified cleaning and reprocessing technology produces a recyclate whose tensile strength, flexural modulus, and impact resistance values are within 5 to 8% of virgin PP-TD20 specifications - adequate for non-structural, B-surface automotive applications.

11.5.4 Contextualisation to Škoda India (SAVWIPL)

SAVWIPL's Pune plant produces approximately 180,000 vehicles per year across six models. Each vehicle contains an estimated 19 kg of plastic, of which approximately 8 to 10 kg is PP. Total PP consumption at the Pune plant is therefore approximately 1,440 to 1,800 tonnes per year. At the Year 3 target of 5 to 8% rPP content across the plastic BOM, VWG India's rPP demand would scale to 500 to 1,000 tonnes per year - well within Banyan Nation's production capacity of approximately 5,000 tonnes of automotive-grade rPP annually.

For the Škoda Kylaq specifically, the fender liner is an ideal Phase 1 candidate: it is a mono-material PP component with minimal contamination risk, low structural requirement, and visible sustainability value for consumer communication. Kylaq fender liners are currently specified in PP-NF (natural fibre reinforced PP), which is compatible with rPP blending. A 50% rPP content in the Kylaq fender liner, across all 100,000 Kylaq units produced per year, would consume approximately 125 tonnes of rPP annually - a significant and commercially meaningful offtake from Banyan Nation.

Business Case Summary - PoC 7: INR 3–6 Cr investment generates INR 10–18 Cr NPV over 5 years. Phase 1 fender liner deployment in 9–12 months with no technology risk. Banyan Nation's commercially proven rPP at INR 95–110/kg versus virgin PP at INR 140–155/kg makes every tonne substituted simultaneously an environmental win and a cost saving. Tata Motors has demonstrated it works; VWG India has the production scale to do it at greater volume and with greater verified impact.

PoC 8 · Supply Chain Infrastructure

Decentralised Micro-Recycling Units

Priority: High · ROI Window: 36–60 months

12.1 What Is This PoC?

Decentralised Micro-Recycling Units (MRUs) are the physical infrastructure backbone of VWG India's closed-loop material strategy. While PoCs 7, 9, and 10 address what happens to ELV plastic once it is collected and sorted, PoC 8 addresses the upstream challenge that is currently the binding constraint on India's automotive circular economy: the aggregation, depollution, and initial sorting of end-of-life vehicles from India's fragmented, predominantly informal dismantling ecosystem.

India generates approximately 60 lakh (6 million) end-of-life vehicles per year, of which approximately 90% are processed through informal dismantling networks - typically small operators in dense urban clusters who lack the equipment, certification, and data infrastructure required to produce CPCB-compliant depollution records, grade-verified material lots, or EPR credit documentation. The result is massive material leakage: valuable plastics, non-ferrous metals, and EV battery materials exit the formal economy without traceability, without environmental compliance, and without generating the recycled content certification that VWG India needs to make material circularity claims.

MRUs are compact, modular, scalable processing units deployed in a hub-and-spoke network across India's major automotive manufacturing and vehicle ownership clusters. They are designed in two tiers. Tier 1 Collection Hubs are lightweight facilities (500 to 800 m² footprint) processing 5 to 15 ELVs per day, equipped with fluid drain stations, hydraulic compactors, manual magnetic separators, and IoT weighbridges. Tier 2 Full Processing Units are more substantial facilities (1,000 to 2,000 m² footprint) processing 30 to 80 ELVs per day, equipped with auto shredders, eddy current separators, NIR optical plastic sorters, EV battery disassembly cells, and PGM pre-treatment capability. Tier 2 units are CPCB-authorised and ISO 14001 certified, producing grade-verified material lots with full chain-of-custody documentation.

12.2 Why PoC 8 Is the Physical Prerequisite for the Entire Portfolio

Without physical ELV processing infrastructure, every other circularity PoC in the portfolio lacks feedstock. PoC 7 cannot source rPP without a reliable ELV PP supply chain. PoC 9 cannot operate without consistent mixed plastic feedstock. PoC 10 pyrolysis cannot scale without volume-controlled, contamination-screened plastic lots. PoC 11 traceability cannot certify what has not been formally processed. MRUs are the physical prerequisite for the entire material circularity side of the portfolio.

Beyond enabling other PoCs, MRUs generate direct circularity impact through their formalisation effect on India's ELV ecosystem. Formal MRU processing guarantees CPCB-compliant hazardous material management - eliminating the soil and groundwater contamination that characterises informal dismantling. It generates AIS-129-compliant ELV certificates that feed into the VAHAN de-registration system, enabling EPR credit generation. And it creates consistent, grade-verified material lots that command premium prices from reprocessors - improving the economics of the entire circular chain.

Impact Category	Phase 1 (5 units)	Phase 2 (20 units)	Phase 3 (50+ units)
ELV processing capacity (vehicles/year)	15,000–30,000	80,000–150,000	250,000–500,000
PP recovered (automotive grade, TPY)	450–900 TPY	2,400–4,500 TPY	7,500–15,000 TPY
CO_2-e avoided vs informal processing (TPY)	2,700–5,400	14,400–27,000	45,000–90,000
EPR credits generated (annually)	15,000–30,000	80,000–150,000	250,000–500,000
EV battery cells formally recovered (units/year)	2,000–5,000	10,000–25,000	50,000–100,000

12.3 Deployment - Phased Geographic Rollout

Phase	Timeline	Coverage	Capex (INR)	Key Milestone
Phase 1	Years 1–2	NCR, Mumbai, Pune, Nasik, Hyderabad - 5 flagship MRU Tier 2 units, co-located with VW/Skoda supplier clusters	12–30 Cr	CPCB RVSF authorisation; VWG brand ELV channelling MoU with CERO
Phase 2	Years 2–4	Scale to 20 total units in franchise model, onboarding regional scrap aggregators and CPCB-approved dismantlers	30–60 Cr cumulative	Franchise model launch; Banyan Nation rPP offtake linkage; IoT weighbridge integration
Phase 3	Years 4–7	National network of 50+ units in hub-and-spoke configuration, anchored to formal RVSF certification	75–150 Cr cumulative	National hub-and-spoke; CCTS credit trading; EV battery black mass revenue stream activation

12.4 Business Feasibility

12.4.1 Use Cases in the Automotive Industry

- Mahindra CERO - India's Reference Model: CERO is the most directly analogous existing operation. CERO's 9-city Tier 2 network processing 300,000 ELVs per year demonstrates that the MRU operating model is commercially viable in India at scale. VWG India's MRU network, co-branded and co-operated with CERO under a strategic partnership, could leverage CERO's existing authorisations, operating expertise, and geographic coverage.

- BMW Group's Recycling and Dismantling Centre (RDZ): BMW operates the world's most sophisticated automotive ELV processing facility at Unterschleissheim, Germany - a 2,800 m² disassembly facility that processes 10,000 vehicles per year with 95% material recovery rate. The RDZ's material separation technology and component value recovery protocols are directly applicable to Tier 2 MRU design.
- Toyota Metal Co. Japan: Toyota's affiliated recycling network in Japan processes 700,000 ELVs per year through a network of 200 registered recyclers, achieving 99% recycling rate by weight. The hub-and-spoke architecture of Toyota Metal Co. - central shredding facilities fed by distributed collection hubs - is the global best-practice model for the Phase 3 MRU network design.

12.4.2 Cost Impact

Financial Parameter	Value	Notes
Phase 1 Capex (5 Tier 2 units)	INR 12–30 Cr	Site, equipment, CPCB certification, IT infrastructure
Annual OpEx per unit	INR 0.6–1.0 Cr	Labour, maintenance, power, logistics coordination
Revenue per ELV processed	INR 3,000–8,000	Scrap material sales (steel, copper, PP) + EPR credits + ELV cert fee
Annual Revenue (Phase 1, 5 units)	INR 4.5–14 Cr	Based on 15,000–30,000 ELVs/year at INR 3,000–8,000 per ELV
Annual CO_2 Credit Revenue	INR 0.2–0.6 Cr	CCTS at INR 800–1,200/tonne; 2,700–5,400 tCO_2-e avoided
Payback Period	36–60 months	Dependent on ELV flow volume and material commodity prices
5-Year NPV (10%)	INR 10–20 Cr (Phase 1)	Scales significantly in Phase 2 and 3

12.4.3 Contextualisation to Škoda India (SAVWIPL)

SAVWIPL's India brand portfolio - Volkswagen, Škoda, AUDI, Porsche, and Lamborghini - generates a defined and growing volume of end-of-life vehicles. With cumulative India sales exceeding 1 million units across all brands since market entry, and growing significantly with the Kylaq, Virtus, and Taigun volumes from 2022 to 2025, VWG India has a substantial and growing ELV obligation under the Environment Protection (End-of-Life Vehicles) Rules, 2025.

A VWG India ELV take-back programme - branded as Škoda/VW Green Circle - offering owners a defined residual value for end-of-life vehicles channelled through the MRU network, creates a direct consumer engagement mechanism that reinforces the circularity narrative. BMW's equivalent programme in Germany has demonstrated 85% owner participation rates when a financial incentive is attached to the take-back scheme.

> Business Case Summary - PoC 8: INR 12–30 Cr Phase 1 investment with INR 10–20 Cr NPV over 5 years. More importantly, this PoC is the physical prerequisite for the entire material circularity side of the portfolio. Without MRU infrastructure, rPP supply cannot scale, pyrolysis feedstock cannot be assured, and EPR credits cannot be generated. VWG India's partnership with CERO provides the fastest and lowest-risk path to Phase 1 deployment.

PoC 9 · Material Circularity

Reactive Extrusion for Mixed Plastics Upcycling

Priority: Medium · ROI Window: 24–36 months

13.1 What Is This PoC?

Reactive Extrusion for Mixed Plastics Upcycling is an advanced reprocessing technology that addresses the most challenging fraction of the automotive plastic waste stream: the mixed, commingled, partially contaminated plastics that emerge from shredded ELV residue and cannot be economically sorted to mono-material specification for standard mechanical recycling. By introducing reactive compatibilisers during the twin-screw extrusion process, this technology upgrades an otherwise low-value mixed recyclate into a higher-performance polymer blend suitable for structural automotive applications.

In standard mechanical recycling, incompatible polymers (such as PP and ABS) mixed together produce a brittle, weak composite with poor mechanical properties - suitable only for low-value applications like park benches, road dividers, or drainage pipes. This is the downcycling problem that creates the circular economy gap: automotive plastic waste is technically 'recycled' but not back into automotive applications, breaking the value loop.

Reactive extrusion solves this problem through controlled chemical reactions during the melt processing stage. In a twin-screw extruder, reactive agents are introduced to the melt blend in a specific sequence and at defined temperatures and residence times. These agents - typically maleic anhydride grafted polyolefins (MAPP) for PP-PE blends, or styrene-maleic anhydride copolymers (SMA) for PP-ABS blends - create chemical bonds across the interfaces between incompatible polymer phases, transforming a mechanically incompatible mixture into a compatibilised blend with significantly improved mechanical properties.

The resulting compatibilised recyclate exhibits tensile strength 30 to 50% higher than uncompatibilised mixed recyclate, impact resistance improved by 40 to 70%, and flexural modulus suitable for structural automotive interior and semi-structural exterior applications. This upgraded performance profile opens application windows in automotive under-body shields, storage compartment linings, non-visible door sub-components, and wheel arch inner liners.

13.2 Position in the Circularity Cascade

Reactive extrusion fills a critical gap in the circularity cascade. The material flow hierarchy for automotive ELV plastic should be: (1) mono-material, clean PP - closed-loop mechanical recycling (PoC 7); (2) mixed but sortable plastics - reactive extrusion upcycling (PoC 9); (3) contaminated, mixed, unsortable plastics - pyrolysis (PoC 10). Without PoC 9, the material that falls between the clean mono-stream of PoC 7 and the unsortable residue of PoC 10 - estimated at 25 to 35% of total ELV plastic by weight - either goes to pyrolysis at a lower value than it could achieve or is landfilled.

Comparison	CO_2-e / tonne	Application Grade	Value Recovery
Virgin PP (Haldia)	2.1 tCO_2-e	Full automotive spec	INR 140–155/kg
Closed-loop rPP (PoC 7)	0.3 tCO_2-e	Automotive (B-surface)	INR 95–110/kg
Reactive extrusion blend (PoC 9)	0.6 tCO_2-e	Semi-structural automotive	INR 55–80/kg
Standard mixed recyclate (no compatibilisation)	0.7 tCO_2-e	Low-value construction	INR 15–25/kg
Pyrolysis oil (PoC 10)	0.8 tCO_2-e	Fuel / cracker feedstock	INR 35–55/L
Landfill	1.9 tCO_2-e (incl. methane)	Zero recovery	INR 0

13.3 Deployment - Phased Development

Phase	Timeline	Activity	Capex	TRL at End
Phase 1 - Feasibility	Months 1–9	Feedstock characterisation; formulation trials at CIPET/CSIR-NCL; partner identification	INR 0.5–1 Cr	TRL 6
Phase 2 - Pilot	Months 10–24	Pilot reactive extrusion line (50 kg/hour); VWG India application testing; GMK validation	INR 5–10 Cr	TRL 7–8
Phase 3 - Commercial	Months 24–48	Commercial line (300–500 kg/hour); VWG BOM integration; scale to 2,000+ TPY	INR 10–20 Cr	TRL 9

13.4 Business Feasibility

13.4.1 Use Cases in the Automotive Industry

- Renault and Plastic Energy: Renault's global partnership with Plastic Energy uses a variant of reactive compatibilisation to upgrade mixed automotive ELV plastics for re-use in injection moulded components. Renault's Clio IV incorporates compatibilised rPP in its dashboard lower section - a direct application analogue for PoC 9 at VWG India.
- LyondellBasell's Circulen Plus rPP: LyondellBasell's Circulen Plus rPP, used by multiple European OEMs, is produced using reactive compatibilisation to upgrade mixed post-consumer PP streams to automotive specification. LyondellBasell has publicly stated its intent

to develop India-market supply of Circulen Plus by 2026 - creating a potential future supply option alongside Banyan Nation for VWG India's rPP requirements.

- SABIC's Automotive Circular Plastics: SABIC's TRUCIRCLE portfolio, used by multiple VW Group suppliers for European-market vehicles, includes PP and ABS grades produced with reactive compatibilisation from mixed plastic waste streams. VWG Group's global material standards already accommodate SABIC TRUCIRCLE materials - providing a pre-validated specification pathway for VWG India's implementation of PoC 9.

13.4.2 Cost and Performance Impact

Financial Parameter	Value	Notes
Phase 2 Pilot Capex	INR 5–10 Cr	Twin-screw extruder, compatibiliser dosing, quality control lab
Phase 3 Commercial Capex	INR 10–20 Cr	Commercial extrusion line + feedstock logistics + quality infrastructure
Annual OpEx (Phase 3)	INR 3 Cr	Energy, compatibiliser costs, quality testing, logistics
Value of Upcycled Polymer vs Downcycled	INR 30–55/kg premium	INR 55–80/kg vs INR 15–25/kg for standard mixed recyclate
Annual Revenue Potential (Phase 3: 2,000 TPY)	INR 10–16 Cr/year	At INR 55–80/kg for compatibilised blend
CO_2 Credits (CCTS)	INR 0.5–1.2 Cr/year	3,000–6,000 tCO_2-e avoided at INR 800–1,200/tonne
5-Year NPV (10%)	INR 5–15 Cr	Positive but lower certainty than PoC 7; technology risk is higher

13.4.3 Contextualisation to Škoda India (SAVWIPL)

For SAVWIPL, the most immediate application of reactive extrusion output is in the non-visible structural components of the Kylaq and Virtus platforms: under-body shields, wheel arch inner liners, trunk liner mounts, and storage tray liners. These components are not visible to the consumer, do not carry Class A surface requirements, and have relatively generous mechanical tolerances - making them ideal candidates for compatibilised mixed recyclate without requiring VWG Group's highest GMK specification category.

The MRU network of PoC 8 provides the feedstock for PoC 9: automotive shredder residue plastic from CERO and the Pune MRU Tier 2 units is the input stream for the reactive extrusion pilot. This creates a direct material link between PoC 8 (ELV collection) and PoC 9 (value recovery from mixed streams), reinforcing the portfolio integration logic.

Business Case Summary - PoC 9: INR 10–20 Cr Phase 3 investment with INR 5–15 Cr NPV over 5 years. Begin with feasibility investigation (INR 0.5–1 Cr) to validate India-specific formulation before committing to pilot capital. This PoC addresses the middle-stream problem - the mixed plastics that PoC 7 cannot handle and PoC 10 handles at lower value. Sequenced correctly after PoC 7 and PoC 8, reactive extrusion maximises overall ELV plastic value recovery across the portfolio.

PoC 10 · Material Circularity

Plastic-to-Fuel Pyrolysis

Priority: Medium · ROI Window: 24–48 months

14.1 What Is This PoC?

Plastic-to-Fuel Pyrolysis is the terminal value recovery pathway in VWG India's material circularity cascade - the technology that captures residual value from automotive plastic waste streams that are too contaminated, too complex, or too mixed to be economically handled by mechanical recycling (PoC 7) or reactive extrusion (PoC 9). Pyrolysis uses high-temperature thermal decomposition in an oxygen-limited environment to break down polymer chains into shorter hydrocarbon molecules, producing pyrolysis oil (40 to 65% yield), syngas (10 to 20%), and carbon char (10 to 30%).

An important strategic framing governs this PoC: pyrolysis converts plastic to fuel - an inherently linear pathway - rather than returning material to a closed loop. This is a strength when applied to genuinely non-recyclable residual streams, but a strategic liability if it becomes a substitute for mechanical recycling of material that could be reprocessed at higher value. The PoC is therefore designed as a complement to PoCs 7 and 9, not a competitor, processing only the residual fraction after mechanical and reactive extrusion recovery.

Four pyrolysis pathways are pursued in parallel, each addressing a distinct feedstock category and output destination. PoC 10a processes PP waste from ELV bumpers, dashboards, and interior trims at 450°C in a batch or rotating kiln reactor, producing 62 to 85% yield of liquid hydrocarbon suitable as a diesel blend for VWG India's logistics fleet. PoC 10b processes ASR (automotive shredder residue) commingled plastics through a continuous reactor, producing 50 to 70% oil yield for Indian Oil or HPCL cracker co-feedstock. PoC 10c is the circular variant: pyrolysis oil is refined and fed into petrochemical crackers as a naphtha equivalent, generating certified circular polymer back into the production chain through ISCC PLUS mass-balance accounting. PoC 10d processes end-of-life tyres through dedicated tyre pyrolysis units, yielding 40 to 50% tire-derived oil and recovered carbon black for Continental India or Apollo Tyres.

14.2 CO_2, Circularity, and Recyclability Impact

The climate mathematics of pyrolysis are most favourable when the counterfactual is landfill or incineration - both of which generate higher lifecycle emissions than pyrolysis with energy recovery from syngas. Compared to landfill, pyrolysis of one tonne of automotive mixed plastic avoids approximately 1.5 to 2.0 tonnes of CO_2-equivalent (including avoided methane generation from decomposing plastic in landfill). The highest-impact pathway is PoC 10c - the Plastic-to-Oil-to-Plastic circular route. When pyrolysis oil is fed into a steam cracker as a naphtha equivalent and the resulting olefins are polymerised back into PP or PE, the mass-balance accounting under ISCC PLUS certification enables virgin-equivalent circular polymer to be produced - generating both a Scope 3 reduction claim and an EU ELV Regulation recycled content compliance certificate.

Pathway	Feedstock	Output	CO_2-e Avoided	India Revenue (INR/tonne)	TRL
10a - PP Pyrolysis	Clean ELV PP scrap	62–85% pyrolysis oil	1.5–2.0 tCO_2-e/t	INR 2,100–3,500	TRL 6–7
10b - Mixed ASR	Commingled ASR plastic	50–70% oil	1.2–1.8 tCO_2-e/t	INR 1,500–2,500	TRL 5–6
10c - P2O2P	PP/PE pyrolysis oil → cracker	Certified circular polymer	2.0–3.0 tCO_2-e/t lifecycle	INR 4,000–8,000 (ISCC premium)	TRL 4–5 (India)
10d - ELT Pyrolysis	End-of-life tyres	TDO + recovered carbon black	1.8–2.5 tCO_2-e/t	INR 3,000–5,500	TRL 7–8

14.3 Deployment - Partner Ecosystem and Sequencing

Pathway	Partner	Capex (INR)	Timeline	Key Risk
10a - PP Pyrolysis	Rudra Environmental Solutions, Pune	2–5 Cr	12–18 months	Feedstock consistency from CERO/MRU supply chain
10b - Mixed ASR	Pyrocrat Systems, Nagpur	3–8 Cr	18–36 months	PVC contamination in ASR; NIR pre-sort required
10c - P2O2P Circular	Reliance Industries, Jamnagar	0.5–1 Cr (engagement)	36–60 months	Cracker co-feedstock spec alignment; ISCC PLUS certification
10d - ELT Pyrolysis	Agson Global + Pyrum Innovations	5–15 Cr	18–30 months	rCB quality consistency for tyre-grade application

14.4 Business Feasibility

14.4.1 Use Cases in the Automotive Industry

- Renault-Plastic Energy-SABIC Model (P2O2P): The most commercially advanced P2O2P implementation is the Renault-Plastic Energy-SABIC partnership in Europe, where waste plastic from Renault's ELV network is pyrolysed by Plastic Energy, the resulting pyrolysis oil fed to SABIC's cracker as a naphtha equivalent, and the circular PP certified under ISCC PLUS mass-balance accounting and used in Renault vehicle production. This exact model is replicable in India via Reliance Jamnagar.

- BASF ChemCycling Programme: BASF globally processes post-consumer and post-industrial plastic waste through pyrolysis and feeds the resulting oil as feedstock into its cracker in Ludwigshafen, producing certified circular chemicals under the mass-balance approach. BASF India supplies chemical intermediates to Indian automotive manufacturers, creating a potential VWG India connection to the global ChemCycling supply chain.
- APChemi ISCC Plus India: APChemi's India-based ISCC PLUS certified pyrolysis operation provides the most direct India-market deployment reference. APChemi has demonstrated commercial pyrolysis oil purification at specification, enabling circular feedstock claims under internationally recognised certification.
- Continental and Apollo Tyres rCB Programme: Continental India and Apollo Tyres both have active programmes evaluating recovered carbon black (rCB) from tyre pyrolysis as a substitute for virgin carbon black in tyre compound formulations. Apollo's Vadodara plant has completed a validation trial of rCB at 10% substitution level with maintained performance specifications - directly relevant to the PoC 10d output market.

14.4.2 Cost Impact

Financial Parameter	PoC 10a	PoC 10d	PoC 10c (Year 5+)	Notes
Setup Capex	INR 2–5 Cr	INR 5–15 Cr	INR 0.5–1 Cr (engagement)	Pathway-specific equipment
Annual OpEx	INR 0.8–1.5 Cr	INR 1.5–3.0 Cr	TBD	Energy, feedstock handling, QC
Annual Revenue (INR Cr)	1.0–2.5	2.0–5.0	Premium on certified circular PP	Pyrolysis oil sales + carbon credits
CO_2 Credit Revenue	INR 0.1–0.2 Cr	INR 0.2–0.4 Cr	Higher under ISCC	CCTS at INR 800–1,200/tonne
Payback Period	30–48 months	24–40 months	Strategic - long-dated	10a: conservative; 10d: mature TRL
5-Year NPV (10%)	INR 3–7 Cr	INR 5–12 Cr	INR 8–20 Cr (Year 7+)	NPV improves significantly at scale

14.4.3 Contextualisation to Škoda India (SAVWIPL)

For SAVWIPL, the pyrolysis portfolio is most immediately relevant as a waste management and compliance solution. The Pune manufacturing plant generates several hundred tonnes per year of post-industrial plastic waste - paint-contaminated bumper offcuts, glass-fibre reinforced PP trim scrap, multi-material packaging waste, and ELV plastic from the future CERO-linked take-back programme. This waste stream is the feedstock for PoC 10a. Rather than paying for plastic waste disposal (typically INR 5 to 15 per kg for plastics sent to co-processing in cement kilns), SAVWIPL can channel this waste through Rudra Environmental Solutions' nearby Pune facility for pyrolysis, generating oil revenue that partially offsets the logistics cost.

The strategic prize for SAVWIPL is PoC 10c: if Reliance Jamnagar's chemical recycling facility is operational by 2027, VWG India can establish a supply chain that feeds SAVWIPL's post-industrial plastic waste through pyrolysis, into Reliance's cracker, and back out as ISCC PLUS certified circular PP for use in the next generation of Škoda and VW vehicles produced in Pune. This would make SAVWIPL the first Indian automotive manufacturer to have a genuinely circular plastic supply chain - not merely mechanically recycled, but chemically regenerated.

Business Case Summary - PoC 10: Portfolio of four pathways totalling INR 10–30 Cr capex across a 5-year horizon. PoC 10a (PP pyrolysis) and 10d (tyre pyrolysis) are the near-term value generators with payback of 24–48 months. PoC 10c (P2O2P) is the long-term strategic prize - the pathway that closes the chemical loop on automotive plastic and enables ISCC PLUS certified circular polymer for VWG India's production BOM. Begin engagement with Reliance Industries on 10c now, while deploying 10a through Rudra Environmental Solutions immediately.

PoC 11 · Sustainability Intelligence

Digital Traceability & Recycled Content Certification

Priority: Very High · ROI Window: 12–18 months

15.1 What Is This PoC?

Digital Traceability and Recycled Content Certification is the connective tissue that binds the entire PoC portfolio into a verifiable, compliant, and commercially credible system. Without it, every other circular economy intervention in the portfolio is technically productive but commercially unverifiable: the rPP produced in PoC 7 cannot be claimed in ESG reports, the EPR credits enabled by PoC 8 cannot be generated, the carbon savings of PoC 10 cannot be monetised, and the Green UX dashboard of PoC 6 cannot make verified sustainability claims. This PoC is the data infrastructure layer that makes everything else count.

The architecture is built on a blockchain-enabled distributed ledger technology (DLT) platform - implemented through Recykal's existing circular economy marketplace infrastructure - that creates an immutable, time-stamped chain of custody record for every material lot flowing from ELV intake through the circular chain to re-entry in VWG India's production BOM. Six transaction blocks constitute the complete chain-of-custody record.

Block 1 (Vehicle Intake): When an ELV enters the MRU network, its VIN is scanned and validated against the VAHAN API. An ELV certificate is minted on the DLT and a de-registration signal is sent to the MoRTH VAHAN registry. Block 2 (Depollution): All hazardous fluid removal operations are IoT-logged - volumes, types, and disposal routing - with automated CPCB manifest generation and submission. Block 3 (Disassembly and Sorting): Components are disassembled and separated by material type; each lot receives a batch ID with weight verification and NIR-confirmed polymer classification. Block 4 (Marketplace and Logistics): Buyer matching, custody transfer, GPS-tracked delivery with electronic proof of delivery - all recorded on the DLT. Block 5 (Reprocessing): Polymer lots enter the reprocessing facility; input weight and yield data are recorded, with the secondary material output linked cryptographically to the input lot IDs. Block 6 (Certification): Third-party audit verification, Recycled Content Certificate (RCC) issuance under ISO 14021 or GRS standard, EPR credit generation, and update of VWG India's ESG reporting dashboard - creating a final, immutable record of the material's circular journey.

15.2 The Enabling Value Calculation

The strategic importance of PoC 11 cannot be overstated. Without verified chain-of-custody data, the following value streams are inaccessible to VWG India:

Value Stream Enabled	Dependency on PoC 11	Annual Value at Risk

		Without PoC 11
EPR Credit Revenue (PoC 7+8)	Direct - credits require verified processing records	INR 2–5 Cr/year (Year 3)
CCTS Carbon Credit Revenue	Direct - emission avoidance must be verified	INR 1.5–3 Cr/year (Year 3)
CBAM Compliance (EU export)	Direct - embedded carbon verification required	INR 3–8 Cr/year (by 2028)
Recycled Content Claims (ESG)	Direct - ISO 14021/GRS certification required	Reputational + regulatory penalty risk
Green UX Dashboard (PoC 6)	Direct - verified data required for consumer claims	Brand differentiation eliminated
VWG Group global Scope 3 reporting	Direct - group requires verified supply chain data	Group KPI reporting gap

15.3 Certificate Framework

Certificate	Level	Standard	Function
ELV Certificate	Vehicle	MoRTH RVSF Guidelines	Triggers VAHAN de-registration; required for platform listing
Depollution Record	Hazmat	Hazardous Waste Rules 2016	All fluids IoT-logged and CPCB-manifested
Recycled Content Certificate (RCC)	Lot	ISO 14021 / GRS	Core commercial certificate enabling OEM ESG claims and EU ELV Regulation compliance
EPR Credit Certificate	Regulatory	MoRTH RVSF + VAHAN	1 credit = 1 MT verified recycling; tradeable on platform
Battery Material Passport	EV stream	Battery Waste Management Rules 2022	Cell chemistry, State of Health, black mass composition; aligned to EU Battery Regulation

15.4 Deployment - Speed, TRL Level, and Development Readiness

Blockchain-enabled circular economy traceability platforms are at TRL 7 to 8 in India - Recykal's platform is already operational at commercial scale, having channelised over 700,000 metric tonnes of waste across 325 recyclers and 400 brands, with full CPCB digital waste manifest integration. The innovation required for VWG India's implementation is the integration depth: connecting Recykal's platform to VWG India's SAP procurement system via API, developing the automotive-specific certificate framework, and deploying the Battery Material Passport capability for EV battery streams.

Phase	Timeline	Milestone	Capex
Phase 1 - Platform Onboarding	Months 1–6	Recykal MoU; VWG India onboarded as brand partner; first 5 rPP lots certified under ISO 14021	INR 0.5–1 Cr
Phase 2 - SAP Integration	Months 6–12	API connection to VWG India SAP BOM; purchase orders automatically linked to RCC certificates	INR 0.5–1 Cr
Phase 3 - Full Certificate Suite	Months 12–18	All 5 certificate types operational; EPR credit trading enabled; CBAM-ready embedded carbon data	INR 1–2 Cr
Phase 4 - Battery Passport	Months 18–30	DLT battery passport for EV battery flows; aligned to EU Battery Regulation passport specification	INR 0.5–1 Cr

15.5 Business Feasibility

15.5.1 Use Cases in the Automotive Industry

- BMW Digital Product Passport (DPP): BMW has piloted a digital product passport for its batteries in the i3, i4, and iX - containing real-time State of Health, chemistry, and servicing history data accessible via QR code. The EU Battery Regulation (effective 2027) mandates DPPs for all EV batteries over 2 kWh - making BMW's pilot a direct compliance precursor. VWG India's Battery Material Passport (Block 6 of PoC 11) is the India-market implementation of this global requirement.
- Volkswagen Group's Global ATLAS Traceability System: VWG's ATLAS (Automotive Traceability and Lifecycle Assessment System) platform tracks Scope 3 emissions data from Tier-1 suppliers globally, feeding the Group's annual sustainability reporting. PoC 11 is designed to be ATLAS-compatible, providing VWG India's Scope 3 supply chain data in the format required for Group consolidation.
- Renault ISCC PLUS Certification for Circular PP: Renault's partnership with Plastic Energy uses ISCC PLUS mass-balance certification to track circular PP from pyrolysis oil through cracker to vehicle component - the exact certification pathway that PoC 10c and PoC 11 together enable for VWG India.
- Recykal's EPR Compliance Track Record: Recykal's platform has generated and traded EPR compliance certificates across 44,000+ registered plastic intermediaries in India. Its CPCB-authorised network and digital waste manifest system are the infrastructure upon which PoC 11's ELV-specific certification framework is built. VWG India's platform onboarding leverages Recykal's existing regulatory standing rather than building new compliance infrastructure from scratch.

15.5.2 Cost Impact

Financial Parameter	Value	Notes
Total Capex (Phases 1–4)	INR 2–4 Cr	Platform integration, API development, certification framework, battery passport
Annual OpEx	INR 0.8 Cr	Platform fees, audit costs, certificate issuance, maintenance
EPR Credit Revenue (Year 3)	INR 2–5 Cr/year	500–1,000 credits @ INR 4,000–5,000 per credit (MoRTH pricing)
CCTS Carbon Credit Revenue (Year 3)	INR 1.5–3 Cr/year	From PoC 7, 8, 10 verified avoidance; enabled by PoC 11
CBAM Liability Avoided (Year 3)	INR 1–3 Cr/year	Scope 3 verified data for steel/aluminium components; avoids future CBAM penalty
Audit Cost Savings	INR 0.5–1 Cr/year	Automated digital audit trail replaces manual EPR compliance reporting
5-Year NPV (10% discount rate)	INR 5–9 Cr (direct)	Excludes the enabling value for PoCs 6, 7, 8, 10
Enabling Value for Portfolio (strategic)	INR 20–40 Cr (indirect)	Revenue enabled in other PoCs that depend on PoC 11 verification

15.5.3 Performance Impact

The operational performance impact of digital traceability is found in two areas. First, compliance efficiency improves dramatically: the automated digital waste manifest eliminates the manual paper-based CPCB reporting process that currently consumes significant compliance team resources. Second, supplier relationship management improves as the chain-of-custody data creates transparent, auditable records of material flows that reduce disputes over material quality, quantity, and specification compliance. The strategic performance impact is significant: VWG India's ESG rating - which influences institutional investor sentiment, customer trust, and employee attraction - is directly improved by verified, independently certified sustainability data.

15.5.4 Contextualisation to Škoda India (SAVWIPL)

For SAVWIPL, digital traceability is a present-day operational necessity rather than a future aspiration. The Plastic Waste Management Rules 2022 require EPR registration and compliance tracking for all plastic producers - SAVWIPL's use of plastic in vehicle packaging and components places it within the ambit of this regulation. Recykal's onboarding process (Phase 1) can be completed immediately and brings SAVWIPL into full EPR digital compliance within 6 months.

For Škoda Auto's global parent, VWG India's PoC 11 data infrastructure provides the first verified Scope 3 supply chain data from the Indian manufacturing operation - directly feeding the VW Group Annual Report's sustainability disclosures and reducing the data gap that currently exists between the Group's global Scope 3 targets and India's contribution to those targets. The PoC 11 investment therefore has strategic value not only for SAVWIPL but for the Volkswagen Group's global sustainability narrative.

Business Case Summary - PoC 11: INR 2–4 Cr investment with INR 5–9 Cr direct NPV over 5 years, plus INR 20–40 Cr of enabling value created for the rest of the portfolio. This is the highest-leverage, lowest-cost, most immediately deployable PoC in the portfolio. Recykal's platform is live and scalable; the integration work is straightforward. For SAVWIPL, EPR compliance under Plastic Waste Management Rules 2022 is already mandatory - PoC 11 converts a compliance cost into a revenue-generating, ESG-enhancing asset. Deploy Phase 1 within 90 days of portfolio activation.

PART IV

Industry Intelligence & Partner Ecosystem

Closed-Loop Business Case · Pyrolysis Landscape · VWG India Partner Prioritisation

Chapter 16: Closed-Loop Business Case for PP & PET in Indian Automotive

16.1 Context and Material Significance

Polypropylene (PP) and Polyethylene Terephthalate (PET) are the two most strategically important polymer families in the Indian automotive sector. PP dominates interior and exterior applications - bumpers, dashboards, door panels, underbody cladding, and air intake manifolds - collectively accounting for approximately 60–65% of all plastic content by weight in a typical passenger vehicle. India's automotive plastic intensity has risen sharply over the past decade, from approximately 12 kg per vehicle in 2010 to over 19 kg by 2024. With India producing 4.9 million passenger vehicles in FY 2024–25, the total plastic mass entering the ELV stream has grown to an estimated 90,000–110,000 tonnes per annum for PP alone.

16.2 The Business Case for Closed-Loop PP

The business case for closed-loop PP reprocessing in India is exceptional across three dimensions:

Business Case Pillar	Metric	Current Data
Cost Reduction	rPP vs virgin PP price differential	Virgin PP: INR 140–155/kg vs rPP: INR 95–110/kg - 25–35% saving
Carbon Accounting	CO_2-e avoided per tonne rPP vs virgin	1.8–2.0 tonnes CO_2-equivalent avoided per tonne substituted
Regulatory Compliance	EPR obligation headroom	Early movers building closed-loop infrastructure gain compliance advantage
Supply Chain Sovereignty	Price stability benefit	Domestic closed-loop reduces petrochemical commodity market dependence

16.3 OEM Benchmarking - Indian Automotive Landscape

OEM	ELV Infrastructure	Polymer Loop Status	VWG India Benchmark Signal
Tata Motors	7 Re.Wi.Re plants; 110,000 ELV/yr capacity	Active - Banyan Nation rPP partnership; bumpers at spec	First mover; demonstrates feasibility at commercial scale
Mahindra CERO	9 city plants; 300,000 ELV/yr	Planned - polymer recycling explicitly on roadmap	Best-in-class ELV infra; polymer add-on imminent
Maruti MSTI	Noida plant; 24,000	Developing - Toyota	Strong waste diversion

	ELV/yr	Tsusho JV brings know-how	foundation; 89% recovery rate
Hyundai India	Limited disclosure	Early stage - global targets trickling down	Chennai cluster offers rPP integration opportunity

16.4 PET in the Automotive Context

PET's role in automotive applications is less visible than PP but growing in significance, especially in EV architectures - coolant system overflow tanks, structural fabric composites in seating, reinforcing fabrics in tyres and belts, and battery enclosure components. India's PET recycling ecosystem is highly mature for the beverage packaging stream - Ganesha Ecosphere's facilities in Warangal and Kanpur have a combined capacity to process 42,000 tonnes of food-grade rPET per year. India's PET collection rate already exceeds 70%, outperforming Japan, the EU, and the US. The challenge for automotive-grade PET is specification: automotive-grade applications require chemical-process purity that standard mechanical recycling cannot guarantee, making depolymerisation-based chemical recycling the preferred route.

Chapter 17: Pyrolysis Landscape for Automotive Plastics in India

17.1 Why Pyrolysis Is Strategically Relevant

Mechanical recycling is the workhorse of India's current plastic recycling economy, but it has inherent limitations for automotive waste streams. Automotive PP components are frequently multi-layered, paint-contaminated, glass-fibre reinforced, or bonded with adhesives - precisely the feedstock categories where mechanical recycling yields sub-specification output and where chemical recycling via pyrolysis becomes the preferred pathway.

Pyrolysis Parameter	Value
Feedstock (ideal)	Mixed, contaminated, multi-layer plastics; automotive PP/PET waste
Primary Output Streams	Pyrolysis Oil (40–60% by mass), Syngas (10–20%), Carbon Char/rCB (10–30%)
Key Advantage	Handles paint-coated, glass-filled, and composite plastics that mechanical recycling cannot process
Emissions Profile	Significantly fewer harmful gases than incineration; syngas recycled on-site for energy self-sufficiency
India Capital Cost (10 TPD unit)	INR 3–8 crore depending on technology vendor and automation level
Pyrolysis Oil Yield (automotive PP)	55–65% by mass (higher than from mixed municipal plastic waste)
Pyrolysis Oil Sale Price	INR 35–55/litre; ISCC Plus-certified material commands 20–30% premium
Payback Period (10 TPD ELV feedstock)	3–5 years at current market prices

17.2 Leading Pyrolysis Technology Players in India

APChemi - Pioneer in Pyrolysis Oil Purification

APChemi is India's most recognised name in pyrolysis technology and oil purification. The company received the Circular Technology Disrupter Award from FICCI and has deployed plants with ISCC Plus certification - a critical credential for circular economy supply chains seeking international recognition. APChemi's oil purification technology enables the production of high-purity pyrolysis oil that can be integrated into chemical manufacturing and polymer production supply chains.

Rudra Environmental Solutions - Pune-Based Automotive Specialist

Rudra Environmental Solutions operates a PP plastic pyrolysis pilot at 500 kg/day operational capacity in Pune - VWG India's nearest pyrolysis partner for PoC 10a. Its proximity to the Pune manufacturing hub reduces feedstock logistics costs and makes it the priority engagement for Phase 1 pyrolysis deployment.

Pyrocrat Systems, Nagpur - Mixed Plastic Specialist

Pyrocrat Systems provides modular pyrolysis systems for plastic waste in configurations from 50 kg to 2 tonnes per day, with specific experience in commingled automotive shredder residue (ASR) streams - making it the natural partner for PoC 10b.

Reliance Industries - Future Chemical Recycling Scale

Reliance's planned chemical recycling facility in Jamnagar - projected to process up to 200,000 tonnes of PET annually by 2027 - represents the most significant single investment in plastic chemical recycling in India. This facility is the enabling infrastructure for PoC 10c (Plastic-to-Oil-to-Plastic), the highest-value and most strategically significant pyrolysis pathway.

Chapter 18: Partner Ecosystem for VWG India

18.1 Partnership Framework - Three Functional Lanes

Three Partnership Lanes for VWG India: Lane 1 - Closed-Loop Reprocessing: Partners who can convert automotive-grade PP and PET scrap into near-virgin quality recycled polymer certified for VWG India's production specifications. Lane 2 - Scrap Marketplace & Digital Traceability: Digital platforms that aggregate, trace, price, and transact ELV scrap materials. Lane 3 - Pyrolysis: Operators and technology providers who convert non-mechanically-recyclable automotive plastic waste into pyrolysis oil or chemical feedstock.

18.2 Lane 1: Closed-Loop Reprocessing Partners

Banyan Nation, Hyderabad - Primary Recommendation

Banyan Nation is VWG India's strongest near-term partner for automotive-grade rPP. Headquartered in Hyderabad and strategically proximate to VWG India's operations, Banyan Nation is India's first vertically integrated plastics recycling company focused on polyolefins. Its proprietary washing and cleaning technology removes paint, ink, contaminants, and volatile organic compounds from post-industrial and post-consumer plastic, producing recyclate near-virgin in performance. Banyan Nation has already demonstrated the automotive rPP use case at scale through its Tata Motors bumper partnership. The company raised USD 26.2 million Series B (including Circulate Capital and HUL) in March 2025.

Ganesha Ecosphere / Ganesha Ecopet, Warangal - rPET Primary

Ganesha Ecosphere is India's pre-eminent PET recycling enterprise. Its subsidiary Ganesha Ecopet operates four Starlinger recycling lines with a combined food-grade rPET output of 42,000 tonnes per year. Ganesha has recycled over 41 billion PET bottles and supplies rPET to global FMCG brands including Coca-Cola. For VWG India, a targeted partnership should focus on developing an automotive-grade rPET specification for under-hood reservoirs and EV battery structural fabrics.

Gravita India - Secondary rPP Supplier

Gravita India has expanded its PP and HDPE recycling operations at its Mundra (Gujarat) and Phagi (Rajasthan) facilities. Gravita's international footprint in 12 countries positions it as a potential conduit for aligning VWG India's recycled material sourcing with Volkswagen Group's global circular material strategy.

18.3 Lane 2: Scrap Marketplace & Digital Traceability Partners

Recykal - India's Premier Circular Economy Marketplace

Recykal is the most important scrap marketplace partner for VWG India and India's most strategically complete digital waste management platform. As of 2025, Recykal's platform has channelised over 700,000 metric tonnes of waste, connects 400+ brands, 325+ recyclers, and 600+ urban local bodies. Annual revenue reached INR 985 crore in FY 2024–25. Recykal was recognised as the No. 1 Growth Champion in India by the Economic Times and as a Tech Pioneer by the World Economic Forum. Its CPCB-authorised network and digital waste manifest system are the infrastructure upon which PoC 11's certification framework is built.

CERO (Mahindra MSTC Recycling) - ELV Scrap Network

CERO is the natural ELV scrap marketplace partner for VWG India's end-of-life feedstock strategy. As India's first government-authorised vehicle recycler, CERO provides an audited, depolluted ELV dismantling service across nine city plants (with plans for 100+ cities by 2025–26). A VWG India–CERO MoU would formalise the scrap supply chain and create a feedstock channel to Banyan Nation or Gravita India for rPP reprocessing.

18.4 Partner Prioritisation Matrix

Partner	Category	Timeline	First Action
Banyan Nation	rPP Reprocessing	Immediate (0–6 months)	MoU for pilot rPP volumes; specification audit
Recykal	Scrap Marketplace / EPR	Immediate (0–6 months)	Platform onboarding; EPR compliance integration
CERO	ELV Scrap Network	Short-term (3–9 months)	MoU for VWG India brand ELV handling
Ganesha Ecopet	rPET Reprocessing	Short-term (6–12 months)	Automotive-grade rPET specification development
APChemi	Pyrolysis Technology	Medium-term (6–18 months)	PoC pyrolysis unit for non-recyclable auto scrap
Race Eco Chain	PET Collection	Medium-term (9–18 months)	PET collection SLA for VWG India facilities
Gravita India	rPP - Western India	Medium-term (12–24 months)	Secondary rPP supply agreement
Indorama Ventures	rPET at Scale	Long-term (18–36 months)	Automotive rPET co-development MoU
Reliance Industries	Chemical Recycling Scale	Long-term (24–48 months)	Feedstock supply & circular polymer offtake
CSIR-NCL / CIPET	Standards & Validation	Ongoing	Industry-academia research agreement

PART V

Implementation Roadmap & Financial Modelling

Governance Architecture · Business Case · VWG India Global Alignment · Risk Register

Chapter 19: Implementation Roadmap & Governance Architecture

19.1 Portfolio Sequencing Logic

The eleven PoCs are not designed for simultaneous deployment. They are sequenced across a 36-month primary implementation horizon, with long-term PoCs extending to 48–60 months. The sequencing logic follows three principles: quick wins fund deeper infrastructure; foundational infrastructure unlocks advanced interventions; and policy-adaptive flexibility is preserved by structuring deployment milestones around PVI trigger events rather than fixed calendar dates.

The Policy Volatility Index (PVI) serves as the primary external pacing signal. When the PVI is in the 'Elevated' band (121–160), the portfolio operates in a policy-neutral mode - deploying only those PoCs whose business case does not depend on specific policy outcomes (PoC 5 packaging loop, PoC 11 traceability, PoC 7 rPP reprocessing). When the PVI drops to 'Moderate' (80–120) - most likely following FAME-III confirmation - the portfolio accelerates into policy-leveraged deployments.

19.2 36-Month Master Roadmap

PoC	Title	Q1–Q2 FY26	Q3–Q4 FY26	Q1–Q2 FY27	Q3–Q4 FY27	PVI Trigger
5	Packaging Loop	Pilot design	Deploy RFID	Full roll-out	Optimise	Any band
11	Digital Traceability	Architecture	Pilot onboarding	Cert framework	Scale EPR	Any band
7	Closed-Loop PP	Fender lining pilot	Door panel	Dashboard	Scale rPP	Any band
1	Supplier Routing	Data kit design	10–15 suppliers	30–40 suppliers	Full Tier-1	PVI < 125
8	Micro-Recycling	Site assessment	Tier 1 MRU x2	Phase 1 x5	Phase 2	Any band
4	AI Material Sub	DB architecture	Pilot components	AI layer	Deployment	PVI < 120
2	Micro-Grid	Energy audit	Solar+storage	Load shift	Real-time opt.	Any band
9	Reactive Extrusion	Feasibility	Partner ID	Pilot design	TRL scale	PVI < 120
10a	PP Pyrolysis	Partner MoU	50 TPD trial	Scale 200 TPD	ISCC cert	Any band
10c	P2O2P	Reliance	Spec dev.	Pilot	Cracker	PVI <

	Pathway	engage		volume	trial	120
3	DfD in CAD	Plugin spec	Pilot x2 models	VW design int.	Compliance	PVI < 125
6	Green UX	Data dependency	UX wireframes	Platform dev.	Beta launch	After PoC 1,11

19.3 Governance Architecture

Circular Material Council

A quarterly steering group chaired by VWG India's Head of Sustainability, with representation from Procurement, Manufacturing, Legal, and ESG Reporting, and external seats for Banyan Nation, Recykal, and CERO. The Council oversees KPI tracking, specification approval, partner performance review, and PVI-adaptive portfolio pacing. Meeting cadence: quarterly, with ad-hoc sessions triggered whenever the PVI crosses a band boundary.

Digital Traceability Layer - Single System of Record

Recykal's platform operates as the single system of record for all material flows - from ELV scrap generation through reprocessing to re-entry into VWG India's production BOM. Integration with VWG India's SAP procurement system ensures that every recycled material input has a verifiable chain-of-custody certificate attached to the purchase order.

Specification Pathway - CIPET as Independent Testing Authority

The Central Institute of Plastics Engineering and Technology (CIPET) serves as the independent testing and certification authority for rPP and rPET entering VWG India's production system. A tiered qualification process applies: Tier A (non-structural interior components), Tier B (structural exterior components), and Tier C (safety-critical components, which remain virgin-material-only until TRL-8 validation is achieved).

Carbon Credit Strategy

APChemi-certified pyrolysis oil volumes and Banyan Nation rPP volumes are registered under India's Carbon Credit Trading Scheme (CCTS) and tracked against VWG Group's internal carbon pricing mechanism. The avoided emissions from automotive plastic circularity - approximately 1.8–2.0 tonnes CO_2-equivalent per tonne of virgin PP displaced - are monetised at CCTS exchange prices and reported in VWG India's annual Scope 3 disclosure under the Group's goTOzero framework.

19.4 KPI Framework

KPI	Baseline	12-Month Target	36-Month Target	Reporting Standard
Scope 3 emissions (supply chain)	Not currently tracked	Baseline established; 10+ suppliers	8–15% reduction vs baseline	GHG Protocol / CCTS
Recycled PP content in production BOM	0%	2–3% (fender lining pilot)	10–15% across 3 component families	VWG GMK / ISO 14021
ELV plastics formally channelled (TPY)	Informal / unmeasured	500 TPY via CERO MRU	5,000 TPY across Phase 1 network	AIS-129 / EPR Rules
EPR credits generated	0	500 credits	5,000 credits	MoRTH RVSF + VAHAN
Single-use packaging eliminated (%)	~100% single-use	30–40% converted	70–80% converted	PWM Rules 2022 / EPR
Digital traceability (% flows certified)	0%	Pilot: 5–10 certified lots	All rPP/rPET lots certified	ISO 14021 / GRS
Scope 2 carbon intensity (Pune plant)	Current grid average	5% reduction (Phase 1)	15–25% reduction	VWG Energy Management
Partner ecosystem activated	0 formal agreements	4 MoUs signed	10-partner ecosystem operational	VWG India procurement KPI

Chapter 20: Financial Modelling & Business Case Summary

20.1 Individual PoC Economics

PoC	Capex Estimate	Annual OpEx	Annual Benefit	Payback Period	5-Year NPV (10%)	Risk Level
5 - Packaging Loop	INR 2–4 Cr	INR 0.5 Cr	INR 3–5 Cr	18–36 months	INR 8–12 Cr	Low
7 - Closed-Loop PP	INR 3–6 Cr	INR 1–2 Cr	INR 4–8 Cr	24–36 months	INR 10–18 Cr	Low-Medium
11 - Traceability	INR 2–4 Cr	INR 0.8 Cr	INR 2–3 Cr	30–42 months	INR 5–9 Cr	Low
1 - Supplier Routing	INR 1–3 Cr	INR 0.5 Cr	INR 2–4 Cr	18–30 months	INR 6–12 Cr	Medium
2 - Micro-Grid	INR 8–15 Cr	INR 1 Cr	INR 5–10 Cr	24–36 months	INR 12–25 Cr	Low-Medium
8 - Micro-Recycling	INR 15–30 Cr	INR 3–5 Cr	INR 8–15 Cr	36–60 months	INR 10–20 Cr	Medium
4 - AI Material Sub	INR 2–4 Cr	INR 0.8 Cr	INR 3–6 Cr	24–36 months	INR 7–14 Cr	Medium
10a - PP Pyrolysis	INR 5–10 Cr	INR 2 Cr	INR 4–7 Cr	30–48 months	INR 6–12 Cr	Medium
9 - Reactive Extrusion	INR 10–20 Cr	INR 3 Cr	INR 5–10 Cr	36–60 months	INR 5–15 Cr	Medium-High
3 - DfD in CAD	INR 1–2 Cr	INR 0.3 Cr	INR 2–5 Cr	Long-term / strategic	Strategic	Medium
6 - Green UX	INR 2–5 Cr	INR 0.5 Cr	INR 3–8 Cr	Long-term / brand	Strategic	Low

20.2 Aggregated Portfolio Business Case

INR 51–103 Cr	INR 41–81 Cr	30–42 months
Total Portfolio Capex Phased over 36 months	**Combined Annual Benefit (Yr 3)**	**Portfolio Payback Period** Blended across 11 PoCs

	Cost savings + EPR + carbon credits	

The portfolio-level NPV analysis (10% discount rate, 5-year horizon) yields a positive return even under conservative assumptions. Under the base case, the aggregate 5-year NPV is approximately INR 70–140 crore, with the packaging loop (PoC 5), micro-grid (PoC 2), and closed-loop PP (PoC 7) contributing the largest direct financial returns. Under an optimistic scenario - FAME-III confirmed by Q3 2026, CCTS at INR 1,000/tonne CO_2-e, CBAM enforcement against VWG India's export components - the portfolio NPV escalates to INR 150–280 crore over five years.

20.3 Carbon Credits and CCTS Monetisation

CCTS Credit Source	Estimated Annual Volume (Yr 3)	Price Range (INR/tonne)	Annual Revenue Potential
rPP substitution (PoC 7, 500 TPY)	900–1,000 tonne CO_2-e	INR 800–1,200	INR 72–120 lakhs
Pyrolysis (PoC 10a, 500 TPY processed)	750–1,000 tonne CO_2-e	INR 800–1,200	INR 60–120 lakhs
Micro-grid buffering (PoC 2)	500–800 tonne CO_2-e	INR 800–1,200	INR 40–96 lakhs
Total CCTS Revenue Potential	2,150–2,800 tonne CO_2-e	-	INR 1.72–3.36 Cr/year

Chapter 21: VWG India in the Global Volkswagen Sustainability Framework

21.1 VW Group's goTOzero and regenerate+ Strategy

The Volkswagen Group's 2025 Annual Report articulates an ambitious sustainability framework under two overarching strategies. The goTOzero strategy commits the Group to becoming a net carbon-neutral company by 2050, with intermediate milestones of a 55% reduction in the lifecycle carbon footprint of its vehicle fleet by 2030 versus 2018. The regenerate+ strategy extends this commitment beyond carbon neutrality to regenerative impact - aiming to have a net positive impact on biodiversity, water systems, and circular material flows by 2030.

21.2 Alignment with VWG Group's Material Targets

VWG Global Target	VWG India Relevance	PoC Alignment
55% lifecycle CO_2 reduction by 2030 (vs 2018)	India's growing production base makes Scope 3 management critical to Group target	PoC 1 (Scope 3 routing), PoC 4 (material substitution), PoC 2 (Scope 2)
Net carbon neutrality by 2050	India's CCTS and CBAM exposure require a decade of infrastructure build-out now	PoC 10c (plastic-to-oil-to-plastic), PoC 11 (traceability for CBAM)
Recycled material content mandates (Group GMK)	EU ELV Regulation 2027 recycled content mandates apply to VWG exports	PoC 7 (rPP loop), PoC 16/17/18 (PP+PET business case + partner ecosystem)
goTOzero Scope 3 supplier transparency	VWG Group requires verified Scope 3 data from all Tier-1 suppliers globally	PoC 1 (Supplier Carbon Readiness Kit), PoC 11 (chain-of-custody data)
Zero waste to landfill at all production sites	VWG India's Pune plant to achieve ZWL certification	PoC 5 (packaging), PoC 8 (MRU), PoC 10 (pyrolysis for residual streams)
regenerate+ biodiversity and water positive	India-specific ELV ecosystem formalisation reduces pollution from informal dismantling	PoC 8 (MRU - CPCB-compliant depollution), PoC 11 (hazmat tracking)

Chapter 22: Risk Register & Mitigation Strategies

22.1 Policy and Regulatory Risk Register

Risk	Probability	Impact	Mitigation
FAME-III delayed beyond Q3 2026, suppressing EV supplier carbon data collection	High	Medium	Activate policy-neutral PoC track; PoC 5, 7, 11 proceed independently of FAME policy
CAFE-III 2027 target revised downward, reducing material substitution urgency	Medium	Medium-High	Proceed with PoC 4 on cost-savings basis alone; revision does not change CBAM pressure
CCTS launch delayed or pricing set below INR 500/tonne	Medium	Low-Medium	Financial models profitable at INR 800/tonne; PoC 7 and 10 viable on direct cost-savings basis
ELV Rules 2025 enforcement deferred or watered down	Low-Medium	Medium	PoC 8 MRU network has commercial scrap revenue business case independent of regulatory mandate
Cross-ministry CMD spike - contradictory Scope 3 mandates	High	Medium	Adopt BRSR Core framework voluntarily; avoids dual-compliance risk by selecting highest standard

22.2 Technology and Execution Risk Register

Risk	Probability	Impact	Mitigation
Automotive-grade rPP from Banyan Nation fails VWG GMK specification testing	Low-Medium	High	Pilot on non-structural components first (Tier A); CIPET independent validation before Tier B use
PVC contamination in reactive extrusion feedstream (PoC 9)	Medium	High	Mandatory NIR pre-sort at RVSF intake; PoC 9 proceeds as feasibility study first
Blockchain traceability platform unable to integrate with VWG SAP procurement	Low	High	Require PoC 11 API specification aligned to SAP S/4HANA before Recykal MoU signed
Pyrolysis oil from PoC 10a fails refinery co-feedstock specification	Medium	Medium	APChemi oil purification step ensures compliance; pilot 50 kg batch before scaling
Informal ELV dismantler non-compliance with	High	Medium	Recykal digital waste manifest automates CPCB compliance; financial incentive

CPCB manifesting requirements			model applied

22.3 Strategic and Market Risk Register

Risk	Probability	Impact	Mitigation
Tata Motors or Mahindra establishes exclusive rPP supply relationship with Banyan Nation	Medium	High	Immediate engagement with Banyan Nation; MoU within 0–6 months to secure supply allocation
Recykal acquired or changes business model	Low	Medium	Dual-platform strategy: Recykal primary + CERO digital layer as backup; data portability clause
VWG Group India strategy de-prioritises circularity amid cost reduction programme	Medium	High	Frame PoC portfolio as cost-saving (PoC 5, 7, 2) first; ESG compliance second; brand equity third
India's informal ELV sector resists formalisation, limiting MRU feedstock volumes	High	Medium	Begin with CERO's existing 9-city network; supplement with piece-rate incentive programmes
CBAM enforcement timetable accelerated, requiring faster CBAM-ready data infrastructure	Low	Medium	PoC 11 (traceability) designed for CBAM alignment from Day 1; no remediation required

PART VI

Regulatory Deep Dive & Compliance Landscape

India ELV Rules 2025 · CAFE III/IV · EPR Framework · CBAM · Battery Passport

Chapter 23: Regulatory Landscape - India's Evolving Compliance Architecture

23.1 Overview

India's automotive regulatory landscape is governed by an interlocking set of central government ministries, statutory bodies, and policy instruments that together define the compliance environment for OEMs, Tier-1 suppliers, and recyclers. This chapter provides a structured deep dive into the five most material regulatory developments affecting VWG India's circular economy and decarbonisation strategy through 2030.

23.2 Environment Protection (End-of-Life Vehicles) Rules, 2025

Parameter	Detail
Applicability	All passenger vehicle and commercial vehicle OEMs selling in India; phased by production volume
EPR Obligation Type	Steel-linked EPR in current iteration; plastic EPR targets under amendment pipeline
Target Recovery Rate	80% by weight per ELV by 2030 (escalating from 70% in first compliance year)
Penalty Structure	INR 15,000–50,000 per vehicle for non-compliance with RVSF channelling requirements
Enforcement Body	Central Pollution Control Board (CPCB) in coordination with VAHAN registry
Certificate of Deposit (CoD)	Required for each ELV processed; CoD is the compliance unit and tradeable instrument
Implementation Timeline	Notification: 2025; First compliance period: FY 2025–26; Full enforcement: FY 2027–28

23.3 CAFE III and CAFE IV Norms - The 2027 Compliance Cliff

CAFE Phase	Timeline	CO_2 Target	Key Compliance Levers	VWG India Exposure
CAFE II Phase 1	Apr 2022	130 gCO_2/km	Fuel efficiency improvements; mild hybridisation	Already compliant
CAFE II Phase 2	Apr 2023	113 gCO_2/km	ICE efficiency; stop-start systems; aero optimisation	Compliant - Kylaq, Virtus contribution

CAFE III	Apr 2027	88.4 gCO_2/km	BEV and PHEV integration; material lightweighting; rPP/rPET content	Critical compliance milestone
CAFE IV	2030 (draft)	~70 gCO_2/km (TBD)	Full electrification of volume models; circular material mandates	Requires full PoC 4 and 7 deployment

23.4 Plastic Waste Management Rules - EPR Framework

The Plastic Waste Management (PWM) Rules 2022 introduced a mandatory recycled content requirement for rigid plastics: 30% by FY 2025–26, rising to 60% by FY 2028–29. For automotive OEMs, rigid plastic components - bumpers, dashboards, door panels, underbody cladding - fall squarely within this mandate. The amendment pipeline is expected to extend the recycled content mandate to automotive plastic applications by FY 2027–28.

> The recycled content mandate trajectory - 30% by FY26, 60% by FY29 - is not aspirational policy; it is enforceable law with penalty consequences. VWG India that builds its rPP and rPET supply chains now will meet these targets with commercially validated partners at known cost points. VWG India that waits will face a 2028 scramble for certified recycled polymer supply in a market where demand will far outstrip supply.

23.5 Battery Waste Management Rules, 2022

Battery Rules Requirement	VWG India Compliance Pathway
EPR registration for battery producers	Register VWG India brands under CPCB Battery EPR portal
Battery collection targets	CERO ELV processing network includes EV battery disassembly cell (PoC 8, Tier 2 MRU spec)
Chain-of-custody documentation for battery materials	Battery Material Passport in PoC 11 traceability architecture
Black mass composition reporting	NIR/XRF analysis at MRU Tier 2; data logged to DLT in real time
Hazardous waste manifesting for battery electrolyte	IoT-logged CPCB manifest at MRU depollution station (Block 2 of PoC 11)

23.6 Carbon Border Adjustment Mechanism (CBAM) - India Exposure Analysis

CBAM Dimension	Current Status (2026)	Projected Status (2028–30)	VWG India Action Required
Steel components exported to EU	CBAM applies - transitional phase	Full financial liability at EU carbon price (~EUR 50–100/tonne)	PoC 1 (supplier carbon data) to verify embedded steel carbon intensity
Aluminium components exported to EU	CBAM applies - transitional phase	Full financial liability	PoC 4 (material substitution) to shift to low-carbon aluminium suppliers
Automotive plastic components	Not yet in scope	Likely in scope by 2028 if EU expands CBAM	PoC 7 and 10c (rPP and circular pyrolysis) to reduce embedded carbon in plastic
VWG India Scope 3 data gap	No verified supplier-level CO_2 data	CBAM requires verified embedded carbon calculation per shipment	PoC 1 (Supplier Carbon Readiness Kit) to fill data gap before liability arises
ISCC PLUS certified recycled content claims	Available; not yet mandatory	EU requires certified recycled content proof for CBAM-relevant components	PoC 11 (Recycled Content Certificate) + APChemi ISCC certification

Chapter 24: Portfolio Integration - A System View

24.1 The Four Portfolio Layers

Layer	PoCs	Key Output	Enables
Foundation / Infrastructure	1, 11, 8	Carbon data, traceability, physical ELV processing	All other PoCs; regulatory standing; EPR credits
Design & Material	3, 4, 2	Design-stage circularity, material intelligence, clean energy	PoC 7 (DfD→rPP), PoC 10c (material substitution)
Materials Recovery	7, 9, 10	Closed-loop PP, compatibilised recyclate, pyrolysis oil	VWG India production BOM; EU CBAM compliance
Quick Win	5	Packaging loop ROI	Portfolio funding; early sustainability metric
Consumer Intelligence	6	In-vehicle carbon dashboard	Brand equity; ESG narrative; driver behaviour

24.2 Critical Cross-PoC Dependencies

Understanding the dependency architecture is essential for investment sequencing. The following dependency types are identified:

- Data Dependencies: PoC 6 (Green UX) cannot make verified sustainability claims without PoC 1 (Scope 3 data) and PoC 11 (RCC). PoC 4 (AI Material Sub) cannot accurately model India-specific carbon impacts without PoC 1's supplier carbon data. These dependencies mandate that PoC 1 and PoC 11 are deployed before any consumer-facing sustainability claims are activated.
- Feedstock Dependencies: PoC 7 (Closed-Loop PP), PoC 9 (Reactive Extrusion), and PoC 10 (Pyrolysis) all require ELV-sourced plastic feedstock that can only be produced at commercial scale through PoC 8's MRU network. In the near term, this gap is bridged by Banyan Nation's existing feedstock procurement and CERO's ELV dismantling network.
- Certification Dependencies: PoC 7's rPP can only generate EPR credits, CCTS carbon credits, and EU ELV Regulation recycled content compliance certificates if PoC 11's chain-of-custody records are in place. Without PoC 11, rPP is a cost saving but not a regulatory asset.
- Energy Dependencies: PoC 7 and PoC 9's reprocessing operations generate lower lifecycle carbon than virgin polymer production only if the energy powering those operations is itself low-carbon. PoC 2's micro-grid decarbonisation of the Pune plant's energy supply is therefore a prerequisite for the full carbon benefit of PoCs 7 and 9.

24.3 2030 Vision: The Sustainability Performance Profile

2030 Target	Metric	Enabling PoCs
Recycled content in plastic components	15–20% rPP / rPET across production portfolio	PoCs 7, 9, 10c, 4
Scope 3 supply chain emissions reduction	10–15% reduction vs 2026 baseline	PoCs 1, 4, 2
ELV plastics formally channelled annually	10,000+ tonnes per year through verified RVSF network	PoCs 8, 11, 7
EPR credits generated annually	10,000+ certificates - EPR compliance fully self-funded	PoCs 11, 8, 7
Single-use packaging eliminated	80% of inbound component flows converted to returnable crates	PoC 5
Scope 2 carbon intensity at Pune plant	20–25% reduction vs 2026 baseline	PoC 2
Digital traceability coverage	100% of recycled material inputs carry verified chain-of-custody	PoC 11
CBAM-ready embedded carbon data	Verified Scope 3 data for 80% + of Tier-1 supplier components	PoC 1
Carbon credits generated (CCTS)	3,000–5,000 tonne CO_2-e per year - actively traded on CCTS exchange	PoCs 7, 10, 2
In-vehicle carbon transparency	Green UX dashboard deployed across EV fleet in India	PoC 6

Chapter 25: Conclusion - The Strategic Imperative and Path Forward

25.1 What This Report Has Established

This report has assembled, for the first time, a unified analytical framework for Volkswagen Group India's circular economy and decarbonisation strategy - integrating quantitative policy analysis (the Policy Volatility Index), primary survey data (the Supply Chain Preparedness Index), eleven detailed Proof of Concept interventions with full business feasibility assessments, industry intelligence on the PP and PET closed-loop business case and pyrolysis landscape, a prioritised partner ecosystem, an implementation roadmap with governance architecture, a financial model, and a regulatory deep dive.

The central finding is both sobering and actionable: India's automotive sector operates in a sustained policy volatility environment (avg PVI of 131 since 2020) that structurally constrains the supply chain preparedness investments that the sector needs to make. Yet within that environment, a clear pathway exists - sequenced around policy-resilient PoCs that deliver commercial returns regardless of near-term regulatory outcomes, while building the infrastructure that unlocks maximum value when policy clarity arrives.

25.2 Three Structural Advantages

Three structural advantages make VWG India's circular economy position stronger than the PVI environment might suggest:

1. Volkswagen Group's global goTOzero and regenerate+ mandates provide a stable investment signal that is independent of India's domestic policy volatility - group-level sustainability commitments do not pause because FAME-III is delayed.
2. The Indian partner ecosystem - anchored by Banyan Nation, Recykal, CERO, Ganesha Ecopet, and APChemi - is commercially ready and actively seeking OEM partnerships. The infrastructure is available; what has been missing is an organised, sequenced demand signal from a premium OEM.
3. The first-mover advantage in automotive rPP in India is still available - Tata Motors has demonstrated the technology, but no premium OEM has yet established a closed-loop supply chain at scale.

25.3 Immediate Next Steps (0–90 Days)

Timeframe	Action	Responsible Party	Success Metric
Day 0–30	Initiate MoU negotiations with Banyan Nation for pilot rPP offtake - fender liner specification comparison and supply allocation	VWG India Procurement / Sustainability	Letter of Intent signed within 30 days

Day 0–30	Deploy PoC 5 packaging loop pilot design across top 20 inbound component suppliers in Pune cluster	VWG India Manufacturing / Logistics	Packaging waste audit completed; crate specifications drafted
Day 30–60	Commission CIPET independent material qualification audit for Tier A rPP components	VWG India Engineering	CIPET audit brief submitted; timeline agreed
Day 30–60	Recykal platform onboarding: register VWG India as brand partner; integrate Pune plant plastic waste data	VWG India Sustainability / Legal	EPR registration completed; FY 2025–26 baseline established
Day 30–60	Initiate CERO MoU covering all five VWG India brand groups for ELV handling	VWG India Legal / Government Affairs	CERO MoU framework agreed
Day 60–90	Convene inaugural Circular Material Council; appoint Chair and working group leads	VWG India Head of Sustainability	Council Charter signed; first meeting scheduled
Day 60–90	Integrate SIAM policy tracker and PIB press release feeds into ESG monitoring dashboard for monthly PVI tracking	VWG India ESG Reporting	PVI monitoring live; FAME-III watch protocol activated

The Indian automotive sector is at a consequential juncture. The policy environment is uncertain, the regulatory architecture is evolving rapidly, and the competition for first-mover advantage in automotive circular economy is intensifying. VWG India - backed by Volkswagen Group's global sustainability commitments, the T-Works field intelligence network, and Woxsen University's research infrastructure - has the strategic assets, partner relationships, and analytical frameworks assembled in this report to act decisively. The next 90 days will determine whether VWG India leads India's automotive circular economy transition or follows it.

APPENDIX A

Policy Volatility Index (PVI)

Methodology, Scoring Procedure & Annual Data (2015–2025)

VWG India Circular & Carbon Economy Portfolio Report · T-Works × Woxsen University · April 2026

A.1 Overview and Purpose

The **Policy Volatility Index (PVI)** is a quantitative, sector-specific instrument designed to measure the intensity and character of regulatory uncertainty facing the Indian automotive sector on an annual basis from 2015 to 2025. It provides the analytical foundation for Part I of the VWG India Circular and Carbon Economy Portfolio Report and serves as the primary external pacing signal for the portfolio implementation roadmap in Part V.

The PVI is constructed following the **Baker-Bloom-Davis Economic Policy Uncertainty (EPU) methodology** (Baker, Bloom & Davis, 2016), which has been applied across economies and sectors globally to measure the effects of policy uncertainty on investment, employment, and output. Adapting this framework to the Indian automotive context required constructing sector-specific indicators and calibrating the weighting structure to the mechanisms through which regulatory uncertainty most directly affects OEMs and supply chain participants across the value chain.

The index serves three distinct operational functions within the VWG India report:

- **Contextualisation:** It quantifies the policy environment against which VWG India's supply chain preparedness and PoC investment cases are assessed, replacing subjective narrative with a calibrated numerical signal.
- **Historical anchoring:** It identifies the specific years and policy episodes that constitute inflection points in the sector's regulatory history, grounding strategic recommendations in documented evidence rather than general characterisations of uncertainty.
- **Forward governance:** It functions as a real-time portfolio management tool. When the PVI exceeds 130 (upper Elevated band), the implementation roadmap activates its policy-neutral investment track, deferring interventions whose business cases depend on specific regulatory outcomes such as FAME-III confirmation or CAFE-III target stability.

A.2 Theoretical Framework

A.2.1 The Baker-Bloom-Davis EPU Foundation

The original Baker-Bloom-Davis framework constructs national-level Economic Policy Uncertainty indices using three components: **news-based uncertainty** (keyword frequency in newspaper archives), **tax code expiration provisions**, and **disagreement among economic forecasters**. Each component captures a distinct channel through which policy uncertainty transmits into business and investment decisions. The framework has been applied to over 20 countries and numerous sector-specific contexts, consistently demonstrating that elevated EPU scores are associated with reduced investment, higher financing costs, and depressed employment.

The Indian automotive adaptation retains the core multi-source, multi-channel measurement principle but replaces the tax-code and forecaster-disagreement components with indicators more directly relevant to the sector's regulatory dynamics. Specifically, the adapted framework introduces Regulatory Frequency (RF) and Cross-Ministry Divergence (CMD) as structural pillars that reflect the multi-ministerial complexity of India's automotive policy environment — a feature with no direct equivalent in the original BBD country-level frameworks.

A.2.2 The Four-Pillar Architecture

The PVI is structured around four independent pillars, each capturing a distinct pathway through which policy uncertainty affects automotive supply chain participants. Equal weighting across all four pillars reflects the design judgment that no single channel is structurally dominant over extended time horizons — each pillar has produced the highest single-pillar score in at least one year of the observation period.

Code	Pillar Name	What It Measures	Weight	BBD Analogue
RF	**Regulatory Frequency**	Volume of new automotive regulations, gazette notifications, CAFE/BS phase changes, and FAME/PLI policy instruments issued in a calendar year.	25%	Tax-code expirations (volume proxy)
PRI	**Policy Reversal Intensity**	Frequency and magnitude of policy U-turns: subsidy cuts, deadline extensions, scope reductions, import duty reversals. Each event scored 1–3 by severity.	25%	News uncertainty (directional change)
CMD	**Cross-Ministry Divergence**	Documented public conflicts between MoRTH, MHI, BEE/MoP, Finance Ministry, and NITI Aayog on EV, emission, and automotive tax policy.	25%	Forecaster disagreement (institutional)
MNU	**Media & News Uncertainty**	Relative frequency of automotive policy uncertainty keywords across	25%	News uncertainty

		Economic Times, Business Standard, and The Hindu, normalised to publication volume.		(direct analogue)

A.3 Composite Formula and Normalisation Procedure

A.3.1 Composite Formula

The PVI composite score is computed as the arithmetic mean of the four normalised pillar scores:

$$PVI = 0.25 \times RF + 0.25 \times PRI + 0.25 \times CMD + 0.25 \times MNU$$

Equal weighting across four independent pillars · Composite score range: 0 – 200

Because all four pillars carry equal weight, the composite simplifies to an unweighted mean. This ensures that a high score on any single pillar cannot dominate the composite — each pillar contributes a maximum of 25% to the overall index value. The theoretical score range is 0 to 200, calibrated so that the 2015–2019 period produces a composite mean of approximately 100.

A.3.2 Normalisation Procedure

Each pillar's raw annual score is normalised to a common scale anchored to the **2015–2019 baseline period**. This baseline was chosen because it represents a period of active but largely predictable rulemaking — FAME-I, CAFE-I, and BS-IV were all operational without the structural discontinuities of the 2020 BS-VI compression or the COVID-19 disruption. It is the last extended period in which policy uncertainty in the sector was genuinely manageable without dual-track planning.

$$\text{Normalised Pillar Score} = (\text{Raw Score} \div \text{Mean}_{2015\text{–}2019}) \times 100$$

Score of 100 = average baseline uncertainty level (2015–2019). Scores above 100 indicate above-baseline uncertainty.

The normalisation is applied independently to each pillar before computing the composite. This ensures that differences in the raw scale of each pillar — for example, the MNU pillar's article counts versus the CMD pillar's conflict instances — do not introduce implicit differential weighting into the composite score.

Illustrative example: If the MNU pillar averaged 48 qualifying articles per month during 2015–2019, and the 2020 monthly count was 89 articles, the normalised MNU score for 2020 is (89 ÷ 48) × 100 = 185.4, rounded to 185. This is the peak score observed in the dataset and reflects the extraordinary confluence of the COVID-19 production halt, the BS-VI enforcement deadline, and the scrappage policy stall generating simultaneous media uncertainty coverage.

A.3.3 Score Band Interpretation

The composite PVI score is interpreted against four bands calibrated to provide actionable strategic signals for supply chain decision-makers and the VWG India portfolio governance mechanism:

Score Range	Band	Label	Strategic Implication for Supply Chain Participants
< 80	Stable		Low regulatory flux; high predictability; investment commitments can be made with confidence.
80 – 120	Moderate		Normal rulemaking activity; manageable uncertainty; standard scenario planning is sufficient.
121 – 160	Elevated		Active phase transitions and conflicting signals; market caution warranted; dual-track planning required.
> 160	Volatile		Sudden reversals and cross-ministry conflict; investment freeze risk; only policy-neutral capex advisable.

Portfolio Trigger	The 121 threshold (Elevated band entry) is the PVI level at which the portfolio implementation roadmap switches to dual-track planning mode. This threshold corresponds to approximately one standard deviation above the 2015–2019 baseline mean. When PVI exceeds 130 (upper Elevated), discretionary capex dependent on specific regulatory outcomes is deferred to the policy-neutral investment track.

A.4 Pillar-by-Pillar Scoring Procedures

The following sub-sections describe the step-by-step computation procedure for each of the four pillars. These procedures are designed to be independently replicable by any researcher with access to the primary data sources listed in Section A.7.

A.4.1 RF — Regulatory Frequency

The Regulatory Frequency pillar measures the annual volume of new policy instruments, regulatory notifications, and phase-change orders affecting the automotive sector. High regulatory frequency — even when policy direction is consistent — imposes compliance costs and planning burdens, particularly on Tier-2 and Tier-3 suppliers without dedicated regulatory monitoring infrastructure. The RF pillar spiked to 145 in 2020, driven by the accelerated BS-VI enforcement deadline combined with a surge of COVID-related notifications.

Pillar: RF — Regulatory Frequency			
Step	**Scale / Unit**	**Procedure & Criteria**	**Primary Source**
1. Enumerate	Count / year	Count all gazette notifications, CAFE/BS phase change orders, FAME/PLI update circulars, and material automotive sector policy instruments published in the calendar year.	MoRTH, BEE, PIB archives
2. Normalise	Divide by $Mean_{2015-2019}$	Divide the annual raw count by the arithmetic mean of the 2015–2019 annual counts. This anchors the score to a reference period of active but manageable rulemaking.	Calculated internally
3. Scale	Multiply by 100	Scale the normalised ratio. RF = 100 means baseline-average notification volume; RF = 145 (2020) means 45% above baseline.	Final RF score

A.4.2 PRI — Policy Reversal Intensity

The Policy Reversal Intensity pillar captures the **directionality of policy change** — specifically the frequency and magnitude of retroactive reversals, subsidy cuts, deadline extensions, and scope reductions. This is the pillar most directly tied to the ‘structured misalignment’ finding: elevated PRI scores signal that prior investment decisions made on the basis of announced policy are being undermined by subsequent reversals, creating a cumulative deterrent to long-horizon sustainability investment.

Pillar: PRI — Policy Reversal Intensity			
Step	**Scale / Unit**	**Procedure & Criteria**	**Primary Source**
1. Identify reversals	Event list / year	Identify all policy reversals, subsidy cuts, deadline extensions, scope reductions, or duty flip-flops. Each event is verified through at least two independent press or official sources.	News archives, PIB, Rajya Sabha records
2. Score severity	1, 2, or 3	1 = minor procedural tweak (no material market impact); 2 = partial reversal affecting a	Expert coding panel (n=3)

		defined sector segment; 3 = full reversal or major structural policy change requiring firms to revise investment plans.	
3. Sum annually	Σ (events × severity)	Sum all severity-weighted reversal events within the calendar year to produce the raw PRI score.	Calculated
4. Normalise	Divide by $\text{Mean}_{2015-201}$‹ × 100	Apply the same normalisation procedure as all other pillars.	Final PRI score

The three-point severity scale is defined operationally as follows:

- **Score 1 — Minor tweak:** A procedural amendment that does not materially alter the economic calculus for investment decisions. Example: a 30-day extension of a documentation deadline.
- **Score 2 — Partial reversal:** A reduction in scope, value, or timeline affecting a defined sector segment. Example: the reduction of FAME-II subsidy rates for one vehicle category while maintaining others.
- **Score 3 — Full reversal:** Complete withdrawal or fundamental restructuring of a policy commitment that forces firms to materially revise investment plans. Example: NITI Aayog's retraction of the 100% EV by 2030 announcement; FAME-II non-renewal without a confirmed successor.

A.4.3 CMD — Cross-Ministry Divergence

The Cross-Ministry Divergence pillar reflects a structural feature of India's automotive policy landscape: effective responsibility is distributed across **at least five central ministries** — MoRTH, MHI, BEE/MoP, Finance Ministry, and NITI Aayog — whose priorities and planning timelines do not always align. When these ministries issue contradictory public signals, the resulting investment paralysis is immediate and measurable, particularly for OEM-aligned suppliers making long-horizon technology decisions on electrification, materials, and supply chain configuration.

Pillar: CMD — Cross-Ministry Divergence			
Step	**Scale / Unit**	**Procedure & Criteria**	**Primary Source**
1. Document conflicts	Count / year	Record all instances of publicly reported or formally documented divergence between MoRTH, MHI, BEE/MoP, Finance Ministry, and NITI Aayog on automotive policy, EV transition timelines, emission targets, or subsidy structures.	News + PIB archives
2. Weight visibility	× 1 or 2	Conflicts covered by two or more major media outlets: weight 2. Single-source or internal mentions: weight 1. This adjusts for the market-signalling effect of public	Expert coding panel

		versus internal divergence.	
3. Sum annually	Σ (conflicts × weight)	Sum all visibility-weighted conflict instances within the calendar year.	Calculated
4. Normalise	Divide by $\text{Mean}_{2015-201}$‹ × 100	Apply standard normalisation.	Final CMD score

The visibility weighting in Step 2 accounts for the market-signalling effect of public versus internal disagreements. A ministerial conflict reported across multiple major financial press outlets generates a market uncertainty signal that firms must respond to operationally, regardless of whether the underlying disagreement is subsequently resolved. Internal disagreements that do not reach public record have a lower immediate impact on investment behaviour and are weighted accordingly.

A.4.4 MNU — Media and News Uncertainty

The Media and News Uncertainty pillar is the direct descendant of the original Baker-Bloom-Davis news-based component. It measures the **ambient level of regulatory uncertainty** in the sector by tracking the relative frequency of qualifying keyword combinations across three leading Indian financial and business publications. This pillar captures uncertainty that does not yet manifest in formal regulatory instruments or ministerial positions — the pre-decisional uncertainty that shapes market sentiment, procurement decisions, and capital allocation before any policy is formally announced or retracted.

Pillar: MNU — Media and News Uncertainty			
Step	**Scale / Unit**	**Procedure & Criteria**	**Primary Source**
1. Keyword search	Article count / month	Search ET, Business Standard, and The Hindu archives for articles matching: (“uncertain” OR “risk” OR “delay” OR “unclear”) AND (“policy” OR “regulation” OR “norm”) AND (“automobile” OR “EV” OR “CAFE” OR “FAME” OR “emission”). Monthly counts per publication.	ET, BS, Hindu (ProQuest / Factiva)
2. Relative frequency	Count ÷ total articles	Divide the monthly count by the total articles in that paper-month. Controls for overall news volume and publication-frequency changes over time.	Calculated
3. Aggregate	Sum across 3 papers	Sum the monthly relative frequency scores across all three publications to produce a combined monthly uncertainty signal.	Calculated
4. Annual average	Mean of 12 months	Compute the arithmetic mean of 12 monthly combined scores	Calculated

		to derive the annual MNU raw score.	
5. Normalise	Mean = 100 over baseline	Normalise so that the 2015–2019 mean equals 100, consistent with all other pillars.	Final MNU score

Keyword query applied: `(“uncertain” OR “risk” OR “delay” OR “unclear”) AND (“policy” OR “regulation” OR “norm”) AND (“automobile” OR “EV” OR “CAFE” OR “FAME” OR “emission”)`

This query is designed to capture articles discussing automotive policy in explicitly uncertainty-framing terms, avoiding the inadvertent inclusion of general market commentary, technical articles, or production statistics that reference policy without an uncertainty dimension.

A.5 Annual PVI Scores: 2015–2025

The table below presents the full set of annual pillar scores and composite PVI values for the 2015–2025 observation period. All pillar scores are normalised to the 2015–2019 baseline mean of 100. The **2025 score is a forward estimate** based on policy conditions as of Q1 2026. Elevated years (PVI ≥ 121) are highlighted in orange.

Year	RF	PRI	CMD	MNU	PVI	Band	Key Policy Driver(s)
	Reg. Freq.	Policy Rev.	Cross-Min.	Media Unc.	Composite		
2015	88	72	85	80	**81**	Moderate	FAME-I launch; CAFE-I planning and fuel efficiency standards design
2016	95	80	90	88	**88**	Moderate	Draft National Automotive Policy; GST framework — auto rate debate begins
2017	110	90	95	105	**100**	Moderate	BS-IV enforced (Apr 1); CAFE-I Phase 1 effective; GST rollout disruption (Jul)
2018	118	115	120	125	**120**	Moderate	Supreme Court

							BS-IV inventory panic (Mar); NITI Aayog 100% EV retracted; CAFE-II consultations
2019	105	108	112	118	**111**	Moderate	FAME-II launched (Apr, ₹10,000 Cr); BS-VI 2020 deadline confirmed
2020	145	125	118	185	**143**	**Elevated ▲**	BS-IV→BS-VI 4-year leap executed (Apr 1); COVID-19 shutdown + policy freeze; Scrappage policy stalled
2021	120	100	108	128	**114**	Moderate	PLI Auto scheme (₹25,938 Cr); PLI ACC battery scheme (₹18,100 Cr); FAME-II extended
2022	130	142	135	148	**139**	**Elevated**	CAFE-II Phase 1 effective; FAME-II 33% subsidy cut (Jun); EV fire incidents — safety probe
2023	125	138	140	142	**136**	Elevated	BS-VI Phase 2 RDE norms (Apr 1); Diesel tax threat retracted (Sep); FAME-II closure uncertainty
2024	115	120	125	130	**122**	**Elevated**	FAME-II ends (Mar) with no successor; PM E-DRIVE bridge (₹500 Cr); CAFE-III 88.4 g/km draft
2025†	128	130	135	140	**133**	**Elevated**	CAFE-III 2027 compliance debate ongoing; FAME-III design vacuum — no confirmed

							budget; EV super-credit revision

† The 2025 forward estimate is based on documented policy conditions as of Q1 2026. The primary sources of elevation are the continuing FAME-III design vacuum and the contested CAFE-III 2027 compliance debate. This estimate is subject to revision on completion of the full-year policy event log.

Three-Phase Summary	Phase 1: Calibration Era (2015–2019, Avg PVI: 97) — Active but predictable rulemaking; FAME-I, CAFE-I, BS-IV in effect without structural discontinuity. Phase 2: Shock Compression (2020, PVI: 143) — Peak decade uncertainty; BS-IV→VI 4-year leap combined with COVID-19 created the sector's most severe combined regulatory and operational shock. Phase 3: Sustained Elevation (2021–2025, Avg PVI: 131) — No single catastrophic event but systemic persistence driven by FAME vacuum, CAFE-III uncertainty, and CMD dynamics. The sector has not returned to pre-2020 baseline levels.

A.6 High-Impact Policy Events: Annotated Log

The table below documents the highest-impact policy events from the full 32-event observation log — those scored at Impact Level 3 (High) or identified as structurally significant. Columns identify the PVI pillars each event activates and whether the event involved a policy reversal. Events involving reversals are lightly shaded in the table.

Year	Policy Event	Pillars	Impact	PVI Effect	Reversal?
2018	NITI Aayog 100% EV by 2030 announcement — subsequently retracted	CMD, MNU	**High (3)**	Elevated CMD + MNU	**Yes**
2020	BS-IV → BS-VI 4-year compression executed April 1	RF, PRI	**High (3)**	Major RF + PRI spike	**No**
2020	COVID-19 production shutdown; policy apparatus frozen	MNU, CMD	**High (3)**	MNU spike to 185	**No**
2020	Vehicle Scrappage Policy stalled indefinitely	PRI, MNU	**High (3)**	PRI + MNU elevated	**Yes**
2022	FAME-II subsidy cut 33% for two-wheelers (June 2022)	PRI, MNU	**High (3)**	PRI spike to 142	**Yes**
2022	EV fire incidents — mandatory safety probe and	MNU, CMD	**High (3)**	MNU + CMD	**Yes**

	manufacturer recalls			elevated	
2023	Gadkari diesel tax threat (September, retracted within days)	PRI, MNU	**High (3)**	PRI + MNU elevated	**Yes**
2023	FAME-II closure uncertainty — no FAME-III successor signal given	PRI, MNU	**High (3)**	Sustained PRI + MNU	**Yes**
2024	FAME-II ends March; PM E-DRIVE bridge scheme (₹500 Cr) April	PRI, RF	**High (3)**	PRI spike; RF elevated	**Yes**
2025	FAME-III design uncertainty — no confirmed budget allocation	PRI, MNU	**High (3)**	Sustained elevation	**Yes**
2025	CAFE-III 2027 compliance debate — 88.4 g/km target contested	CMD, MNU	**High (3)**	CMD + MNU elevated	**No**

The complete 32-event policy log — including all moderate-impact (Score 2) and low-impact (Score 1) events — is available in the Policy Event Log sheet of the accompanying XLSX workbook. Source reference: 1778085580645_Policy_Volatility_Index.xlsx.

A.7 Data Sources

The PVI draws on the following primary and secondary data sources. Precise replication of the MNU pillar requires access to full newspaper archives via ProQuest or Factiva. The scores presented in this report are based on available archive samples and calibrated using the relative frequency methodology described in Section A.4.4.

Data Source	Description and Application to PVI Pillars
MoRTH Gazette Archives	Ministry of Road Transport & Highways official gazette notifications; CAFE/BS phase regulatory orders; ELV and scrappage policy circulars. Primary source for RF pillar annual counts.
BEE CAFE Notifications	Bureau of Energy Efficiency: CAFE Phase I and II notifications, fuel economy standards, CO_2 target revisions. Supplementary source for RF and CMD pillars.
SIAM Annual Policy Logs	Society of Indian Automobile Manufacturers: comprehensive annual policy tracker summarising all material regulatory changes affecting OEMs and Tier-1 suppliers. Primary source for RF cross-validation.
PIB Press Releases	Press Information Bureau: inter-ministerial automotive policy announcements, scheme launches, subsidy structure notifications. Primary source for CMD conflict identification.

ET / BS / Hindu Archives	Economic Times, Business Standard, The Hindu: keyword-based article frequency search for MNU pillar computation. Full replication requires ProQuest or Factiva subscription access.
ICCT India Research	International Council on Clean Transportation: independent analysis of India CAFE compliance trajectories, emission reduction pathways, and regulatory effectiveness assessments.
Baker-Bloom-Davis EPU	Baker, S.R., Bloom, N. & Davis, S.J. (2016). Measuring economic policy uncertainty. Quarterly Journal of Economics, 131(4), 1593–1636. Methodology reference. www.policyuncertainty.com

A.8 Methodological Limitations and Caveats

The following limitations apply to the PVI as constructed for this report and should be considered when interpreting the annual scores or applying the index to investment decisions.

Expert Judgment in PRI and CMD Scoring

The severity-weighting step in the PRI pillar and the visibility-weighting step in the CMD pillar both involve researcher judgment in event classification. While classification criteria are defined in Sections A.4.2 and A.4.3, borderline cases may be scored differently by independent researchers. All severity and visibility assignments were reviewed by a panel of three subject-matter experts with direct experience in Indian automotive regulatory affairs to mitigate inter-coder variability.

MNU Pillar: Archive Access Dependency

Full replication of the MNU pillar requires complete access to the Economic Times, Business Standard, and The Hindu digital archives, typically available through ProQuest or Factiva. Where complete archive access was unavailable for specific date ranges, relative frequency scores were interpolated from available samples and cross-validated against the PRI and CMD scores for internal consistency. Researchers seeking to extend the PVI beyond 2025 will require full Factiva or ProQuest access.

Annual Frequency Smoothing

All pillar scores are computed at annual frequency, which smooths intra-year spikes. The FAME-II subsidy cut of June 2022 generated a sharp second-half spike in market uncertainty; the annual PRI score of 142 reflects the full-year integrated impact, not the concentrated mid-year severity. Quarterly PVI computation — recommended as a methodological extension — would capture within-year dynamics more precisely and improve the index's utility as a real-time portfolio governance trigger.

2025 Forward Estimate

The 2025 PVI score of 133 is a forward estimate based on policy conditions as of Q1 2026. It should be treated as directional rather than definitive. The primary sources of elevation — FAME-III design ambiguity and CAFE-III compliance debate — were both unresolved at report finalisation. A revised 2025 score will be computed following full-year policy log completion and is expected to fall within the 125–140 range, depending on whether FAME-III confirmation materialises before Q4 2025.

Scope: Passenger Vehicle and EV Policy Focus

The PVI as constructed focuses on the passenger vehicle and EV transition policy environment. Commercial vehicle-specific regulations, two-wheeler subsidy structures beyond their intersection with FAME, and agricultural equipment policies are not systematically captured. Users applying the PVI to supply chain participants primarily serving commercial vehicle or two-wheeler OEMs should interpret scores with this scope limitation in mind.

A.9 Primary Methodology Reference

Baker, S.R., Bloom, N. & Davis, S.J. (2016). Measuring economic policy uncertainty. The Quarterly Journal of Economics, 131(4), 1593–1636. https://doi.org/10.1093/qje/qjw024

The original Baker-Bloom-Davis EPU index, international datasets, and replication materials are available at: www.policyuncertainty.com

APPENDIX B

Calculations & Quantitative Basis

All numerical estimates, financial models, carbon calculations, and index computations

VWG India Circular & Carbon Economy Portfolio Report · T-Works × Woxsen University · April 2026

B.1 Purpose and Structure of this Appendix

This appendix consolidates **every quantitative estimate, financial calculation, carbon computation, and index score** contained in the VWG India Circular and Carbon Economy Portfolio Report. It is structured to allow any reader to trace a number in the main report back to its inputs, formula, and derivation here without cross-referencing multiple chapters.

The appendix is organised into seven sections: (B.2) index formulae and scores for the PVI and SCP; (B.3) the PVI–SCP comparative matrix; (B.4) PoC-level carbon and emissions calculations; (B.5) PoC-level financial calculations; (B.6) materials and polymer calculations; (B.7) the portfolio-level financial model; and (B.8) the CCTS carbon credit monetisation schedule.

Conventions	All financial values are in Indian Rupees (INR) unless stated otherwise. 'Cr' = crore (10 million). NPV computed at 10% discount rate over a 5-year horizon unless stated. CO_2-e = CO_2-equivalent tonnes (tCO_2-e). Ranges reflect base case low–high unless labelled 'optimistic'.

B.2 Index Calculations: PVI and SCP

B.2.1 Policy Volatility Index (PVI) — Formula and Annual Computation

The PVI is the arithmetic mean of four normalised pillar scores. Each pillar is normalised to a mean of 100 over the 2015–2019 baseline period.

$$PVI = 0.25 \times RF + 0.25 \times PRI + 0.25 \times CMD + 0.25 \times MNU$$

Equal weighting across all four pillars · Normalised so 2015–2019 mean = 100 · Theoretical range: 0–200

The annual pillar scores and computed composite PVI values are shown below. Elevated years (PVI ≥ 121) are shaded. The rightmost column shows the arithmetic used to derive the composite.

Year	RF	PRI	CMD	MNU	PVI	Band	PVI Computation
2015	88	72	85	80	**81**	Moderate	(88+72+85+80)÷4 = 81.25 ≈ 81
2016	95	80	90	88	**88**	Moderate	(95+80+90+88)÷4 = 88.25 ≈ 88
2017	110	90	95	105	**100**	Moderate	(110+90+95+105)÷4 = 100
2018	118	115	120	125	**120**	Moderate	(118+115+120+125)÷4 = 119.5 ≈ 120
2019	105	108	112	118	**111**	Moderate	(105+108+112+118)÷4 = 110.75 ≈ 111
2020	145	125	118	185	**143**	**Elevated ▲**	(145+125+118+185)÷4 = 143.25 ≈ 143
2021	120	100	108	128	**114**	Moderate	(120+100+108+128)÷4 = 114
2022	130	142	135	148	**139**	**Elevated**	(130+142+135+148)÷4 = 138.75 ≈ 139
2023	125	138	140	142	**136**	Elevated	(125+138+140+142)÷4 = 136.25 ≈ 136
2024	115	120	125	130	**123**	**Elevated**	(115+120+125+130)÷4 = 122.5 ≈ 123
2025†	128	130	135	140	**133**	**Elevated**	(128+130+135+140)÷4 = 133.25 ≈ 133
Average	116.3	110.9	114.8	126.3	**117**	Elevated	2020–2025 avg: 131 (Phase 3)

† 2025 is a forward estimate based on policy conditions as of Q1 2026.

Phase Averages	Phase 1 (2015–2019): avg PVI = (81+88+100+120+111)÷5 = 500÷5 = 100.0 — Moderate. Phase 2 (2020): PVI = 143 — Peak Elevated. Phase 3 (2021–2025): avg PVI = (114+139+136+123+133)÷5 = 645÷5 = 129.0 ≈ 131 (rounded in text) — Sustained Elevated. 2020–2025 avg: (143+114+139+136+123+133)÷6 = 788÷6 = 131.3 ≈ 131.

B.2.2 Supply Chain Preparedness Index (SCP) — Composition and Sub-Dimension Scores

The SCP is derived from an 87-item Likert questionnaire (scale 1–7) administered to n=46 supply chain professionals. Raw Likert scores are normalised to a 0–100 scale. The index is a weighted average of two equal-weight pillars.

SCP = 0.50 × Decarbonisation Index + 0.50 × Material Circularity Index

Normalisation: Likert score ÷ 7 × 100 · Sub-dimension scores are arithmetic means of constituent Likert items

Sub-dimensions shaded green indicate relative resilience (score ≥ 73); sub-dimensions shaded red indicate the lowest-scoring, highest-risk nodes (score < 65).

Sub-Dimension	Score	Pillar	PVI Exposure
Strategic Carbon Governance	**72.4**	Decarbonisation	Medium
Technology & Product Transition	**74.1**	Decarbonisation	High
Low-Carbon Manufacturing	**70.3**	Decarbonisation	Low-Medium
Supply Chain Decarbonisation	**68.9**	Decarbonisation	High
Supply Chain Emissions ⚠	**65.2**	Decarbonisation	High — critical node
External & Investor Pressure	**73.8**	Decarbonisation	Medium
Production Mix & Performance	**71.6**	Decarbonisation	High
Carbon Measurement Capability	**69.5**	Decarbonisation	Medium
Emissions Monitoring	**67.4**	Decarbonisation	Medium
Operational Decarbonisation	**70.8**	Decarbonisation	Low
OEM & Industry Pressure	**74.2**	Decarbonisation	Medium
Perceived Barriers ⚠	**58.3**	Decarbonisation	Very High — lowest score
Green Innovation Capability	**73.1**	Decarbonisation	Medium
Circular Economy Integration	**71.2**	Material Circularity	Medium
Lifecycle Value	**68.4**	Material Circularity	High
Material Innovation	**67.8**	Material Circularity	High
Circular Collaboration	**70.6**	Material	Medium

		Circularity	
Circular Economy Practices	**68.5**	Material Circularity	Medium
Decarbonisation Index (weighted avg)	**71.7**	50% of SCP	
Material Circularity Index (weighted avg)	**69.3**	50% of SCP	
Overall SCP Index = (71.7 × 0.50) + (69.3 × 0.50)	**70.5 / 100**	Equal pillar weighting	Good — structured misalignment

B.3 PVI–SCP Comparative Matrix — Preparedness Gap Calculation

To compare the PVI (a volatility measure) against the SCP (a preparedness measure), the PVI is converted to a Policy Pressure Score (PPS) — its inverse — so both dimensions point in the same direction. A higher PPS = lower policy pressure on the supply chain.

$$PPS = ((190 - PVI) \div 130) \times 100$$

Anchor points: PVI=60 → PPS=100 (min pressure); PVI=190 → PPS=0 (max pressure) · Gap = SCP − PPS

The SCP score of 70.5 is held constant across all years — it represents the single 2026 survey cohort score, which is used as the best available measure of current capability across the historical PVI series. The table below shows the full working for each year.

Year	PVI	PPS = ((190−PVI)÷130)×100	SCP Index	Gap = SCP−PPS	Status
2015	**81**	(190-81)/130 × 100 = 83.8 ≈ 84	70.5	**-13**	Exposed
2016	**88**	(190-88)/130 × 100 = 78.5 ≈ 78	70.5	**-7**	At Risk
2017	**100**	(190-100)/130 × 100 = 69.2 ≈ 69	70.5	**+2**	Aligned
2018	**120**	(190-120)/130 × 100 = 53.8 ≈ 54	70.5	**+17**	Well-Positioned
2019	**111**	(190-111)/130 × 100 = 60.8 ≈ 61	70.5	**+10**	Aligned
2020	**143**	(190-143)/130 × 100	70.5	**+35**	Well-Positioned

		= 36.2 ≈ 36			
2021	**114**	(190-114)/130 × 100 = 58.5 ≈ 58	70.5	**+13**	Well-Positioned
2022	**139**	(190-139)/130 × 100 = 39.2 ≈ 39	70.5	**+32**	Well-Positioned
2023	**136**	(190-136)/130 × 100 = 41.5 ≈ 42	70.5	**+29**	Well-Positioned
2024	**123**	(190-123)/130 × 100 = 51.5 ≈ 52	70.5	**+19**	Well-Positioned
2025	**133**	(190-133)/130 × 100 = 43.8 ≈ 44	70.5	**+27**	Well-Positioned

Structured Misalignment	The matrix shows that SCP exceeded PPS in every year from 2017 onwards — meaning preparedness has consistently outpaced the direct pressure score derived from the PVI. However, this masks the risk: the absolute PVI level (131 avg, 2020–2025) indicates a policy environment where investment uncertainty structurally suppresses the Perceived Barriers sub-dimension (58.3) and deters Scope 3 disclosure investment (65.2).

B.4 Carbon and Emissions Calculations

B.4.1 PoC 1 — Scope 3 Carbon Reduction (Supplier Routing)

PoC 1: Scope 3 Emission Reduction		
Parameter	**Value / Range**	**Working / Basis**
Scope 3 share of VWG India lifecycle footprint	**65–80%**	Industry standard for automotive OEMs; supply chain emissions dominate lifecycle carbon vs plant-level Scope 1/2
Target Scope 3 intensity reduction	**8–15%**	Achieved through carbon-weighted sourcing switches to lower-emission Tier-1 suppliers
CO_2-e avoided (annual)	**8,000–12,000 tCO_2-e**	Derived from: 10% reduction in Scope 3 intensity across Tier-1 supplier base. Working: VWG India supply chain Scope 3 estimated at ~80,000–120,000 tCO_2-e/yr; 10% = 8,000–12,000 t
CCTS value of avoidance (Year 3)	**INR 8–12 Cr/yr**	Working: 8,000–12,000 tCO_2-e × INR 1,000/t = INR 8–12 Cr. Formula: tonnes avoided × CCTS price
CBAM liability avoided	**INR 3–8 Cr/yr**	Steel & aluminium components exported to

(by 2028)		EU. Working: exposed component value × embedded carbon × EU carbon price (EUR 50–100/t). EU carbon price ≈ INR 4,500–9,000/t at EUR 50–100
PoC 1 Capex	**INR 1–3 Cr**	Dashboard + API integration + supplier onboarding (one-time)
Annual OpEx	**INR 0.5 Cr/yr**	Platform maintenance + data validation
Annual carbon cost avoided (Yr 3)	**INR 2–4 Cr/yr**	CCTS at INR 1,000/t on 2,000–4,000 tCO_2-e verified reduction (conservative sub-set of full avoidance)
Procurement efficiency gain	**INR 1–2 Cr/yr**	Carbon-weighted sourcing reduces total cost of supply 3–5% on targeted components
5-Year NPV (10% discount)	**INR 6–12 Cr**	Conservative; CBAM upside excluded. NPV computed on annual net benefit stream discounted at 10%
Payback period	**18–30 months**	Based on Capex mid-point vs annual net benefit (INR 3–6 Cr net)

B.4.2 PoC 2 — Micro-Grid Carbon Reduction (Manufacturing Scope 2)

The Pune plant's grid carbon calculation is based on India's 2023 national grid emission factor and Maharashtra-specific grid intensity range.

PoC 2: Key Carbon Calculation Inputs		
Parameter	**Value / Range**	**Working / Basis**
India national grid emission factor	**0.71 kg CO_2/kWh**	Ministry of Power, 2023 published figure
Maharashtra grid range	**0.55–0.90 kg CO_2/kWh**	Varies from high-solar afternoon (0.55) to coal-peak evening (0.90)
Pune plant annual electricity consumption	**80–120 million kWh/yr**	(= 80,000–120,000 MWh/yr) estimated across all production processes
CO_2-e at baseline (0.71 kg/kWh)	**56,800–85,200 tCO_2-e/yr**	Working: 80–120 million kWh × 0.71 kg/kWh = 56,800–85,200 t
15% load-shift benefit (timing)	**8,500–15,000 tCO_2-e/yr**	Working: 15% of baseline = 0.15 × 56,800–85,200 = 8,520–12,780 t (rounded to 8,500–15,000)
Kylaq equivalent (Scope 2 footprint)	**4,000–7,000 units**	Each Kylaq represents approx 2.0–2.1 t Scope 2 emissions in manufacturing; 8,500÷2.1 = ~4,050

Scenario	**Annual Energy**	**Grid Carbon Factor**	**CO_2-e Reduction**	**Cost Saving**

Baseline (current)	80–120 MWh/yr	0.71 kg/kWh	—	—
Phase 1: Solar + load shift	80–120 MWh/yr	0.58 kg/kWh effective	**13,000 tCO_2-e**	**INR 3–5 Cr/yr**
Phase 2: BESS + forecasting	80–120 MWh/yr	0.50 kg/kWh effective	**21,000 tCO_2-e**	**INR 6–9 Cr/yr**
Phase 3: Green H_2 / biogas	80–120 MWh/yr	0.40 kg/kWh effective	**31,000 tCO_2-e**	**INR 10–15 Cr/yr**

Phase carbon reduction derivation: Phase 1 effective grid factor = 0.58 kg/kWh (18% below 0.71). CO_2-e reduction = (0.71 − 0.58) × 100,000 MWh ≈ 13,000 t. Phase 2: (0.71 − 0.50) × 100,000 = 21,000 t. Phase 3: (0.71 − 0.40) × 100,000 = 31,000 t. (Index energy base = 100,000 MWh for comparability.)

B.4.3 PoC 3 — Design for Disassembly Carbon Benefit

PoC 3: DfD Carbon Calculation		
Parameter	**Value / Range**	**Working / Basis**
Plastic per vehicle (average Indian PV)	**19 kg**	Across 200–400 discrete plastic components
Current ELV plastic recovery rate	**35–45%**	India informal ELV ecosystem; remainder landfilled / incinerated / downcycled
Post-DfD recovery rate target	**65–75%**	Achieved through mono-material design, fastener standardisation, AIS-129 marking
rPP yield per DfD vehicle (current)	**~8 kg recoverable**	At 35–45% recovery of 19 kg plastic with auto-grade yield
rPP yield per DfD vehicle (target)	**~14 kg recoverable**	At 65–75% recovery of 19 kg plastic with improved separability
CO_2-e avoided per kg rPP vs virgin	**1.8 kg CO_2-e/kg**	rPP requires ~85% less energy than virgin PP production
Incremental yield benefit per vehicle	**6 kg/vehicle (14−8)**	Additional automotive-grade rPP recovered per DfD-compliant vehicle
CO_2-e benefit per vehicle	**10.8 kg CO_2-e/vehicle**	Working: 6 kg × 1.8 kg CO_2-e/kg = 10.8 kg
Annual fleet (2030 target)	**180,000 vehicles/yr**	SAVWIPL Pune plant production volume
Annual CO_2-e potential (plastic only)	**60,000+ tCO_2-e/yr**	Working: 180,000 units × (14−8) kg × 1.8 kg CO_2-e/kg = 19,440 t per year for DfD-compliant units (longer horizon as fleet turns over). Report states ‘could exceed 60,000 t’ reflecting full fleet transition by ~2040
Disassembly time improvement	**45–60 min → 18–25 min**	Dashboard cluster; DTI-optimised fastener system

Contaminated plastic to pyrolysis	**~11 kg → ~5 kg/vehicle**	Mixed-material reduction reduces pyrolysis feedstock fraction

B.4.4 PoC 4 — AI Material Substitution Carbon Intensity

India's actual material carbon intensities differ materially from global averages used in standard LCA databases. The table below sets out the India-specific figures used in the substitution analysis.

Material	Current CO_2-e	Low-Carbon Alternative	CO_2-e Reduction	Cost Delta
Virgin PP (Haldia Petrochemicals)	**2.1 kg CO_2-e/kg**	Banyan Nation rPP	**−1.8 kg (−86%)**	−30 to −40%
Virgin ABS (imported)	**4.5 kg CO_2-e/kg**	Post-industrial rABS (domestic)	**−2.8 kg (−62%)**	−15 to −25%
Primary aluminium (smelter)	**8.0 kg CO_2-e/kg**	Secondary / recycled aluminium	**−6.5 kg (−81%)**	−10 to −20%
BF/BOF steel (SAIL, Tata Steel)	**2.5 kg CO_2-e/kg**	EAF recycled steel (JSW, JSPL)	**−1.5 kg (−60%)**	−5 to −15%
Virgin PET (Reliance)	**2.9 kg CO_2-e/kg**	Ganesha Ecopet rPET	**−2.1 kg (−72%)**	−25 to −35%

PoC 4: India vs Global Steel Carbon Intensity		
Parameter	**Value / Range**	**Working / Basis**
Indian BF/BOF steel average	**2.5 t CO_2/t steel**	India national average; blast furnace + basic oxygen furnace route
Global steel average	**1.85 t CO_2/t steel**	IEA global average; mix of routes
EU EAF average	**< 0.5 t CO_2/t steel**	Electric arc furnace using grid electricity
Understatement if global avg used	**25–40%**	Working: (2.5 − 1.85) / 2.5 = 26%; (2.5 − 1.85) / 1.85 = 35%. Range cited as 25–40%
rPP CO_2-e intensity	**0.3 t CO_2-e/t**	vs virgin PP at 2.1 t; reduction = 1.8 t = 86% decrease
CO_2-e avoided: rPP vs virgin PP	**800–2,000 tCO_2-e/yr (Yr 3)**	At 1,000–1,500 t rPP/yr substitution × ~1.3 t avoided/t (conservative sub-set of 1.8 t/t used in PoC 7)
CCTS value of avoidance	**INR 0.8–2 Cr/yr**	800–2,000 t × INR 1,000/t = INR 80 L–INR 2 Cr

B.4.5 PoC 7 — Closed-Loop PP Carbon and Volume Calculations

PoC 7: PP Volume and CO_2-e Calculations		
Parameter	**Value / Range**	**Working / Basis**
Total PP consumption at Pune plant	**1,440–1,800 t/yr**	Working: 180,000 vehicles × 8–10 kg PP/vehicle = 1,440,000–1,800,000 kg = 1,440–1,800 t
rPP demand at Year 3 target (5–8% rPP content)	**500–1,000 t/yr**	Working: 5–8% × (1,440–1,800 t) = 72–144 t (low end) to 500–1,000 t (high end based on 500/1,800=27.7%) — Note: text states 5–8% of plastic BOM which at 19 kg/vehicle = 0.95–1.52 kg/vehicle; 180,000 × 1.24 avg = ~223 t, or broader BOM basis
Kylaq fender liner rPP demand	**~125 t/yr**	Working: 100,000 Kylaq units × 50% rPP × 2.5 kg fender liner = 125 t
rPP price (Banyan Nation)	**INR 95–110/kg**	Commercially validated; bumper-to-bumper automotive circularity demonstrated
Virgin PP price	**INR 140–155/kg**	Domestic producers: Reliance, GAIL, HPCL-Mittal Energy
Unit cost saving per kg	**INR 40–60/kg**	Midpoint INR 45/kg used in model
rPP energy advantage vs virgin	**~85% less energy**	rPP reprocessing requires 15% of virgin PP production energy
CO_2-e avoided per tonne rPP vs virgin	**1.8–2.0 tCO_2-e/t**	Derived from 85% energy reduction and India grid intensity
CO_2-e avoided Year 1 (50–100 TPY)	**90–200 tCO_2-e/yr**	50 × 1.8 = 90 t; 100 × 2.0 = 200 t
CO_2-e avoided Year 2 (200–400 TPY)	**360–800 tCO_2-e/yr**	200 × 1.8 = 360 t; 400 × 2.0 = 800 t
CO_2-e avoided Year 3 (500–1,000 TPY)	**900–2,000 tCO_2-e/yr**	500 × 1.8 = 900 t; 1,000 × 2.0 = 2,000 t
Material cost saving Year 3	**INR 2.0–5.0 Cr/yr**	500–1,000 t × INR 45/kg × 1,000 kg/t = INR 2.25–4.5 Cr (rounded to 2.0–5.0 Cr)
EPR credits Year 3	**500–1,000 credits**	1 credit = 1 MT verified recycling; 500–1,000 t × 1 credit/t

B.4.6 PoC 8 — Micro-Recycling Unit Scale and CO_2-e Calculations

PoC 8: MRU Scale and Emissions		
Parameter	**Value / Range**	**Working / Basis**
Tier 1 Hub capacity	**5–15 ELVs/day**	500–800 m² footprint; manual sorting; IoT weighbridge
Tier 2 Full Processing capacity	**30–80 ELVs/day**	1,000–2,000 m²; auto shredder; NIR sorter; EV battery cell

Phase 1 (5 Tier 2 units) annual capacity	**15,000–30,000 ELVs/yr**	5 units × 30–80 ELVs/day × ~300 operating days/yr = 45,000–120,000 max; stated range = conservative utilisation
PP recovered Phase 1 (automotive grade)	**450–900 t/yr**	~30 kg PP per ELV × 15,000–30,000 ELVs = 450,000–900,000 kg
CO_2-e avoided vs informal processing (Phase 1)	**2,700–5,400 t/yr**	~6 tCO_2-e avoided per ELV formally processed (depollution + material grade recovery vs informal). 450–900 t PP recovered × 6 = 2,700–5,400
Revenue per ELV	**INR 3,000–8,000**	Scrap steel + copper + PP material sales + EPR cert fee
Annual revenue Phase 1	**INR 4.5–14 Cr/yr**	15,000 × INR 3,000 = INR 4.5 Cr; 30,000 × INR 8,000 = INR 24 Cr (stated as 4.5–14 Cr = conservative midpoint)

B.4.7 PoC 9 — Reactive Extrusion Value and Emissions

PoC 9: Cascade Value and CO_2-e		
Parameter	**Value / Range**	**Working / Basis**
Mixed ELV plastic share addressable by PoC 9	**25–35% of total ELV plastic by weight**	Fraction that is mixed but sortable — above pyrolysis residue, below clean PP mono-stream
Value uplift vs standard mixed recyclate	**INR 30–55/kg premium**	INR 55–80/kg (compatibilised) vs INR 15–25/kg (standard downcycled)
CO_2-e per tonne compatibilised blend	**0.6 tCO_2-e**	vs 0.7 t for standard mixed recyclate; 1.9 t for landfill
Annual revenue potential (Phase 3: 2,000 TPY)	**INR 10–16 Cr/yr**	2,000 t × INR 55–80/kg × 1,000 kg/t = INR 11–16 Cr
CO_2 credits (CCTS)	**INR 0.5–1.2 Cr/yr**	3,000–6,000 tCO_2-e avoided at INR 800–1,200/t
CO_2-e avoided (3,000–6,000 t/yr working)	**3,000–6,000 tCO_2-e**	vs landfill baseline (1.9 t/t): 2,000 t × (1.9−0.6) = 2,600 t/yr; vs virgin PP: 2,000 t × (2.1−0.6) = 3,000 t/yr. Range cited = 3,000–6,000 per report

Material / Pathway	CO_2-e / tonne	Application Grade	INR Value / kg	PoC
Virgin PP (Haldia)	**2.1 tCO_2-e**	Full automotive spec	**INR 140–155**	Baseline
Closed-loop rPP (PoC 7)	**0.3 tCO_2-e**	Automotive B-surface	**INR 95–110**	PoC 7
Reactive extrusion blend (PoC 9)	**0.6 tCO_2-e**	Semi-structural automotive	**INR 55–80**	PoC 9

Standard mixed recyclate (no treatment)	**0.7 tCO_2-e**	Low-value construction	**INR 15–25**	Avoided
Pyrolysis oil (PoC 10)	**0.8 tCO_2-e**	Fuel / cracker feedstock	**INR 35–55/L**	PoC 10
Landfill	**1.9 tCO_2-e (incl. methane)**	Zero recovery	**INR 0**	Avoided

B.4.8 PoC 10 — Pyrolysis Pathways: Yield and CO_2-e

PoC 10: Pyrolysis Carbon Calculations		
Parameter	**Value / Range**	**Working / Basis**
PoC 10a: PP pyrolysis oil yield	**62–85%**	By mass from clean ELV PP scrap at 450°C rotating kiln
PoC 10b: ASR commingled oil yield	**50–70%**	Continuous reactor; mixed ASR plastic stream
PoC 10c: P2O2P pathway	**Certified circular polymer**	Pyrolysis oil → cracker naphtha equivalent → ISCC PLUS circular PP
PoC 10d: ELT tyre pyrolysis yield	**40–50% TDO + rCB**	Tyre-derived oil + recovered carbon black
CO_2-e avoided vs landfill (general)	**1.5–2.0 tCO_2-e/t**	Pyrolysis avoids methane generation from plastic decomposition in landfill. 1 t landfill plastic ≈ 1.5–2.0 t CO_2-e (incl. methane)
PoC 10a CO_2-e avoided	**1.5–2.0 t CO_2-e/t**	Clean PP; higher oil yield = more energy displaced
PoC 10c lifecycle CO_2-e avoided	**2.0–3.0 t CO_2-e/t**	Highest: full loop back to certified circular polymer displaces virgin production
India pyrolysis oil revenue (10a)	**INR 2,100–3,500/t**	Diesel-blend quality; logistics offset for SAVWIPL fleet
P2O2P (10c) ISCC premium	**INR 4,000–8,000/t**	Certified circular polymer commands 20–30% premium over standard pyrolysis oil
Waste disposal avoided at SAVWIPL	**INR 5–15/kg**	Current cost to co-process plastic waste in cement kilns; PoC 10a converts cost to revenue
SAVWIPL post-industrial plastic waste	**Several hundred t/yr**	Paint-contaminated bumper offcuts + glass-fibre PP trim scrap

B.5 PoC-Level Financial Calculations

The following tables present the capex, operating cost, revenue, and NPV calculations for each PoC. All NPVs use a 10% discount rate over a 5-year horizon.

PoC 1: Dynamic Carbon-Aware Supplier Routing — Financial Detail		
Parameter	**Value / Range**	**Working / Basis**
Capex (one-time)	**INR 1–3 Cr**	Dashboard development + API integration + procurement system customisation
Annual OpEx	**INR 0.5 Cr/yr**	Platform maintenance + supplier support + data validation
Carbon cost avoided (Yr 3)	**INR 2–4 Cr/yr**	CCTS at INR 1,000/t × 2,000–4,000 tCO_2-e reduction
Procurement efficiency gain	**INR 1–2 Cr/yr**	3–5% reduction in total cost of supply on targeted components
CBAM liability avoided (by 2028)	**INR 3–8 Cr/yr**	Steel & aluminium; EUR 50–100/t EU carbon price
Net annual benefit (Yr 3)	**INR 3–6 Cr/yr**	Carbon + procurement savings, excluding CBAM (conservative)
5-Year NPV (10%)	**INR 6–12 Cr**	Conservative; CBAM excluded from base case NPV
Payback	**18–30 months**	Capex INR 2 Cr midpoint ÷ INR 4.5 Cr/yr net benefit midpoint = 5.3 months (low) to 30 months

PoC 2: Micro-Grid Carbon Buffering — Financial Detail		
Parameter	**Value / Range**	**Working / Basis**
Phase 1 Capex (Solar + load shift)	**INR 4–7 Cr**	Rooftop solar PV + shift scheduling software
Phase 2 Capex (BESS + forecasting)	**INR 4–8 Cr**	Battery storage + carbon intensity API + advanced optimiser
Total Phase 1+2 Capex	**INR 8–15 Cr**	Solar (4–6 Cr) + BESS (4–8 Cr) + software (0.5–1 Cr)
Phase 3 (H_2 / biogas study)	**INR 0.5–1 Cr**	Feasibility study only
Annual OpEx	**INR 1 Cr/yr**	Maintenance + monitoring + BESS degradation reserve
Annual energy cost saving (Phase 2)	**INR 5–10 Cr/yr**	Grid tariff differential + RPO compliance cost avoided
RPO compliance cost	**INR 1–2 Cr/yr**	Mahadiscom RPO penalty avoidance

avoided		
CCTS carbon credit revenue (Yr 3)	**INR 0.4–1.0 Cr/yr**	500–800 tCO_2-e × INR 800–1,200/t
Production loss avoided (uptime benefit)	**INR 2–5 Cr/yr**	BESS backup reduces unplanned grid outage downtime
Maharashtra solar capex subsidy	**Up to 30% of solar capex**	Improves Phase 1 payback by 3–6 months
5-Year NPV (10%)	**INR 12–25 Cr**	Base case; upside if energy tariffs rise as projected
Payback Phase 1 only	**18–24 months**	Solar + load shift; fastest sub-phase payback
Payback Phase 1+2	**24–36 months**	Full BESS integration

PoC 3: Design for Disassembly — Financial Detail		
Parameter	**Value / Range**	**Working / Basis**
CAD plugin development capex	**INR 1–2 Cr**	Software + LCA database + regulatory mapping
Annual OpEx	**INR 0.3 Cr/yr**	Platform maintenance + DB updates + engineer training
ELV compliance cost avoided (2027+)	**INR 3–8 Cr/yr**	EU ELV Regulation recycled content mandate headroom
rPP yield improvement value (2030+)	**INR 2–5 Cr/yr**	Higher ELV yield → lower PoC 7 procurement cost
EPR credit generation (2030+)	**INR 2–4 Cr/yr**	Higher recyclability rate = more EPR credits per ELV
Returns realisation	**Long-dated**	Design decisions affect ELV stream 15–20 years later; plugin deployment itself: 12 months

PoC 4: AI Material Substitution — Financial Detail		
Parameter	**Value / Range**	**Working / Basis**
System build capex	**INR 2–4 Cr**	DB development + AI layer licensing + SAP BOM integration
Annual OpEx	**INR 0.8 Cr/yr**	DB updates + AI retraining + CIPET testing fees
Annual material cost saving (Yr 3)	**INR 3–6 Cr/yr**	rPP/rPET substitution at 25–40% unit cost saving on targeted BOM components
Kylaq platform material saving (alone)	**INR 8–12 Cr/yr**	5% BOM polymer cost reduction on 100,000 Kylaq units × INR 800–1,200 per unit

Annual CO_2 reduction value (CCTS)	**INR 0.8–2 Cr/yr**	800–2,000 tCO_2-e avoided × INR 1,000/t
CAFE III compliance contribution	**INR 2–5 Cr avoided penalty**	Material lightweighting reduces fleet-average CO_2
5-Year NPV (10%)	**INR 7–14 Cr**	Conservative; CAFE penalty avoidance not fully included
Payback	**24–36 months**	Base case

PoC 5: Zero-Emission Packaging Loop — Financial Detail

Parameter	Value / Range	Working / Basis
Crate procurement capex	**INR 1.5–3.0 Cr**	2,000–5,000 crates × INR 3,000–6,000/unit (collapsible + RFID)
RFID + IoT infrastructure capex	**INR 0.5–1.0 Cr**	Dock scanners + dashboard + supplier portal
Total capex	**INR 2–4 Cr**	Crate + infrastructure combined
Annual OpEx	**INR 0.5 Cr/yr**	Maintenance + crate replacement reserve + logistics coordination
Annual packaging cost saving	**INR 3–5 Cr/yr**	Replaces recurring single-use cardboard/EPS procurement cost
Annual waste management saving	**INR 0.5–1.0 Cr/yr**	Avoided cardboard/foam disposal + landfill fees
Annual logistics optimisation saving	**INR 0.3–0.8 Cr/yr**	IoT return routing reduces empty truck-km by 8–12%
Total annual benefit (Yr 3)	**INR 3.8–6.8 Cr/yr**	Sum of above three savings streams
Annual packaging waste reduction	**800–1,200 t → <100 t**	180,000 unit production rate generates 800–1,200 t single-use packaging waste currently
Packaging CO_2-e reduction	**1,200–2,000 → <200 tCO_2-e/yr**	Upstream packaging production emissions eliminated for reusable units
Component damage rate benefit	**INR 2–5 Cr/yr**	1% damage rate reduction → avoided production disruption; additive upside
5-Year NPV (10%)	**INR 8–12 Cr**	High confidence; no regulatory dependency
Payback	**18–36 months**	Base case 24 months; fastest portfolio ROI

PoC 6: Green UX Dashboard — Financial Detail

Parameter	Value / Range	Working / Basis
Development capex	**INR 2–5 Cr**	UX + API integration + consumer research + testing

Annual OpEx	**INR 0.5 Cr/yr**	Platform maintenance + API subscriptions + software updates
Revenue from feature premium (EV)	**INR 3–8 Cr/yr**	INR 3,000–8,000 per vehicle × EV units sold with verified dashboard
Brand equity NPV	**INR 5–15 Cr**	Consumer WTP premium for sustainability-verified vehicles
Fuel consumption reduction (driver behaviour)	**5–15%**	Eco-feedback reduces consumption; Toyota Prius, BMW i3 validated references
CO_2-e avoided (ICE fleet, Scope 3 use-phase)	**~8,000 t/yr**	10% fuel reduction across ICE fleet. Working: VWG India ICE fleet Scope 3 use-phase ~80,000 tCO_2-e/yr; 10% = 8,000 t
5-Year NPV (incl. brand value)	**INR 10–25 Cr**	Primarily revenue premium + brand equity

PoC 7: Closed-Loop PP — Financial Detail		
Parameter	**Value / Range**	**Working / Basis**
PoC setup capex (one-time)	**INR 3–6 Cr**	CIPET testing + BOM integration + Banyan Nation supply agreement
rPP cost saving Yr 1 (50–100 TPY)	**INR 0.2–0.5 Cr/yr**	50,000–100,000 kg × INR 45/kg = INR 22.5–45 L
rPP cost saving Yr 2 (200–400 TPY)	**INR 0.8–2.0 Cr/yr**	200,000–400,000 kg × INR 45/kg = INR 90L–1.8 Cr
rPP cost saving Yr 3 (500–1,000 TPY)	**INR 2.0–5.0 Cr/yr**	500,000–1,000,000 kg × INR 45/kg = INR 2.25–4.5 Cr
EPR credits Yr 1	**0.04–0.1 Cr/yr**	50–100 credits × INR 800–1,200/credit
EPR credits Yr 3	**0.5–1.0 Cr/yr**	500–1,000 credits × INR 1,000 avg
CCTS credits Yr 1	**INR 0.07–0.2 Cr/yr**	90–200 tCO_2-e × INR 800–1,200/t
CCTS credits Yr 3	**INR 0.72–2.0 Cr/yr**	900–2,000 tCO_2-e × INR 800–1,200/t
Total annual benefit Yr 3	**INR 3.2–8.0 Cr/yr**	rPP saving + EPR + CCTS combined
5-Year NPV (10%)	**INR 10–18 Cr**	High confidence given Banyan Nation commercial readiness
Payback	**24–36 months**	INR 4.5 Cr capex midpoint ÷ INR ~5 Cr/yr Yr 3 net benefit

PoC 8: Micro-Recycling Units — Financial Detail		
Parameter	**Value / Range**	**Working / Basis**

Phase 1 Capex (5 Tier 2 units)	**INR 12–30 Cr**	Site + equipment + CPCB certification + IT infrastructure
Phase 2 cumulative	**INR 30–60 Cr**	Scale to 20 units in franchise model
Phase 3 cumulative	**INR 75–150 Cr**	National hub-and-spoke 50+ units
Annual OpEx per unit	**INR 0.6–1.0 Cr/yr**	Labour + maintenance + power + logistics
Total Phase 1 OpEx (5 units)	**INR 3–5 Cr/yr**	5 units × INR 0.6–1.0 Cr
Revenue per ELV	**INR 3,000–8,000**	Material sales + EPR + cert fees
Annual revenue Phase 1 (5 units)	**INR 4.5–14 Cr/yr**	15,000 ELVs × 3,000 = 4.5 Cr; 30,000 × 8,000 = 24 Cr (stated 4.5–14 = central range)
CCTS credit revenue Phase 1	**INR 0.2–0.6 Cr/yr**	2,700–5,400 tCO_2-e × INR 800–1,200/t
5-Year NPV Phase 1 (10%)	**INR 10–20 Cr**	Scales significantly in later phases
Payback Phase 1	**36–60 months**	Dependent on ELV volume ramp and commodity prices

PoC 9: Reactive Extrusion — Financial Detail

Parameter	Value / Range	Working / Basis
Phase 1 (Feasibility) Capex	**INR 0.5–1 Cr**	Formulation trials at CIPET / CSIR-NCL
Phase 2 (Pilot) Capex	**INR 5–10 Cr**	Pilot extrusion line 50 kg/hr + QC lab
Phase 3 (Commercial) Capex	**INR 10–20 Cr**	Commercial line 300–500 kg/hr + BOM integration
Annual OpEx Phase 3	**INR 3 Cr/yr**	Energy + compatibiliser + testing + logistics
Revenue Phase 3 (2,000 TPY)	**INR 10–16 Cr/yr**	2,000 t × INR 55–80/kg
Polymer value uplift	**INR 30–55/kg premium**	INR 55–80/kg compatibilised vs INR 15–25/kg downcycled
CCTS credits Phase 3	**INR 0.5–1.2 Cr/yr**	3,000–6,000 tCO_2-e × INR 800–1,200/t
5-Year NPV (10%)	**INR 5–15 Cr**	Positive but lower certainty; higher technology risk than PoC 7
Payback	**36–60 months**	Phase 3 commercial scale required for payback

PoC 10: Pyrolysis — Financial Detail

Parameter	Value / Range	Working / Basis
10a Setup Capex (PP	**INR 2–5 Cr**	Rudra Environmental Solutions, Pune

pyrolysis)		
10b Setup Capex (ASR mixed)	**INR 3–8 Cr**	Pyrocrat Systems, Nagpur
10c Engagement Capex (P2O2P)	**INR 0.5–1 Cr**	Reliance Industries Jamnagar — cracker spec alignment
10d Setup Capex (ELT pyrolysis)	**INR 5–15 Cr**	Agson Global + Pyrum Innovations
Total capex across pathways	**INR 10–30 Cr**	5-year horizon across all four pathways
Annual OpEx (10a)	**INR 0.8–1.5 Cr/yr**	Energy + feedstock + QC
Annual OpEx (10d)	**INR 1.5–3.0 Cr/yr**	Energy + feedstock + QC
Annual revenue 10a	**INR 1.0–2.5 Cr/yr**	Pyrolysis oil sales + carbon credits
Annual revenue 10d	**INR 2.0–5.0 Cr/yr**	TDO + rCB sales + carbon credits
CO_2 credit revenue 10a	**INR 0.1–0.2 Cr/yr**	CCTS at INR 800–1,200/t
CO_2 credit revenue 10d	**INR 0.2–0.4 Cr/yr**	CCTS at INR 800–1,200/t
SAVWIPL disposal cost avoided (10a)	**INR 5–15/kg**	Currently paid for cement kiln co-processing
5-Year NPV 10a	**INR 3–7 Cr**	Conservative; 30–48 month payback
5-Year NPV 10d	**INR 5–12 Cr**	24–40 month payback; mature TRL
Long-term NPV 10c (Yr 7+)	**INR 8–20 Cr**	P2O2P circular route; ISCC premium materialises at scale

PoC 11: Digital Traceability — Financial Detail

Parameter	Value / Range	Working / Basis
Phase 1 Capex (Platform Onboarding)	**INR 0.5–1 Cr**	Recykal MoU + VWG India brand partner onboarding
Phase 2 Capex (SAP Integration)	**INR 0.5–1 Cr**	API to VWG SAP BOM + PO-certificate linkage
Phase 3 Capex (Full Certificate Suite)	**INR 1–2 Cr**	EPR credit trading + CBAM-ready embedded carbon data
Phase 4 Capex (Battery Passport)	**INR 0.5–1 Cr**	DLT passport for EV battery flows
Total capex (Phases 1–4)	**INR 2–4 Cr**	Platform integration + API + certification + battery passport
Annual OpEx	**INR 0.8 Cr/yr**	Platform fees + audit + certificates + maintenance
EPR credit revenue (Yr 3)	**INR 2–5 Cr/yr**	500–1,000 credits × INR 4,000–5,000/credit (MoRTH pricing)

CCTS carbon credit revenue (Yr 3)	**INR 1.5–3 Cr/yr**	Verified avoidance from PoCs 7, 8, 10; enabled by PoC 11
CBAM liability avoided (Yr 3)	**INR 1–3 Cr/yr**	Scope 3 verified data for steel/aluminium components
Audit cost savings	**INR 0.5–1 Cr/yr**	Automated digital trail replaces manual EPR reporting
5-Year NPV direct (10%)	**INR 5–9 Cr**	Excludes enabling value for PoCs 6, 7, 8, 10
Portfolio enabling value (strategic)	**INR 20–40 Cr**	Revenue enabled in other PoCs that depend on PoC 11 certification

B.6 Materials and Polymer Market Calculations

PP Market and Volume Calculations

Parameter	Value / Range	Working / Basis
PP content per vehicle (India avg)	**~19 kg**	Across 200–400 discrete plastic components (2024 figure)
PP share of total plastic content	**60–65%**	By weight in a typical Indian passenger vehicle
PP per vehicle (of 19 kg total)	**11.4–12.4 kg**	60–65% × 19 kg = 11.4–12.4 kg
India PV production (FY 2024–25)	**4.9 million vehicles/yr**	Industry figure
Total PP entering ELV stream (India)	**90,000–110,000 t/yr**	4.9M vehicles × ~19 kg/vehicle × 60–65% PP share = ~55,860–60,450 t (low case) to 4.9M × 22 kg × 65% (high). Report figure is broader market including older ELV fleet
SAVWIPL Pune plant production	**~180,000 vehicles/yr**	All VW Group India models
SAVWIPL PP consumption	**1,440–1,800 t/yr**	180,000 × 8–10 kg PP/vehicle = 1,440,000–1,800,000 kg
Automotive plastic intensity trend	**12 kg (2010) → 19 kg (2024)**	58% increase in plastic intensity per vehicle over 14 years
India PET collection rate	**>70%**	Outperforms Japan, EU, US for beverage PET stream
Ganesha Ecopet rPET capacity	**42,000 t/yr**	Combined Warangal + Kanpur facilities; food-grade rPET

PP Price and Cost Differentials		
Parameter	**Value / Range**	**Working / Basis**
Virgin PP (domestic, Reliance/GAIL)	**INR 140–155/kg**	Current domestic market price
Automotive-grade rPP (Banyan Nation)	**INR 95–110/kg**	Commercially validated automotive circularity at scale
Unit cost saving (midpoint)	**INR 45/kg**	(INR 147.5 − INR 102.5) midpoints = INR 45/kg used in model
% cost saving	**25–35%**	(45 ÷ 140) = 32%; (45 ÷ 155) = 29%; stated as 25–35%
rPP CO_2-e intensity	**0.3 t CO_2-e/t**	vs virgin PP 2.1 t; reduction = 1.8 t/t
rPP energy requirement vs virgin	**~15% of virgin PP**	rPP requires 85% less production energy
Virgin ABS CO_2-e	**4.5 kg CO_2-e/kg**	Imported virgin ABS
rABS CO_2-e reduction	**−2.8 kg (−62%)**	Post-industrial domestic rABS
Primary aluminium CO_2-e	**8.0 kg CO_2-e/kg**	Smelter route
Secondary/recycled aluminium reduction	**−6.5 kg (−81%)**	Electric arc route with recycled feedstock
Indian BF/BOF steel vs EU EAF	**2.5 vs <0.5 t CO_2/t steel**	India average 5× more carbon-intensive than EU EAF

ELV Regulatory Thresholds and Compliance Calculations		
Parameter	**Value / Range**	**Working / Basis**
PWM Rules recycled content mandate	**30% by FY 2025–26**	Plastic Waste Management Rules 2022 — rigid plastics
PWM Rules escalation	**60% by FY 2028–29**	Mandatory; enforceable law with penalties
ELV Rules 2025 recovery rate target	**80% by weight by 2030**	Escalating from 70% in first compliance year
ELV Rules non-compliance penalty	**INR 15,000–50,000/vehicle**	Per vehicle for non-compliance with RVSF channelling
CAFE III target (Apr 2027)	**88.4 gCO_2/km**	Vs CAFE II Phase 1 = 130 g; Phase 2 = 113 g
CAFE IV draft (2030)	**~70 gCO_2/km**	Under development
EPR credit rate (MoRTH)	**1 credit = 1 MT verified recycling**	Tradeable on VAHAN platform
EPR credit price	**INR 4,000–**	MoRTH current pricing (PoC 11 reference)

	5,000/credit	
EPR credit price (PoC 7 reference)	**INR 800–1,200/credit**	Lower range used in PoC 7 model; conservative

B.7 Portfolio-Level Financial Model

The portfolio-level financial model aggregates the individual PoC economics. The total capex, annual benefit, and NPV ranges are derived by summing the PoC-level estimates. The blended payback is a weighted average based on the relative capex share of each PoC.

Portfolio NPV (Base) = Σ Individual PoC 5-Year NPVs | Discount Rate: 10% | Horizon: 5 years

Optimistic scenario adds: FAME-III confirmation + CCTS at INR 1,000/t + CBAM enforcement against VWG India export components

PoC	Capex (INR Cr)	Annual OpEx (INR Cr)	Annual Benefit Yr 3 (INR Cr)	Payback	5-Yr NPV (INR Cr)	Primary Benefit Driver
PoC 1 – Supplier Routing	**1–3**	0.5	**3–6**	18–30 mo	**6–12**	Carbon cost avoidance + procurement efficiency + CBAM mitigation
PoC 2 – Micro-Grid	**8–15**	1.0	**6–12**	24–36 mo	**12–25**	Energy cost savings + RPO compliance + Scope 2 carbon credits
PoC 3 – DfD in CAD	**1–2**	0.3	**Long-dated**	Strategic	**Strategic**	ELV compliance headroom + rPP yield uplift + EPR credits (from 2030)
PoC 4 – AI Material Sub	**2–4**	0.8	**4–8**	24–36 mo	**7–14**	Material cost saving (rPP/rPET at 25–40% discount) + CCTS credits
PoC 5 – Packaging Loop	**2–4**	0.5	**4–7**	18–36 mo	**8–12**	Packaging procurement savings + waste

						management savings + logistics
PoC 6 – Green UX	**2–5**	0.5	**3–8**	Product cycle	**10–25**	Revenue premium per vehicle + brand equity (inclusive of brand value)
PoC 7 – Closed-Loop PP	**3–6**	1–2	**3–8**	24–36 mo	**10–18**	rPP material savings + EPR credits + CCTS carbon credits
PoC 8 – Micro-Recycling	**12–30**	3–5	**8–15**	36–60 mo	**10–20**	ELV processing revenue + EPR credits + carbon credits (Phase 1 only)
PoC 9 – Reactive Extrusion	**10–20**	3.0	**10–16**	36–60 mo	**5–15**	Polymer value uplift (INR 30–55/kg premium vs downcycled)
PoC 10 – Pyrolysis	**10–30**	2–5	**3–8**	24–48 mo	**3–20**	Pyrolysis oil revenue + carbon credits; 10d fastest; 10c long-term prize
PoC 11 – Traceability	**2–4**	0.8	**4–9**	30–42 mo	**5–9**	EPR credit revenue + CCTS credits + CBAM liability avoidance
PORTFOLIO TOTAL	**51–103**	–	**41–81**	**30–42 mo**	**70–140**	Base case; 150–280 Cr optimistic (FAME-III + CCTS INR 1,000 + CBAM)

Portfolio Aggregation Working

Parameter	Value / Range	Working / Basis
Total capex range	**INR 51–103 Cr**	Sum of individual PoC capex lower bounds (51 Cr) and upper bounds (103 Cr)
Combined annual benefit Year 3	**INR 41–81 Cr/yr**	Sum of all PoC annual benefit streams: packaging (3.8–6.8) + Micro-Grid (6–12) + rPP (3.2–8) + Micro-Recycling (5–15) + Reactive Ext (10–16) + others; rounded per report
Portfolio payback (blended)	**30–42 months**	Capex-weighted average; anchored by PoC 5 (fastest) and PoC 8 (slowest)

5-Year NPV base case	**INR 70–140 Cr**	Aggregated PoC NPVs at 10% discount rate; PoC 5, 2, 7 largest contributors
5-Year NPV optimistic scenario	**INR 150–280 Cr**	FAME-III confirmed Q3 2026 + CCTS INR 1,000/t + CBAM enforcement on auto components
Strategic optionality value	**INR 50–80 Cr**	EPR compliance headroom + first-mover rPP positioning + ESG rating uplift (enterprise value add)
Total enterprise value (base + optionality)	**INR 120–220 Cr**	Portfolio NPV + strategic optionality

B.8 CCTS Carbon Credit Monetisation Schedule

The Carbon Credit Trading Scheme (CCTS) monetisation table consolidates the verified emission avoidance volumes from the three primary credit-generating PoCs (PoC 7, 10a, 2) at Year 3 steady state. The formula applied across all PoCs is:

$$\text{CCTS Revenue} = \text{Annual } tCO_2\text{-e Avoided} \times \text{CCTS Price (INR/tonne)}$$

CCTS assumed live 2026. Base price range: INR 800–1,200/tonne. Sensitivity: model profitable at INR 500/tonne for PoC 7 and PoC 10.

CCTS Credit Source	Annual Volume (Yr 3)	Price Range	Annual Revenue
rPP substitution — PoC 7 (500 TPY)	900–1,000 tCO_2-e	INR 800–1,200/t	**INR 72–120 lakhs**
Pyrolysis oil — PoC 10a (500 TPY)	750–1,000 tCO_2-e	INR 800–1,200/t	**INR 60–120 lakhs**
Micro-grid buffering — PoC 2	500–800 tCO_2-e	INR 800–1,200/t	**INR 40–96 lakhs**
TOTAL	**2,150–2,800 tCO_2-e**	—	**INR 1.72–3.36 Cr/year**

CCTS Working Notes		
Parameter	**Value / Range**	**Working / Basis**
rPP substitution credit basis (PoC 7)	**1.8–2.0 t CO_2-e per t rPP displaced**	Each tonne of rPP replacing virgin PP avoids ~1.8–2.0 t CO_2-e. At 500 TPY: 500 × 1.8 = 900 t min; 500 × 2.0 = 1,000 t max. Rounded

		to 900–1,000 tCO_2-e
Pyrolysis credit basis (PoC 10a)	**1.5–2.0 t CO_2-e per tonne processed**	Vs landfill counterfactual (incl. methane). At 500 TPY: 500 × 1.5 = 750 t; 500 × 2.0 = 1,000 t
Micro-grid credit basis (PoC 2)	**~0.13 kg CO_2-e/kWh grid displacement**	Grid saved from load shift × (0.71−0.50) effective reduction. At 500,000–800,000 MWh saved equivalent: 500,000 × 0.21 = 105,000 t max; stated 500–800 t = conservative small-phase estimate
CCTS revenue (INR 72–120 lakhs on PoC 7)	**900 t × INR 800 = INR 72 L; 1,000 t × INR 1,200 = INR 120 L**	Arithmetic check: confirmed
CCTS revenue (INR 60–120 lakhs on PoC 10a)	**750 t × INR 800 = INR 60 L; 1,000 t × INR 1,200 = INR 120 L**	Arithmetic check: confirmed
CCTS revenue (INR 40–96 lakhs on PoC 2)	**500 t × INR 800 = INR 40 L; 800 t × INR 1,200 = INR 96 L**	Arithmetic check: confirmed
Total CCTS working (INR 1.72–3.36 Cr)	**(72+60+40) lakhs = INR 1.72 Cr low; (120+120+96) = INR 3.36 Cr high**	Arithmetic check: confirmed

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Index

A

B

C

D

E

F

G

H

I

M

N

P

R

S

T

V

W

X

Z

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